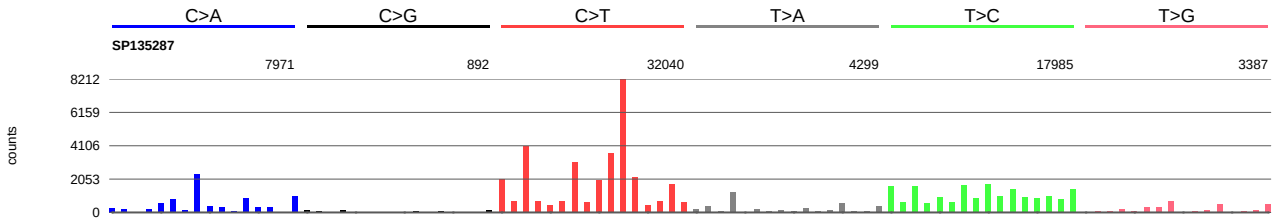
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SP135287
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892
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8212
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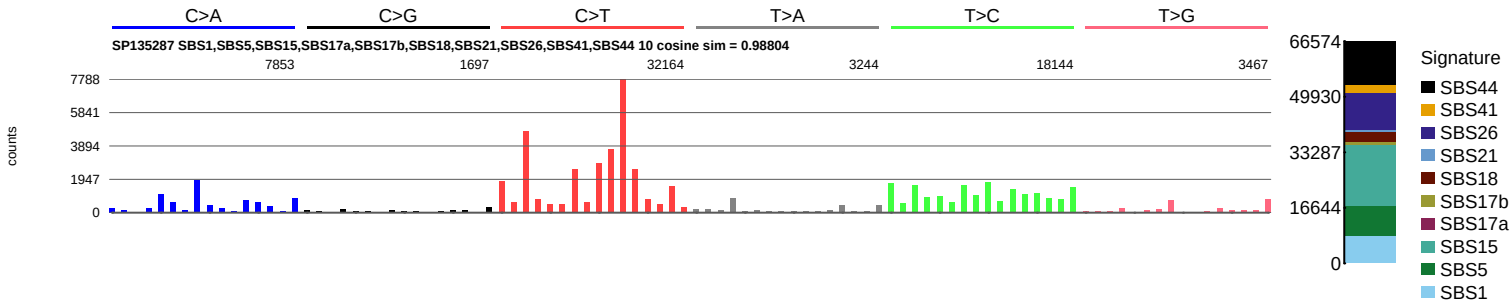
2053
0
counts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SP135287 SBS1,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS41,SBS44 10 cosine sim = 0.98804
7853
1697
32164
3244
18144
3467
7788
5841
3894
1947
0
counts
66574
49930
33287
16644
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SBS18
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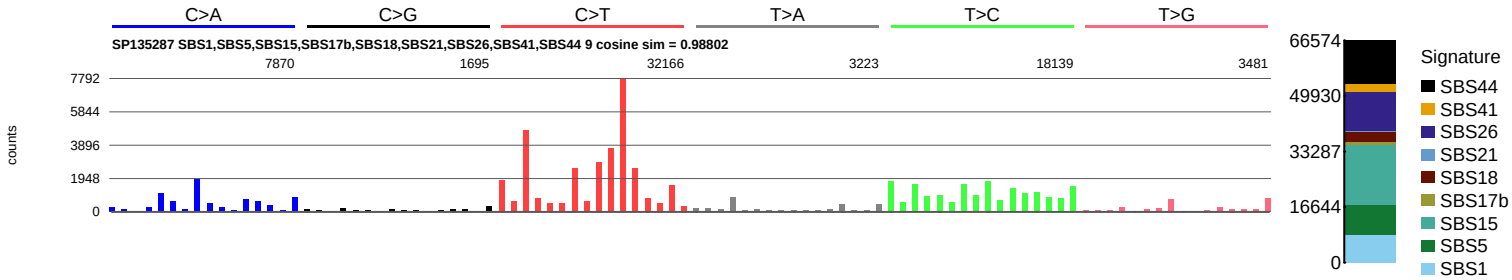
SBS17a
SBS15
SBS5
SBS1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1695
32166
3223
18139
3481
7792
5844
3896
1948
0
counts
66574
49930
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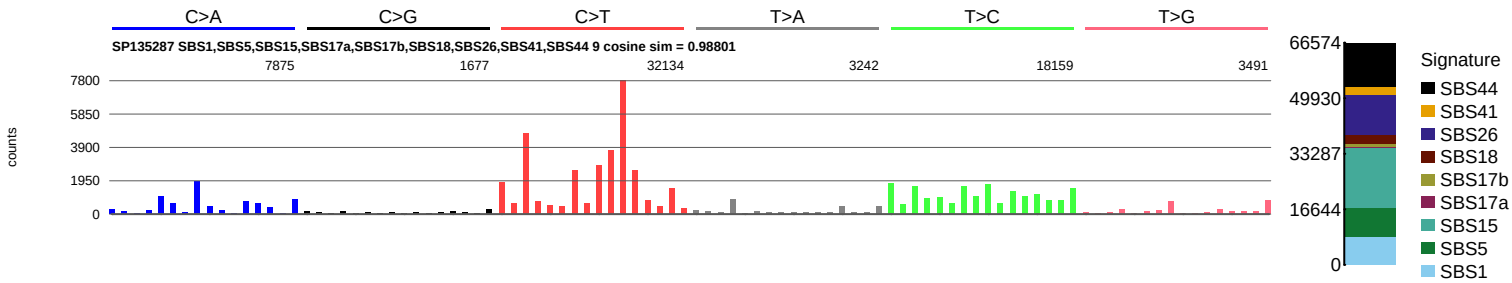
SBS5
SBS1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1677
32134
3242
18159
3491
7800
5850
3900
1950
0
counts
66574
49930
33287
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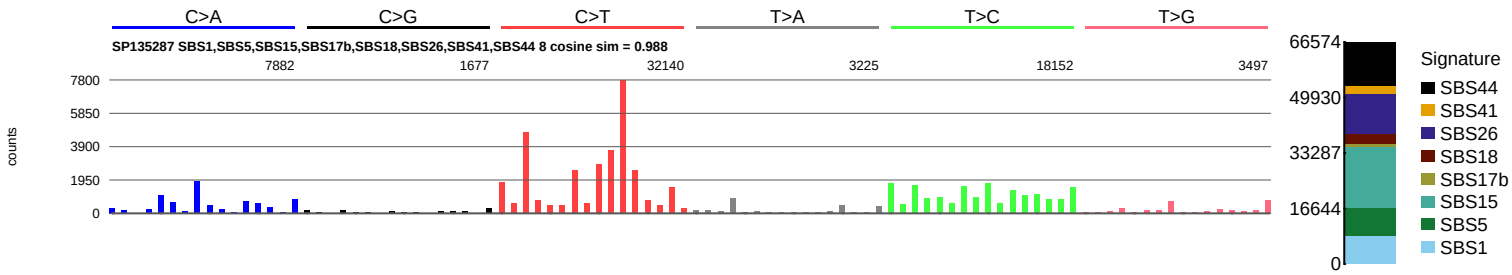
SBS5
SBS1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SP135287 SBS1,SBS5,SBS15,SBS17b,SBS18,SBS26,SBS41,SBS44 8 cosine sim = 0.988
7882
1677
32140
3225
18152
3497
7800
5850
3900
1950
0
counts
66574
49930
33287
16644
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Signature
SBS44
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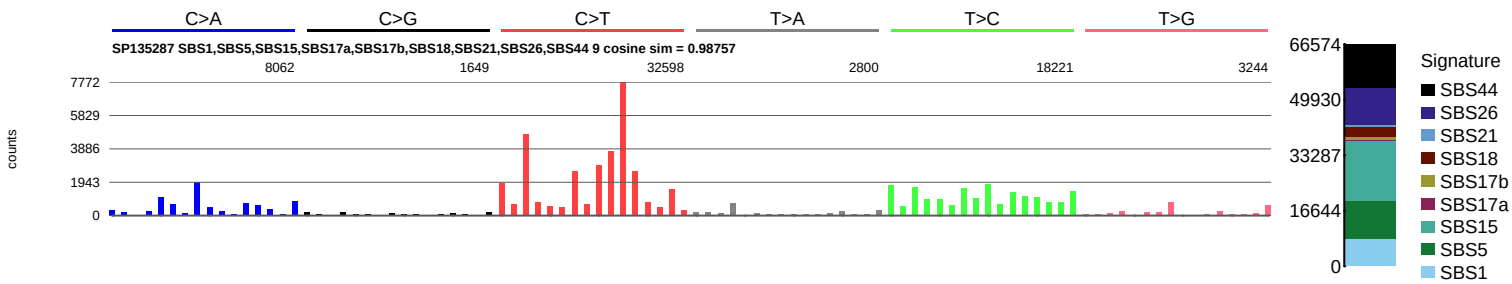
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SP135287 SBS1,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS44 9 cosine sim = 0.98757
8062
1649
32598
2800
18221
3244
7772
5829
3886
1943
0
counts
66574
49930
33287
16644
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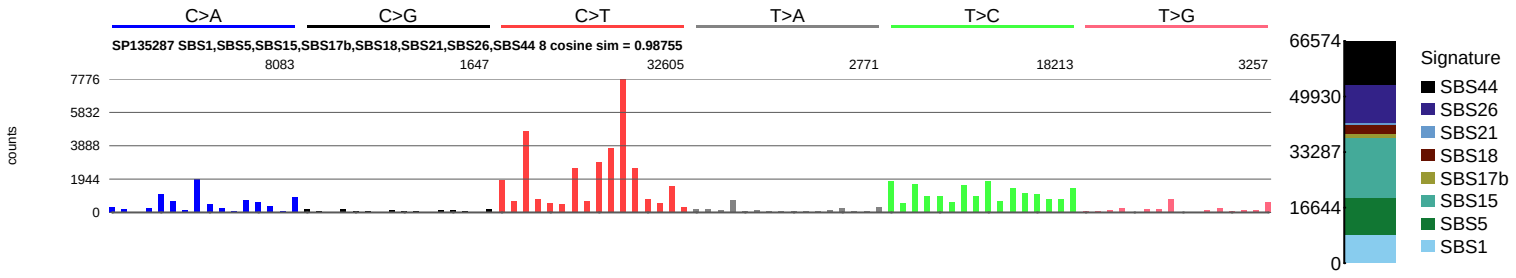
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS1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1647
32605
2771
18213
3257
counts
7776
5832
3888
1944
0
66574
49930
33287
16644
0
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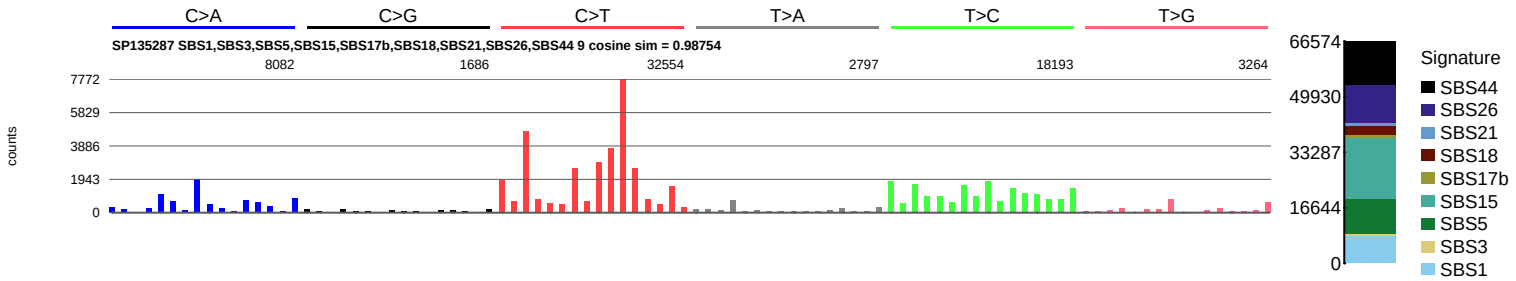
SBS18
SBS17b
SBS15
SBS5
SBS1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T>G
SP135287 SBS1,SBS3,SBS5,SBS15,SBS17b,SBS18,SBS21,SBS26,SBS44 9 cosine sim = 0.98754
8082
1686
32554
2797
18193
3264
counts
7772
5829
3886
1943
0
66574
49930
33287
16644
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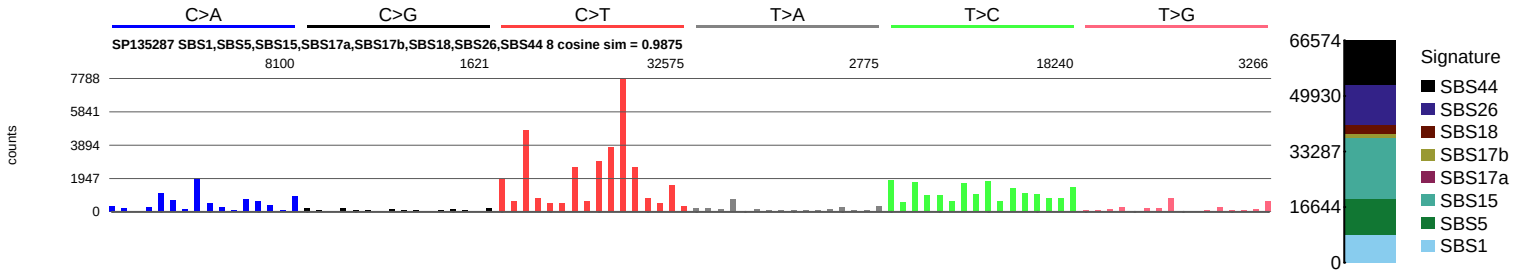
SBS18
SBS17b
SBS15
SBS5
SBS3
SBS1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SP135287 SBS1,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS44 8 cosine sim = 0.9875
8100
1621
32575
2775
18240
3266
counts
7788
5841
3894
1947
0
66574
49930
33287
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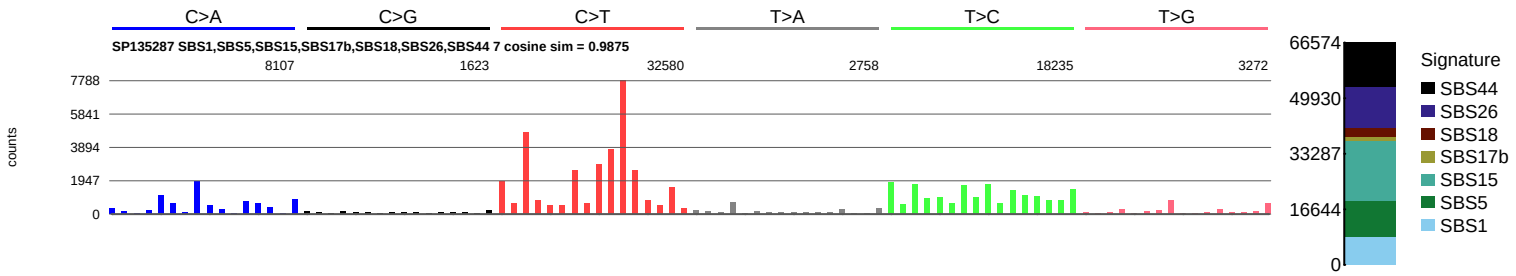
SBS5
SBS1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SP135287 SBS1,SBS5,SBS15,SBS17b,SBS18,SBS26,SBS44 7 cosine sim = 0.9875
8107
1623
32580
2758
18235
3272
counts
7788
5841
3894
1947
0
66574
49930
33287
16644
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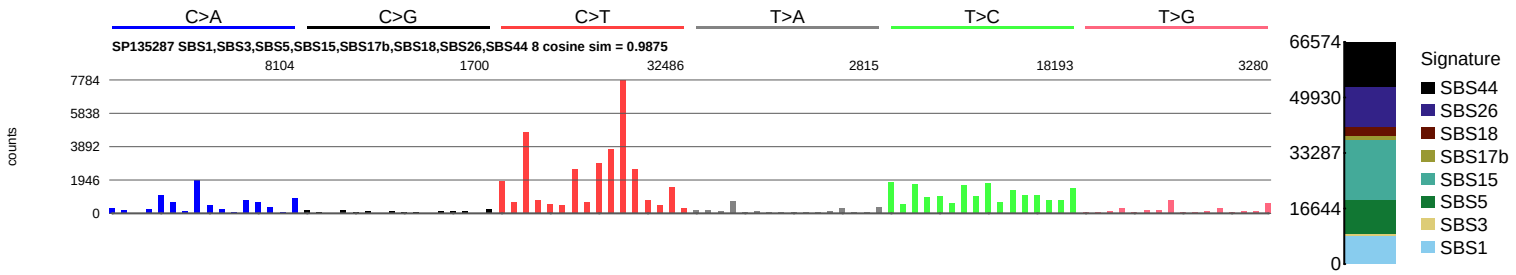
SBS18
SBS17b
SBS15
SBS5
SBS1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C>T
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T>C
T>G
SP135287 SBS1,SBS3,SBS5,SBS15,SBS17b,SBS18,SBS26,SBS44 8 cosine sim = 0.9875
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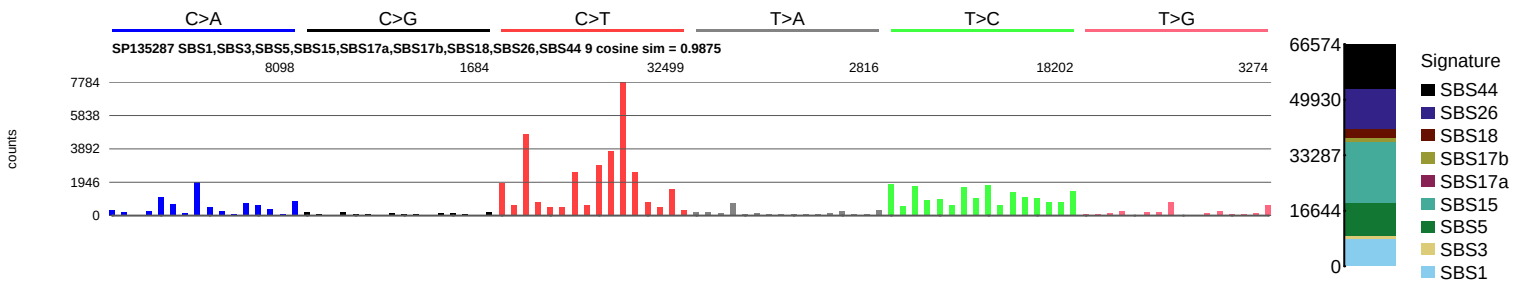
1700
32486
2815
18193
3280
counts
7784
5838
3892
1946
0
66574
49930
33287
16644
0
Signature
SBS44
SBS26
SBS18
SBS17b
SBS15
SBS5
SBS3
SBS1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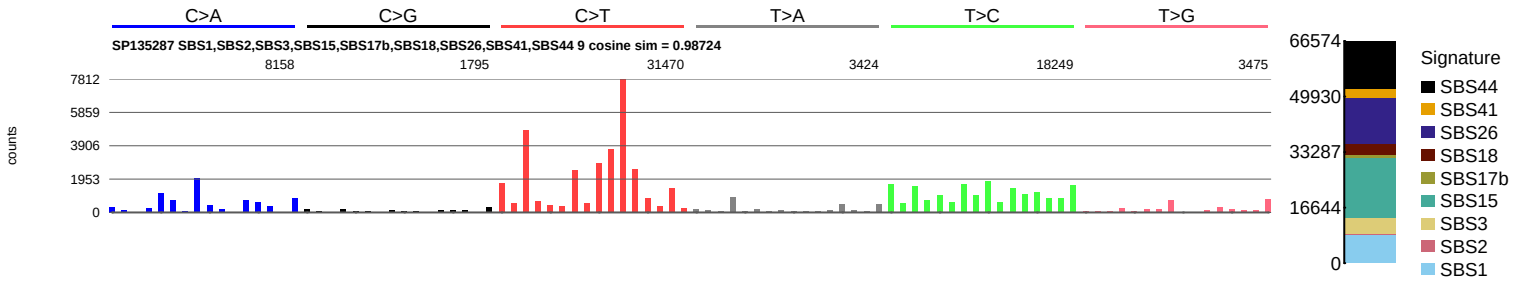
SP135287 SBS1,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS44 9 cosine sim = 0.9875
8098
1684
32499
2816
18202
3274
counts
7784
5838
3892
1946
0
66574
49930
33287
16644
0
Signature
SBS44
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS1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T>C
T>G
SP135287 SBS1,SBS2,SBS3,SBS15,SBS17b,SBS18,SBS26,SBS41,SBS44 9 cosine sim = 0.98724
8158
1795
31470
3424
18249
3475
counts
7812
5859
3906
1953
0
66574
49930
33287
16644
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SBS41
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SBS18
SBS17b
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SBS2
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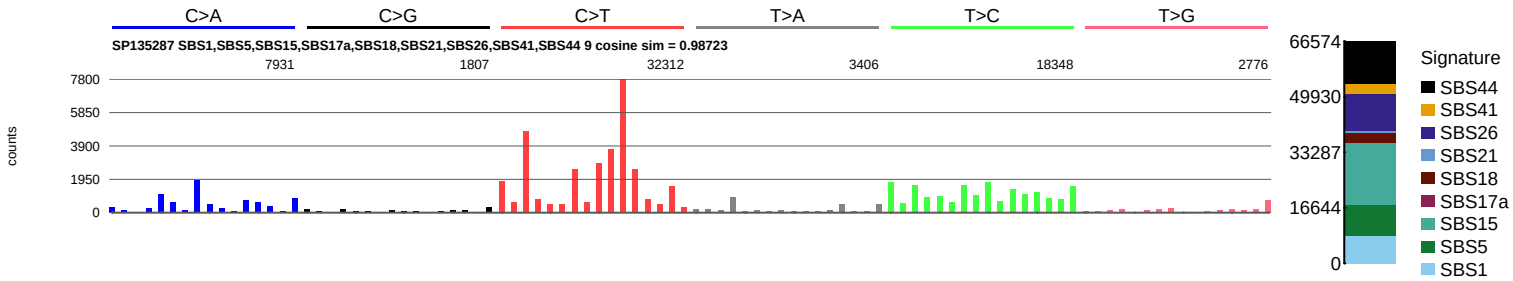
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C>G
C>T
T>A
T>C
T>G
SP135287 SBS1,SBS5,SBS15,SBS17a,SBS18,SBS21,SBS26,SBS41,SBS44 9 cosine sim = 0.98723
7931
1807
32312
3406
18348
2776
counts
7800
5850
3900
1950
0
66574
49930
33287
16644
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Signature
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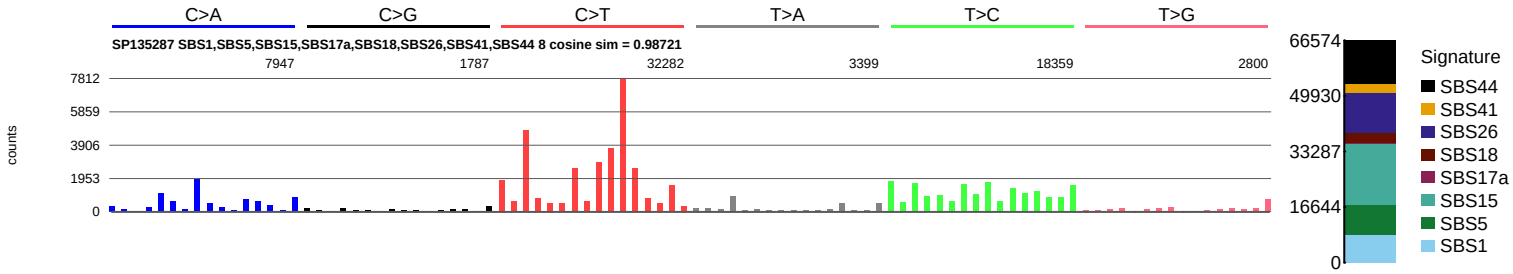
SBS41
SBS26
SBS21
SBS18
SBS17a
SBS15
SBS5
SBS1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C>G
C>T
T>A
T>C
T>G
SP135287 SBS1,SBS5,SBS15,SBS17a,SBS18,SBS26,SBS41,SBS44 8 cosine sim = 0.98721
7947
1787
32282
3399
18359
2800
counts
7812
5859
3906
1953
0
66574
49930
33287
16644
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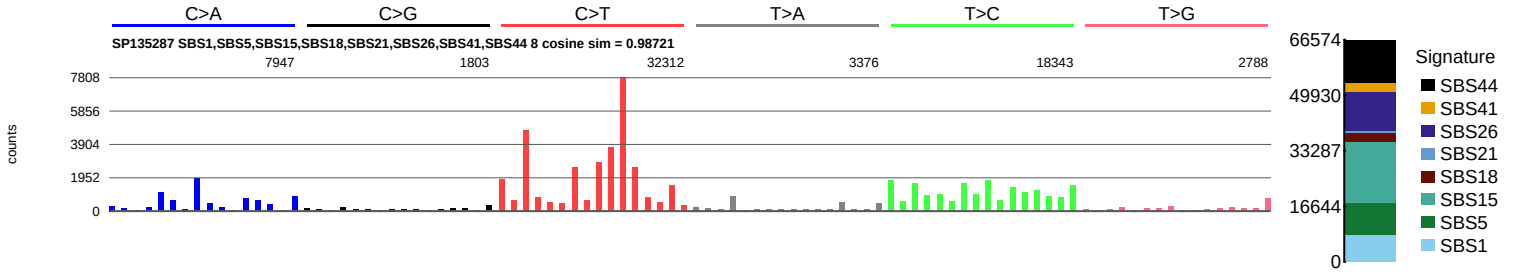
Signature
SBS44
SBS41
SBS26
SBS18
SBS17a
SBS15
SBS5
SBS1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T>A
T>C
T>G
SP135287 SBS1,SBS5,SBS15,SBS18,SBS21,SBS26,SBS41,SBS44 8 cosine sim = 0.98721
7947
1803
32312
3376
18343
2788
counts
7808
5856
3904
1952
0
66574
49930
33287
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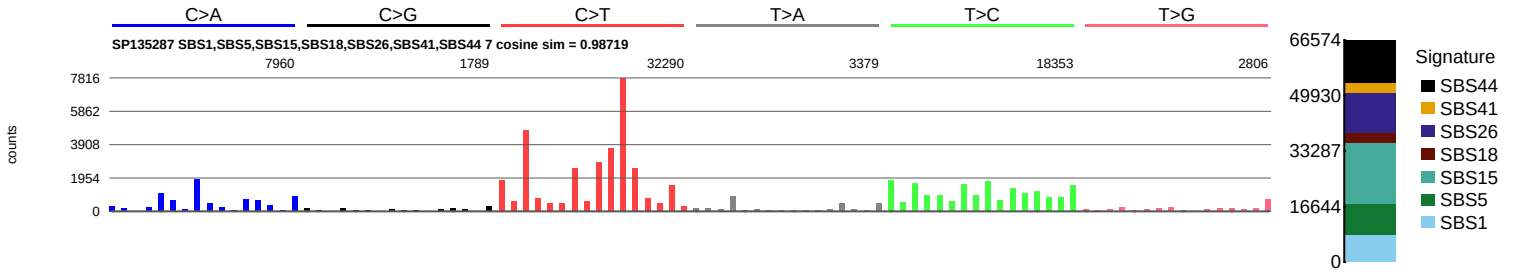
SBS5
SBS1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SP135287 SBS1,SBS5,SBS15,SBS18,SBS26,SBS41,SBS44 7 cosine sim = 0.98719
7960
1789
32290
3379
18353
2806
counts
7816
5862
3908
1954
0
66574
49930
33287
16644
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SBS26
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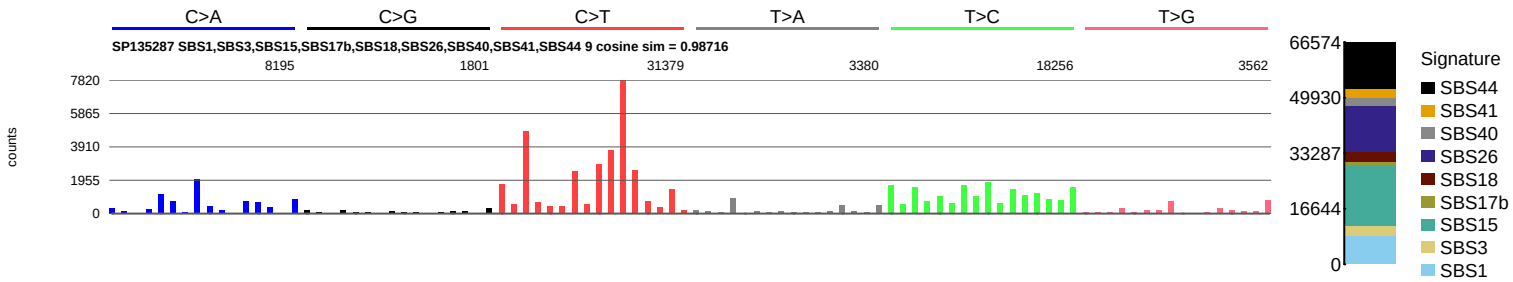
SBS5
SBS1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T>C
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SP135287 SBS1,SBS3,SBS15,SBS17b,SBS18,SBS26,SBS40,SBS41,SBS44 9 cosine sim = 0.98716
8195
1801
31379
3380
18256
3562
counts
7820
5865
3910
1955
0
66574
49930
33287
16644
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SBS41
SBS40
SBS26
SBS18
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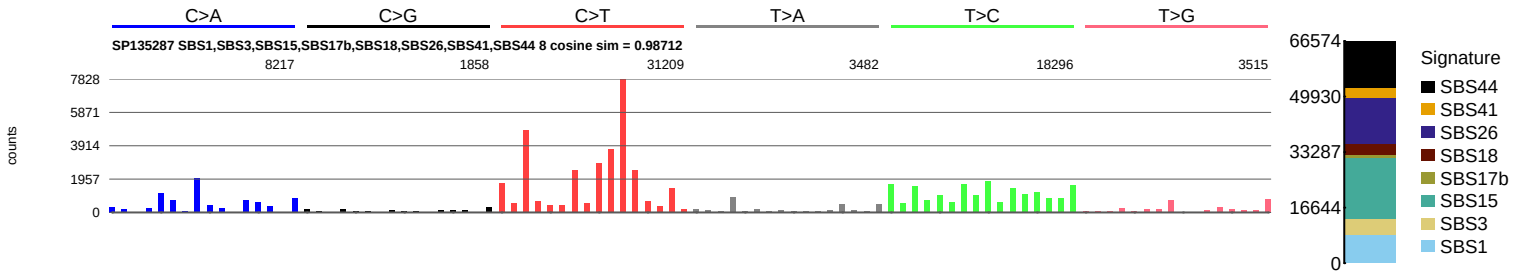
SBS15
SBS3
SBS1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SP135287 SBS1,SBS3,SBS15,SBS17b,SBS18,SBS26,SBS41,SBS44 8 cosine sim = 0.98712
8217
1858
31209
3482
18296
3515
counts
7828
5871
3914
1957
0
66574
49930
33287
16644
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SBS41
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SBS18
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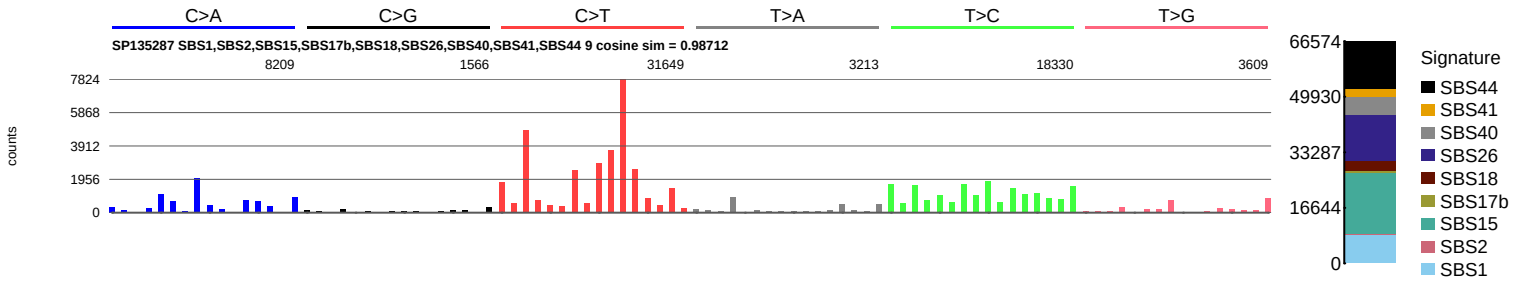
SBS3
SBS1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T>G
SP135287 SBS1,SBS2,SBS15,SBS17b,SBS18,SBS26,SBS40,SBS41,SBS44 9 cosine sim = 0.98712
8209
1566
31649
3213
18330
3609
counts
7824
5868
3912
1956
0
66574
49930
33287
16644
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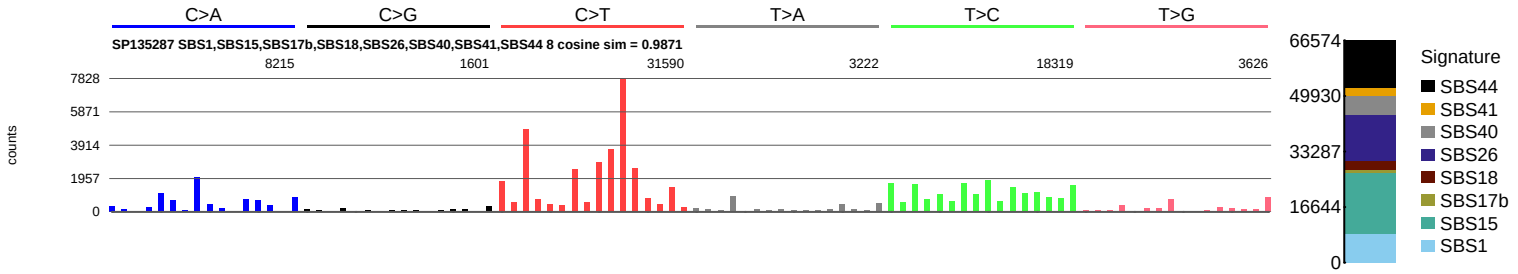
SBS15
SBS2
SBS1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T>G
SP135287 SBS1,SBS15,SBS17b,SBS18,SBS26,SBS40,SBS41,SBS44 8 cosine sim = 0.9871
8215
1601
31590
3222
18319
3626
counts
7828
5871
3914
1957
0
66574
49930
33287
16644
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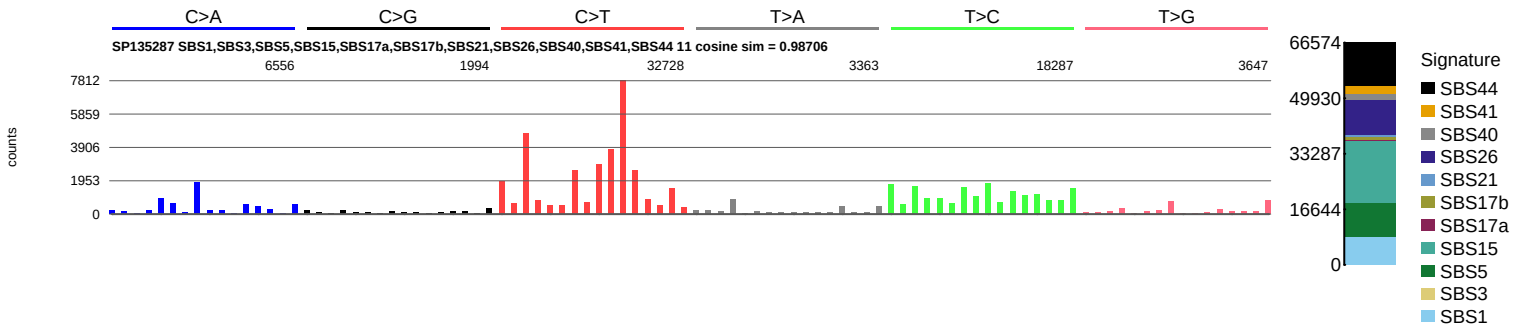
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS15
SBS1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6556
1994
32728
3363
18287
3647
counts
7812
5859
3906
1953
0
66574
49930
33287
16644
0
Signature
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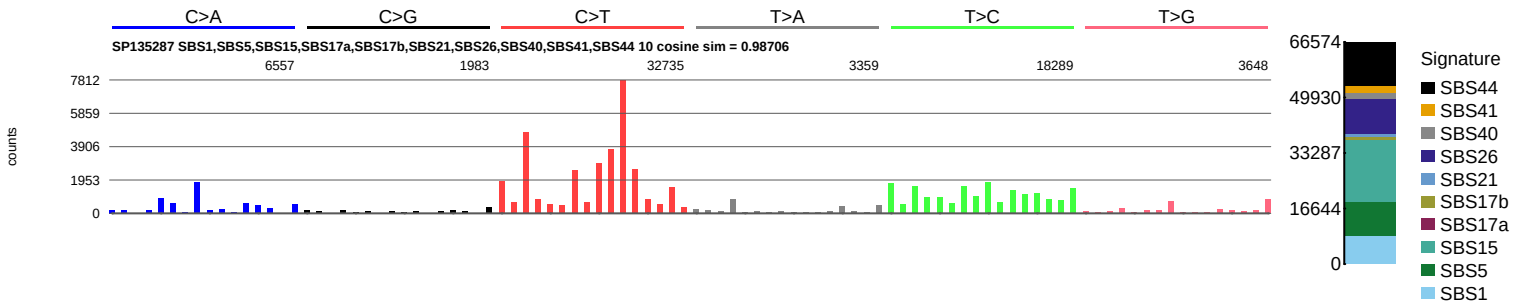
SBS26
SBS21
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS1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T>A
T>C
T>G
SP135287 SBS1,SBS5,SBS15,SBS17a,SBS17b,SBS21,SBS26,SBS40,SBS41,SBS44 10 cosine sim = 0.98706
6557
1983
32735
3359
18289
3648
counts
7812
5859
3906
1953
0
66574
49930
33287
16644
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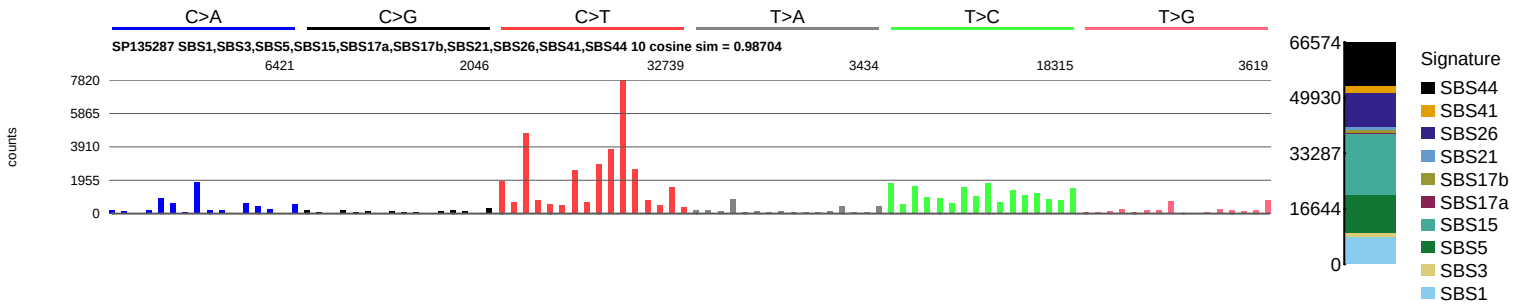
SBS5
SBS1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T>C
T>G
SP135287 SBS1,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS21,SBS26,SBS41,SBS44 10 cosine sim = 0.98704
6421
2046
32739
3434
18315
3619
counts
7820
5865
3910
1955
0
66574
49930
33287
16644
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SBS41
SBS26
SBS21
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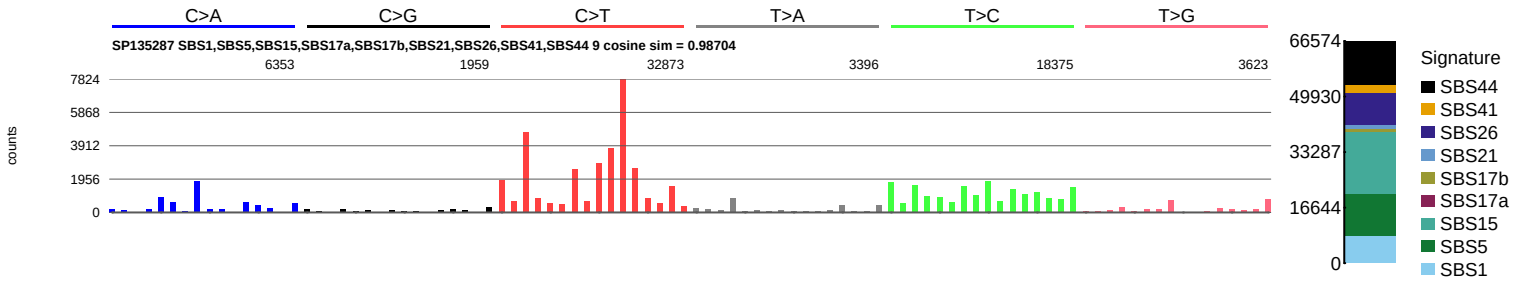
SBS3
SBS1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T>C
T>G
SP135287 SBS1,SBS5,SBS15,SBS17a,SBS17b,SBS21,SBS26,SBS41,SBS44 9 cosine sim = 0.98704
6353
1959
32873
3396
18375
3623
counts
7824
5868
3912
1956
0
66574
49930
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SBS44
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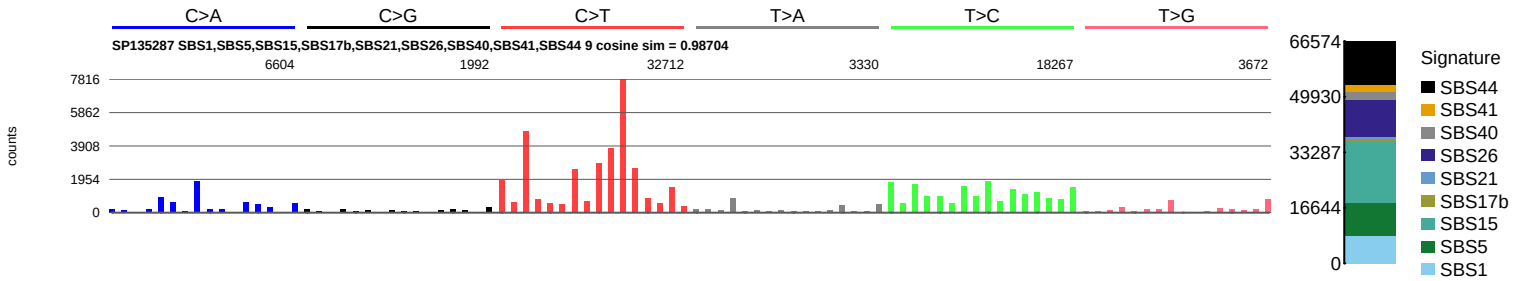
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SP135287 SBS1,SBS5,SBS15,SBS17b,SBS21,SBS26,SBS40,SBS41,SBS44 9 cosine sim = 0.98704
6604
1992
32712
3330
18267
3672
counts
7816
5862
3908
1954
0
66574
49930
33287
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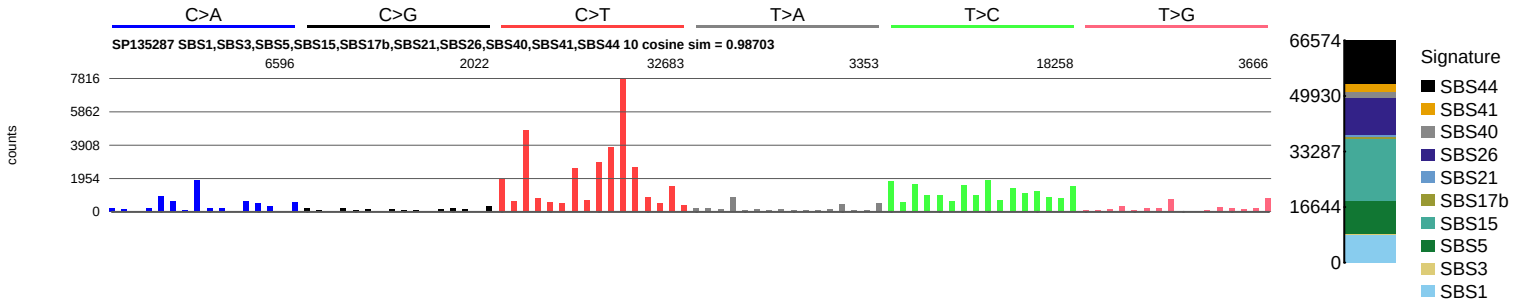
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SP135287 SBS1,SBS3,SBS5,SBS15,SBS17b,SBS21,SBS26,SBS40,SBS41,SBS44 10 cosine sim = 0.98703
6596
2022
32683
3353
18258
3666
counts
7816
5862
3908
1954
0
66574
49930
33287
16644
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SBS17b
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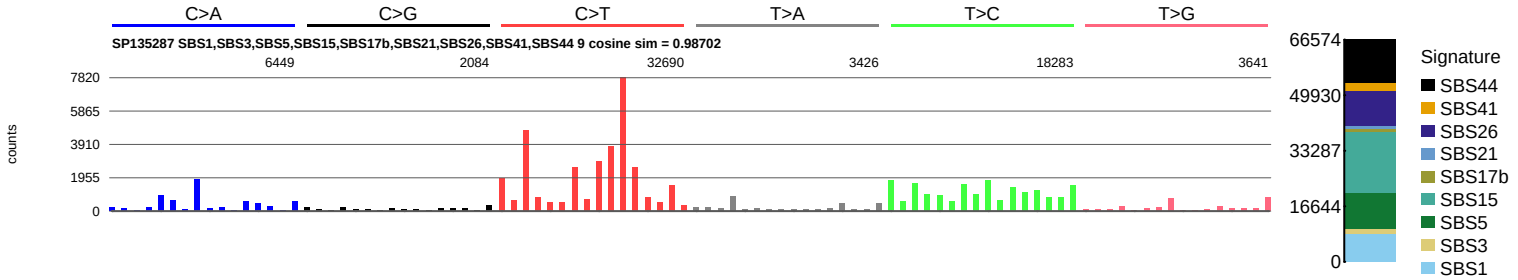
SBS5
SBS3
SBS1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SP135287 SBS1,SBS3,SBS5,SBS15,SBS17b,SBS21,SBS26,SBS41,SBS44 9 cosine sim = 0.98702
6449
2084
32690
3426
18283
3641
counts
7820
5865
3910
1955
0
66574
49930
33287
16644
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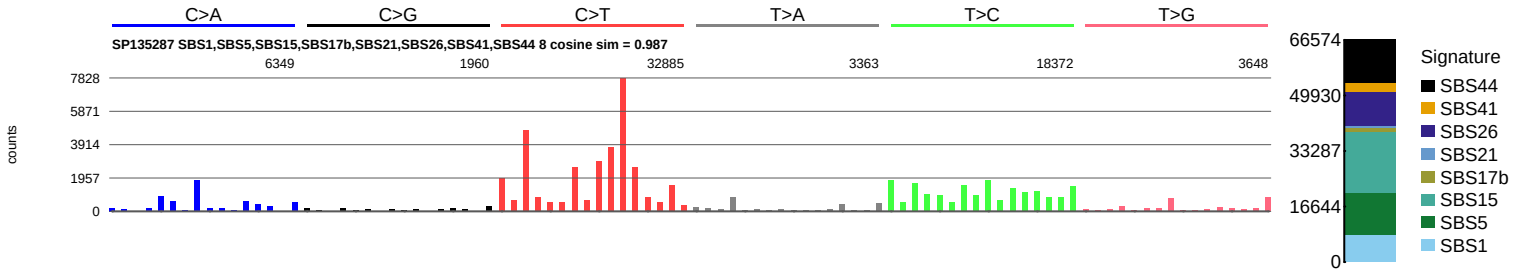
SBS21
SBS17b
SBS15
SBS5
SBS3
SBS1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C>G
C>T
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T>C
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SP135287 SBS1,SBS5,SBS15,SBS17b,SBS21,SBS26,SBS41,SBS44 8 cosine sim = 0.987
6349
1960
32885
3363
18372
3648
counts
7828
5871
3914
1957
0
66574
49930
33287
16644
0
Signature
SBS44
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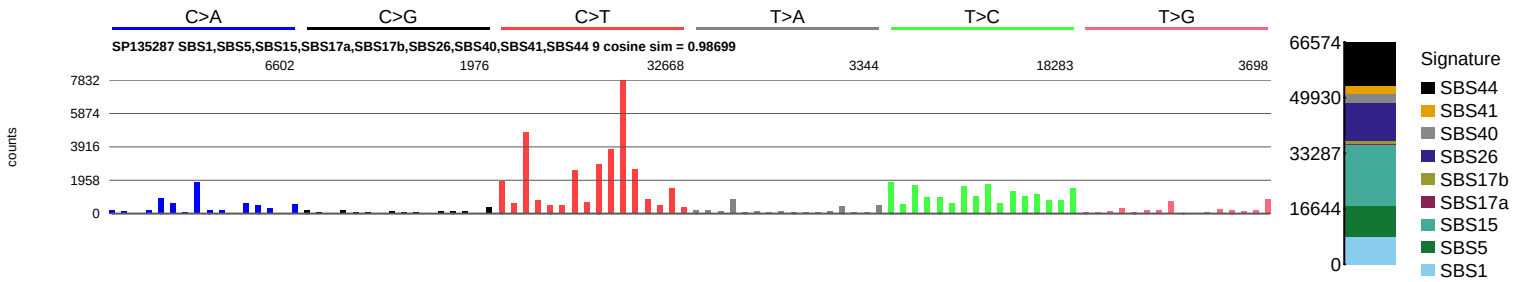
SBS21
SBS17b
SBS15
SBS5
SBS1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C>G
C>T
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SP135287 SBS1,SBS5,SBS15,SBS17a,SBS17b,SBS26,SBS40,SBS41,SBS44 9 cosine sim = 0.98699
6602
1976
32668
3344
18283
3698
counts
7832
5874
3916
1958
0
66574
49930
33287
16644
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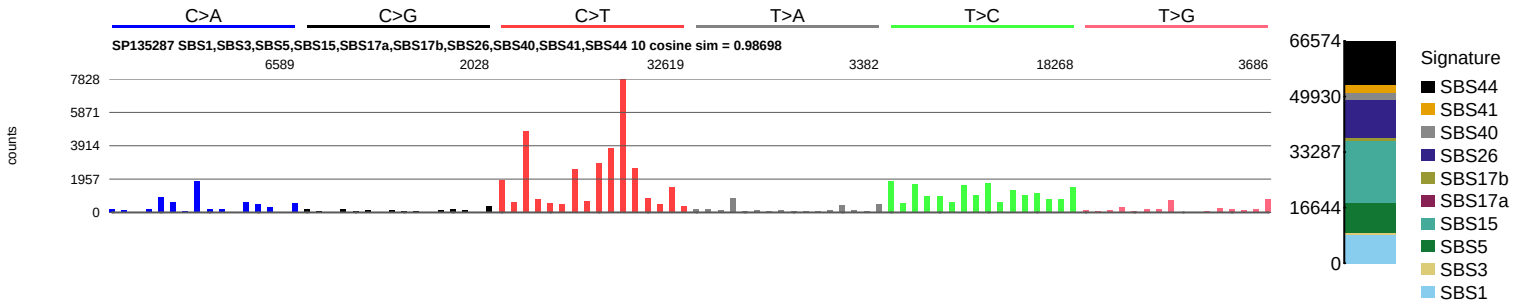
SBS17a
SBS15
SBS5
SBS1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T>C
T>G
SP135287 SBS1,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS26,SBS40,SBS41,SBS44 10 cosine sim = 0.98698
6589
2028
32619
3382
18268
3686
counts
7828
5871
3914
1957
0
66574
49930
33287
16644
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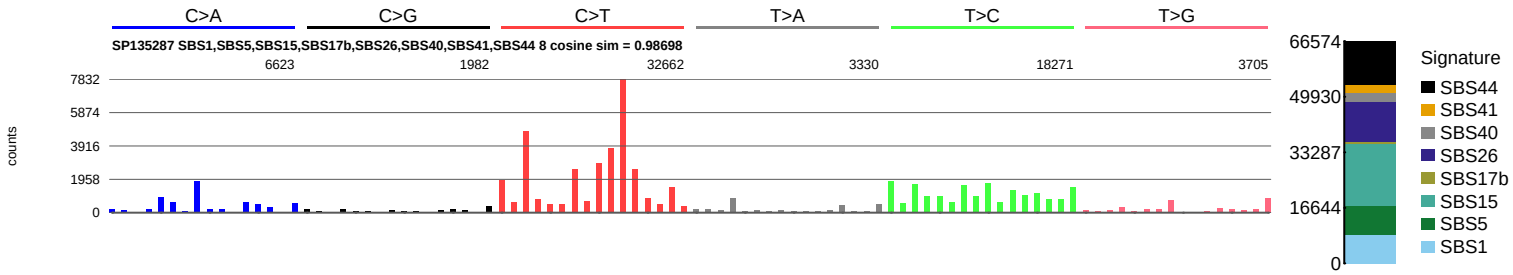
SBS40
SBS26
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS1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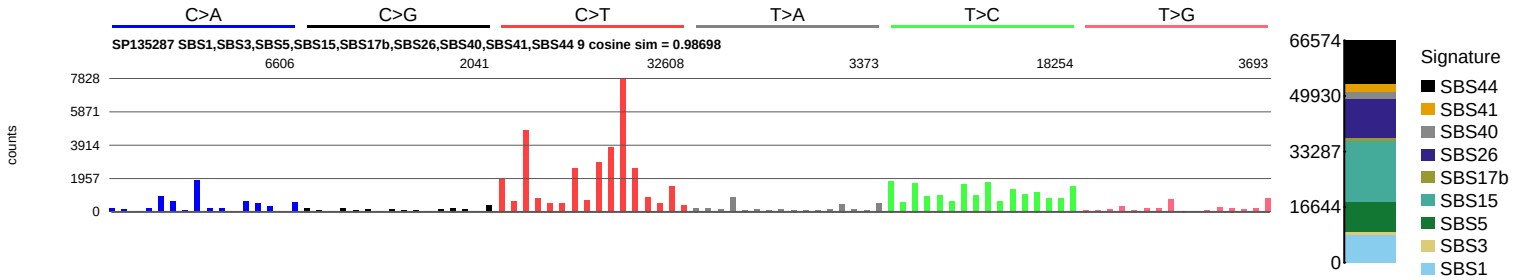
C>G
C>T
T>A
T>C
T>G
SP135287 SBS1,SBS5,SBS15,SBS17b,SBS26,SBS40,SBS41,SBS44 8 cosine sim = 0.98698
6623
1982
32662
3330
18271
3705
counts
7832
5874
3916
1958
0
66574
49930
33287
16644
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS17b
SBS15
SBS5
SBS1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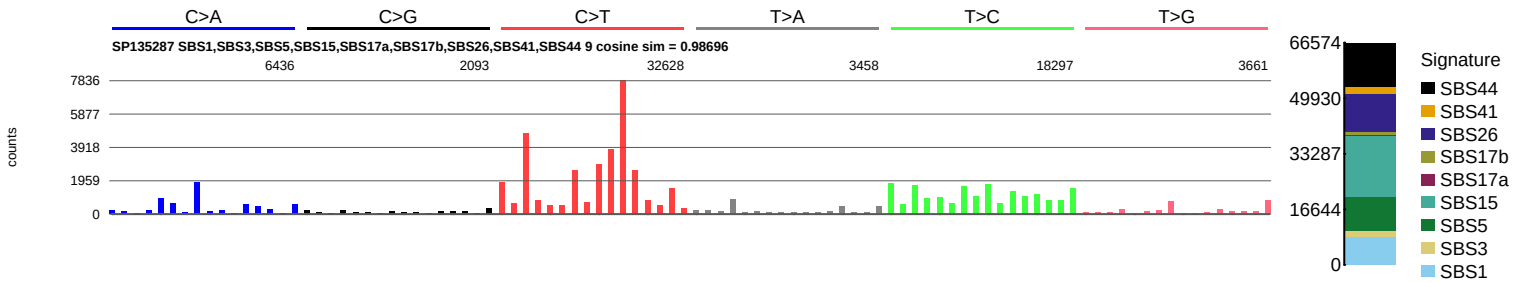
C>G
C>T
T>A
T>C
T>G
SP135287 SBS1,SBS3,SBS5,SBS15,SBS17b,SBS26,SBS40,SBS41,SBS44 9 cosine sim = 0.98698
6606
2041
32608
3373
18254
3693
counts
7828
5871
3914
1957
0
66574
49930
33287
16644
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS17b
SBS15
SBS5
SBS3
SBS1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C>G
C>T
T>A
T>C
T>G
SP135287 SBS1,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS26,SBS41,SBS44 9 cosine sim = 0.98696
6436
2093
32628
3458
18297
3661
counts
7836
5877
3918
1959
0
66574
49930
33287
16644
0
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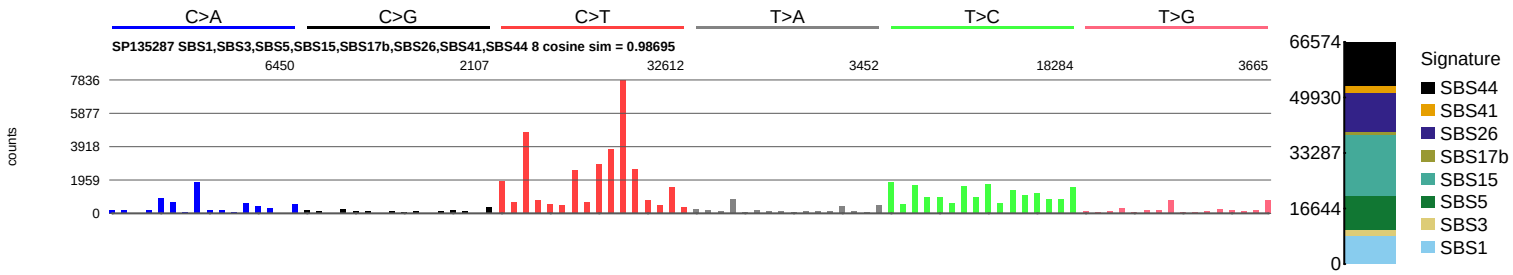
SBS41
SBS26
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS1


C>A
C>G
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T>A
T>C
T>G
SP135287 SBS1,SBS3,SBS5,SBS15,SBS17b,SBS26,SBS41,SBS44 8 cosine sim = 0.98695
6450
2107
32612
3452
18284
3665
counts
7836
5877
3918
1959
0
66574
49930
33287
16644
0
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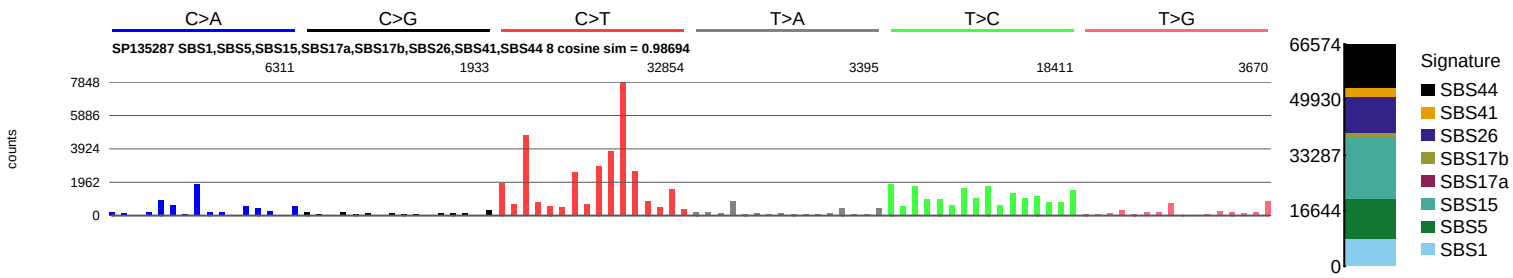
SBS17b
SBS15
SBS5
SBS3
SBS1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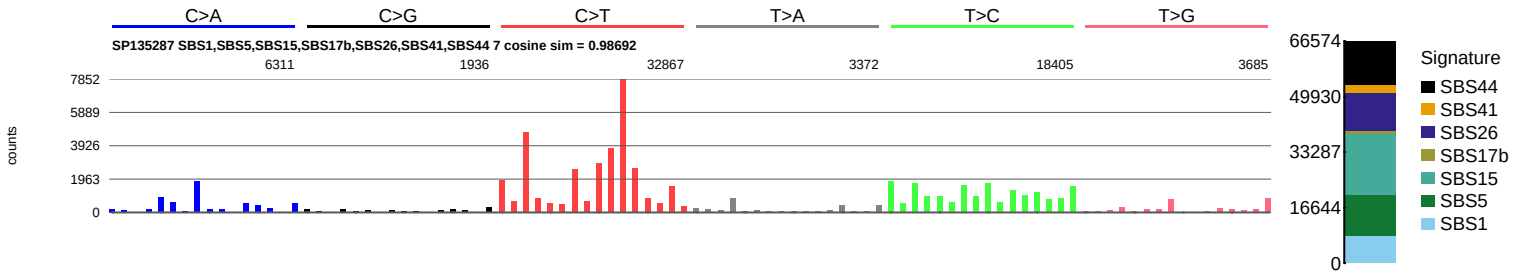
C>G
C>T
T>A
T>C
T>G
SP135287 SBS1,SBS5,SBS15,SBS17a,SBS17b,SBS26,SBS41,SBS44 8 cosine sim = 0.98694
6311
1933
32854
3395
18411
3670
counts
7848
5886
3924
1962
0
66574
49930
33287
16644
0
Signature
SBS44
SBS41
SBS26
SBS17b
SBS17a
SBS15
SBS5
SBS1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C>G
C>T
T>A
T>C
T>G
SP135287 SBS1,SBS5,SBS15,SBS17b,SBS26,SBS41,SBS44 7 cosine sim = 0.98692
6311
1936
32867
3372
18405
3685
counts
7852
5889
3926
1963
0
66574
49930
33287
16644
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SBS41
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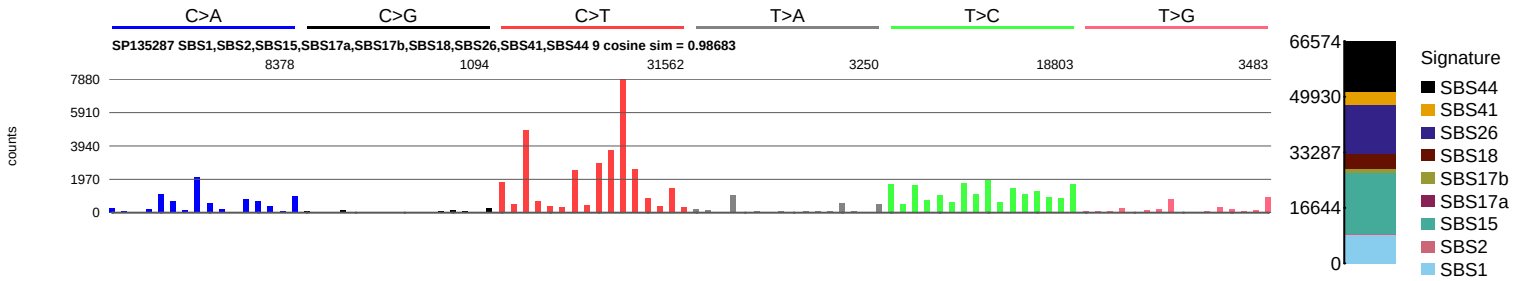
SBS17b
SBS15
SBS5
SBS1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C>G
C>T
T>A
T>C
T>G
SP135287 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS41,SBS44 9 cosine sim = 0.98683
8378
1094
31562
3250
18803
3483
counts
7880
5910
3940
1970
0
66574
49930
33287
16644
0
Signature
SBS44
SBS41
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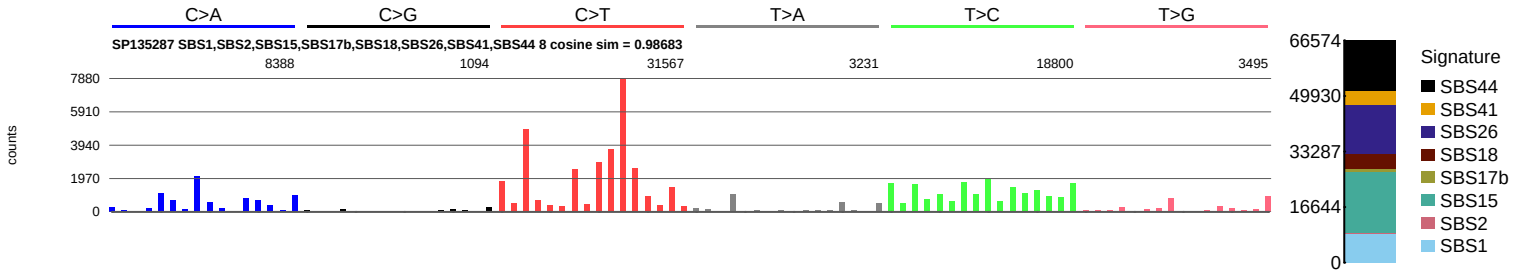
SBS18
SBS17b
SBS17a
SBS15
SBS2
SBS1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C>G
C>T
T>A
T>C
T>G
SP135287 SBS1,SBS2,SBS15,SBS17b,SBS18,SBS26,SBS41,SBS44 8 cosine sim = 0.98683
8388
1094
31567
3231
18800
3495
counts
7880
5910
3940
1970
0
66574
49930
33287
16644
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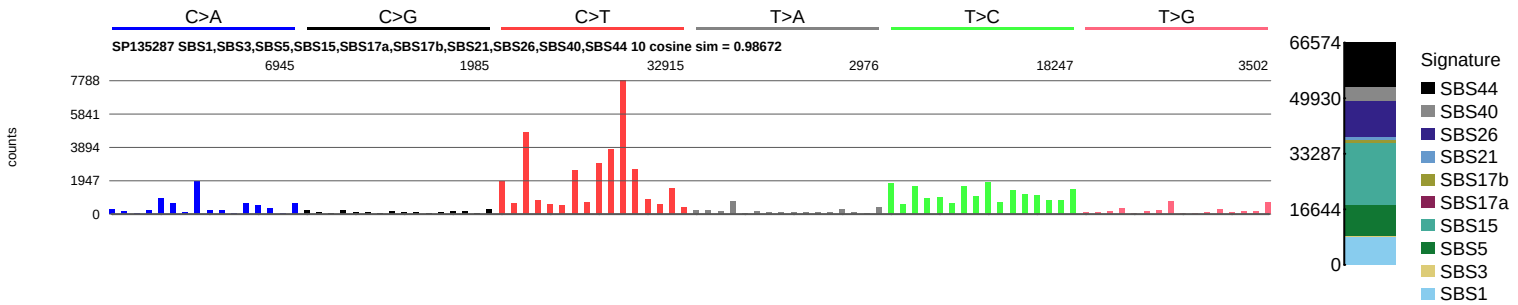
SBS18
SBS17b
SBS15
SBS2
SBS1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T>C
T>G
SP135287 SBS1,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS21,SBS26,SBS40,SBS44 10 cosine sim = 0.98672
6945
1985
32915
2976
18247
3502
counts
7788
5841
3894
1947
0
66574
49930
33287
16644
0
Signature
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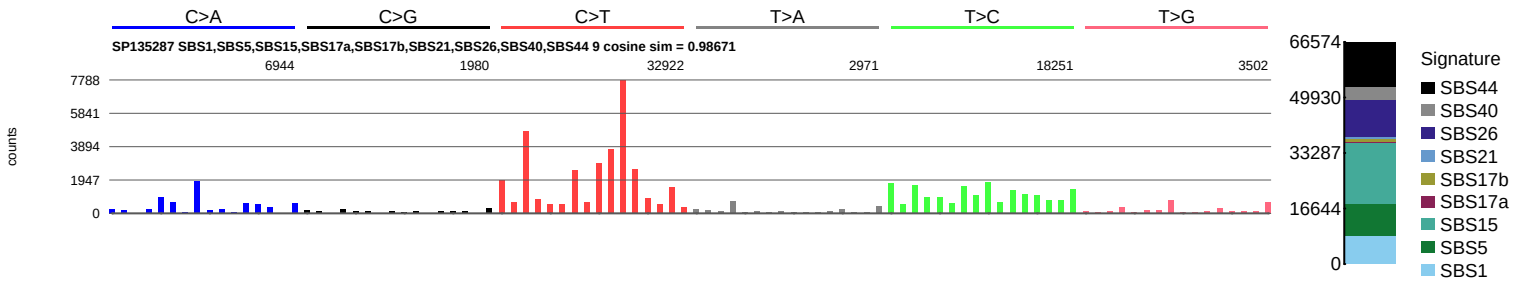
SBS40
SBS26
SBS21
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS1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T>C
T>G
SP135287 SBS1,SBS5,SBS15,SBS17a,SBS17b,SBS21,SBS26,SBS40,SBS44 9 cosine sim = 0.98671
6944
1980
32922
2971
18251
3502
counts
7788
5841
3894
1947
0
66574
49930
33287
16644
0
Signature
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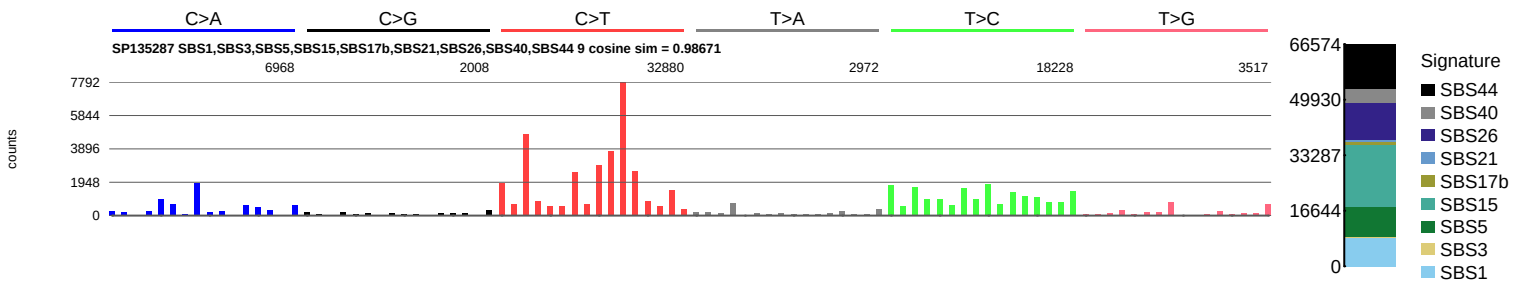
SBS40
SBS26
SBS21
SBS17b
SBS17a
SBS15
SBS5
SBS1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C>G
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T>A
T>C
T>G
SP135287 SBS1,SBS3,SBS5,SBS15,SBS17b,SBS21,SBS26,SBS40,SBS44 9 cosine sim = 0.98671
6968
2008
32880
2972
18228
3517
counts
7792
5844
3896
1948
0
66574
49930
33287
16644
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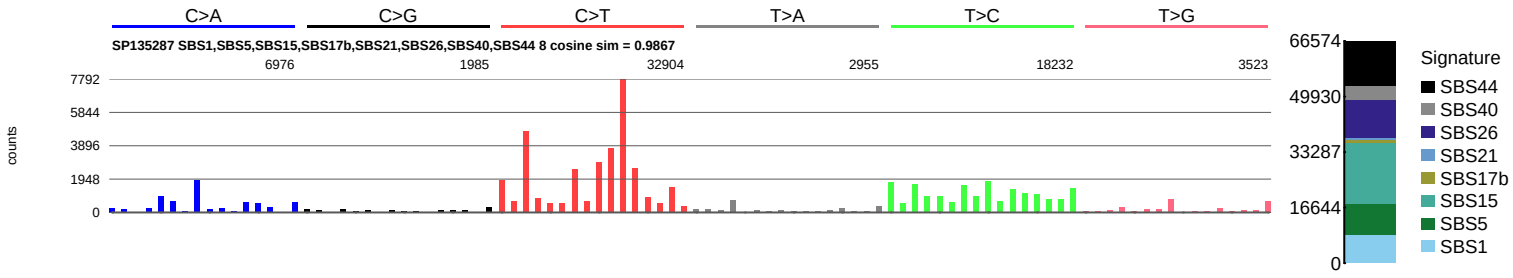
Signature
SBS44
SBS40
SBS26
SBS21
SBS17b
SBS15
SBS5
SBS3
SBS1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T>A
T>C
T>G
SP135287 SBS1,SBS5,SBS15,SBS17b,SBS21,SBS26,SBS40,SBS44 8 cosine sim = 0.9867
6976
1985
32904
2955
18232
3523
counts
7792
5844
3896
1948
0
66574
49930
33287
16644
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Signature
SBS44
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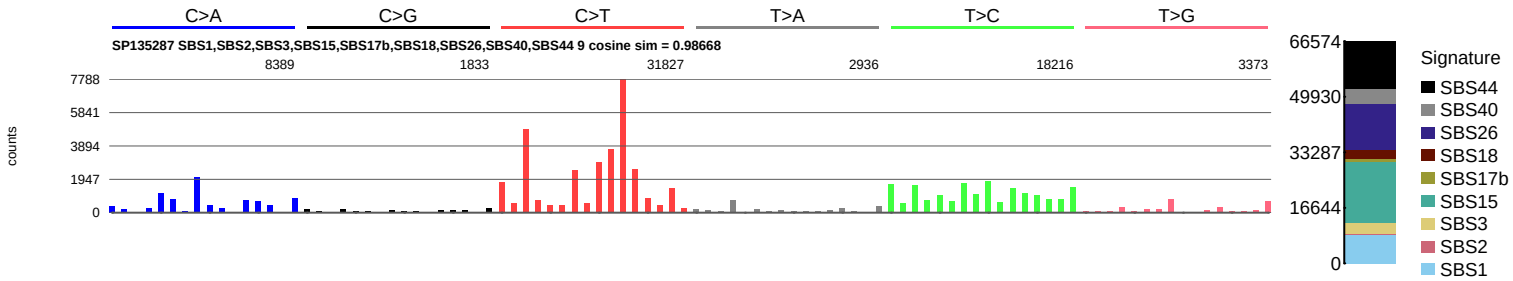
SBS26
SBS21
SBS17b
SBS15
SBS5
SBS1


C>A
C>G
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T>A
T>C
T>G
SP135287 SBS1,SBS2,SBS3,SBS15,SBS17b,SBS18,SBS26,SBS40,SBS44 9 cosine sim = 0.98668
8389
1833
31827
2936
18216
3373
counts
7788
5841
3894
1947
0
66574
49930
33287
16644
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SBS40
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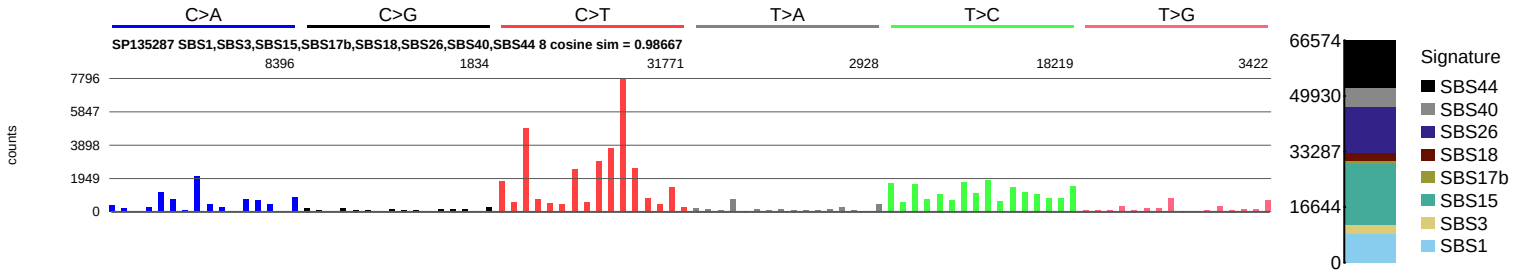
SBS18
SBS17b
SBS15
SBS3
SBS2
SBS1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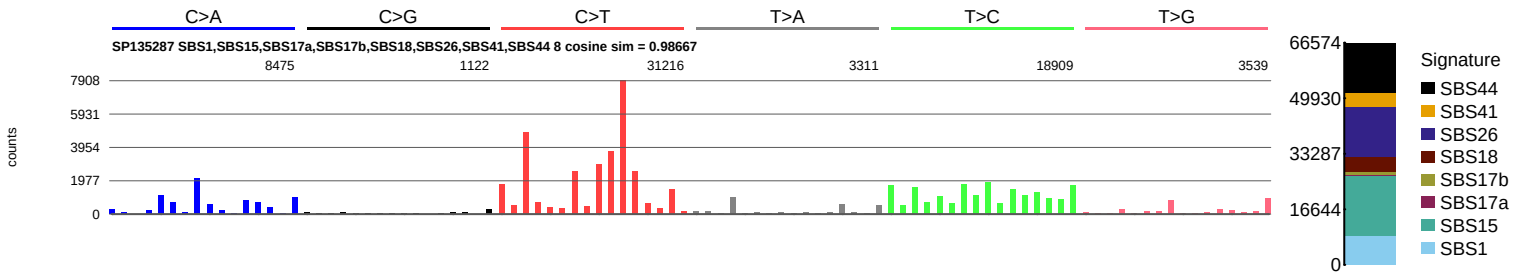
C>G
C>T
T>A
T>C
T>G
SP135287 SBS1,SBS3,SBS15,SBS17b,SBS18,SBS26,SBS40,SBS44 8 cosine sim = 0.98667
8396
1834
31771
2928
18219
3422
counts
7796
5847
3898
1949
0
66574
49930
33287
16644
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS17b
SBS15
SBS3
SBS1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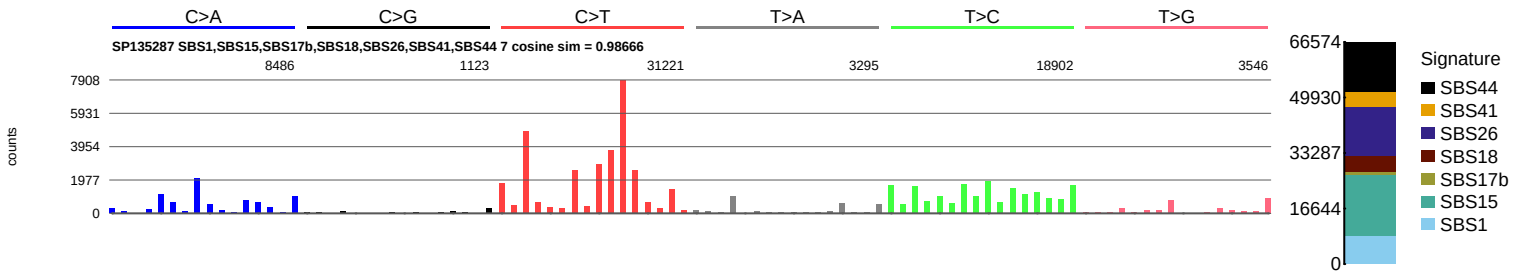
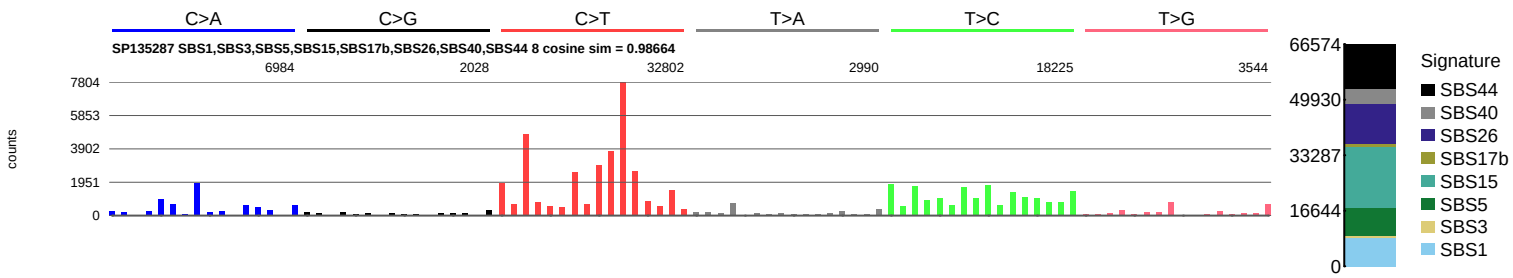
C>G
C>T
T>A
T>C
T>G
SP135287 SBS1,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS41,SBS44 8 cosine sim = 0.98667
8475
1122
31216
3311
18909
3539
counts
7908
5931
3954
1977
0
66574
49930
33287
16644
0
Signature
SBS44
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135287 SBS1,SBS15,SBS17b,SBS18,SBS26,SBS41,SBS44 7 cosine sim = 0.98666
8486
1123
31221
3295
18902
3546
counts
7908
5931
3954
1977
0
66574
49930
33287
16644
0
Signature
SBS44
SBS41
SBS26
SBS18
SBS17b
SBS15
SBS1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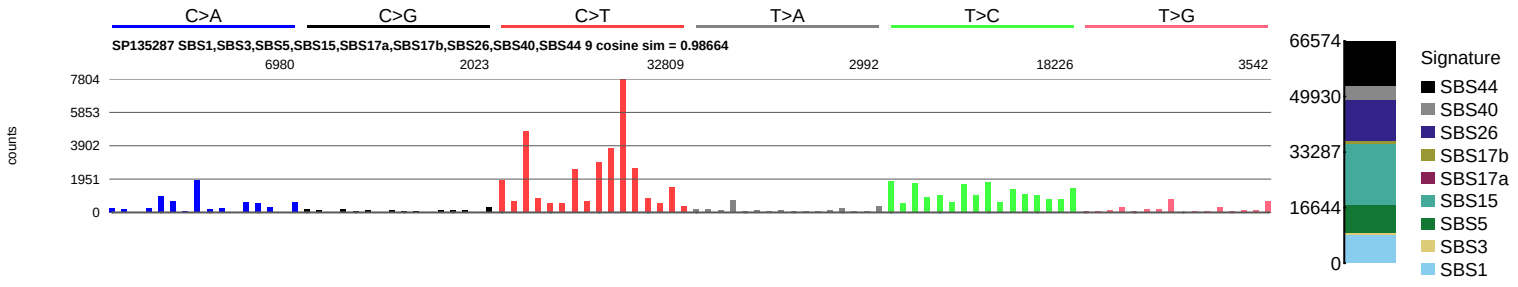
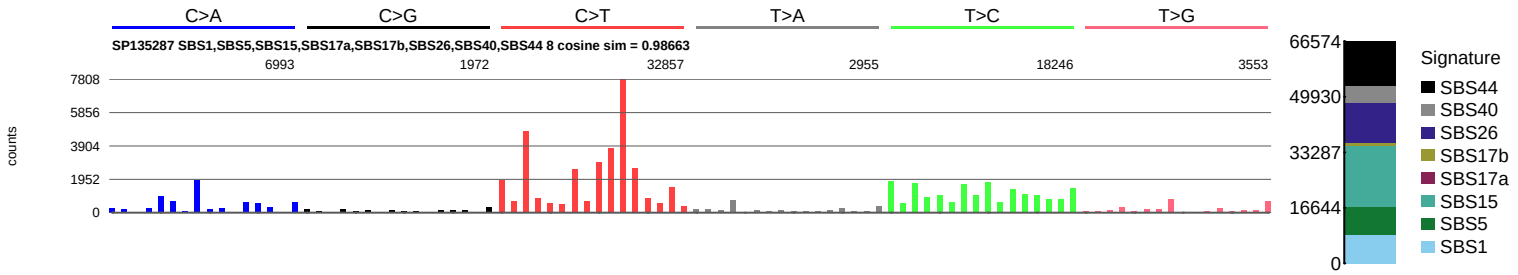
C>G
C>T
T>A
T>C
T>G
SP135287 SBS1,SBS3,SBS5,SBS15,SBS17b,SBS26,SBS40,SBS44 8 cosine sim = 0.98664
6984
2028
32802
2990
18225
3544
counts
7804
5853
3902
1951
0
66574
49930
33287
16644
0
Signature
SBS44
SBS40
SBS26
SBS17b
SBS15
SBS5
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135287 SBS1,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS26,SBS40,SBS44 9 cosine sim = 0.98664
6980
2023
32809
2992
18226
3542
counts
7804
5853
3902
1951
0
66574
49930
33287
16644
0
Signature
SBS44
SBS40
SBS26
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS1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SP135287 SBS1,SBS5,SBS15,SBS17a,SBS17b,SBS26,SBS40,SBS44 8 cosine sim = 0.98663
6993
1972
32857
2955
18246
3553
counts
7808
5856
3904
1952
0
66574
49930
33287
16644
0
Signature
SBS44
SBS40
SBS26
SBS17b
SBS17a
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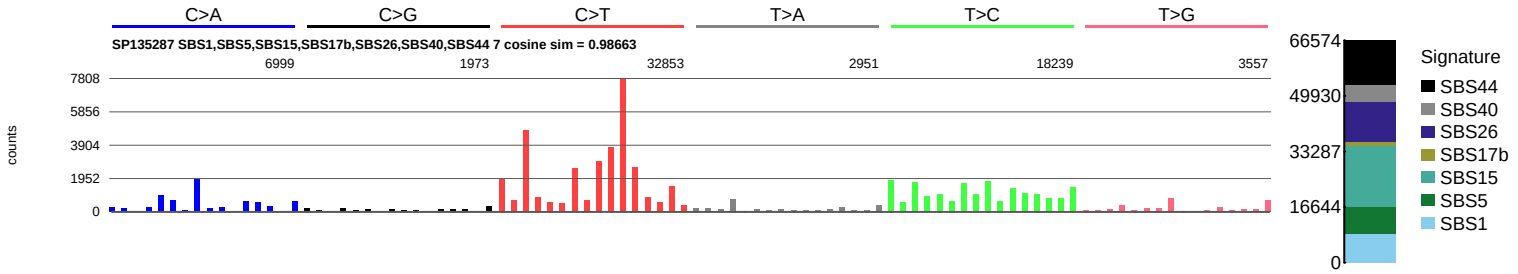
SBS5
SBS1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SP135287 SBS1,SBS5,SBS15,SBS17b,SBS26,SBS40,SBS44 7 cosine sim = 0.98663
6999
1973
32853
2951
18239
3557
counts
7808
5856
3904
1952
0
66574
49930
33287
16644
0
Signature
SBS44
SBS40
SBS26
SBS17b
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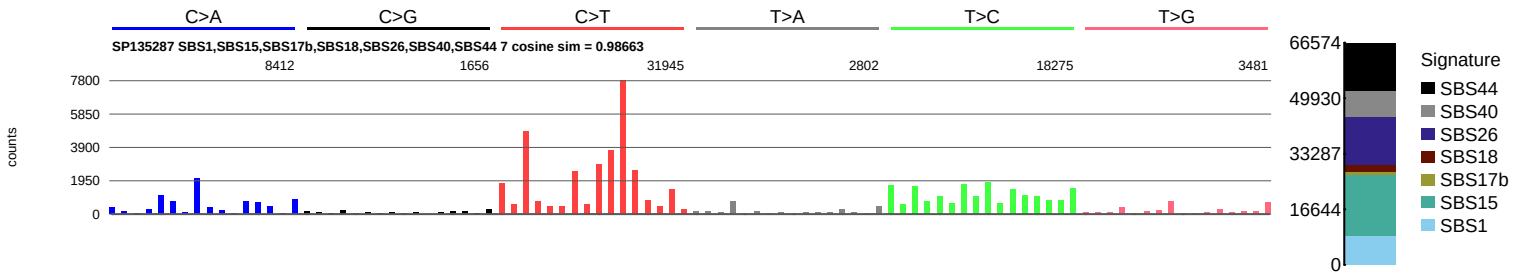
SBS5
SBS1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SP135287 SBS1,SBS15,SBS17b,SBS18,SBS26,SBS40,SBS44 7 cosine sim = 0.98663
8412
1656
31945
2802
18275
3481
counts
7800
5850
3900
1950
0
66574
49930
33287
16644
0
Signature
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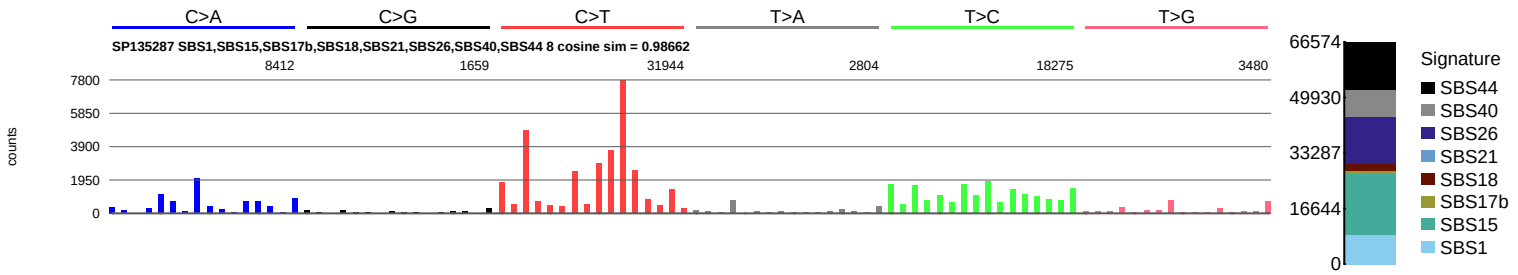
SBS40
SBS26
SBS18
SBS17b
SBS15
SBS1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SP135287 SBS1,SBS15,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS44 8 cosine sim = 0.98662
8412
1659
31944
2804
18275
3480
counts
7800
5850
3900
1950
0
66574
49930
33287
16644
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Signature
SBS44
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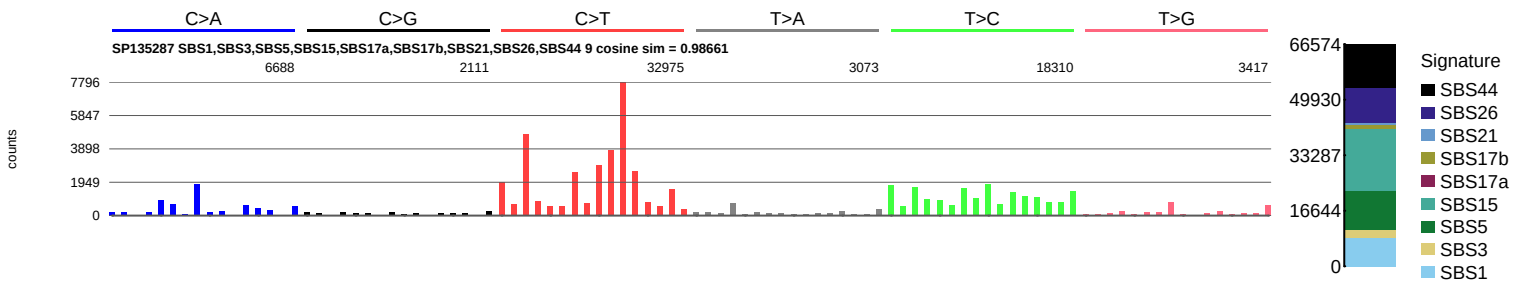
SBS26
SBS21
SBS18
SBS17b
SBS15
SBS1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SP135287 SBS1,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS21,SBS26,SBS44 9 cosine sim = 0.98661
6688
2111
32975
3073
18310
3417
counts
7796
5847
3898
1949
0
66574
49930
33287
16644
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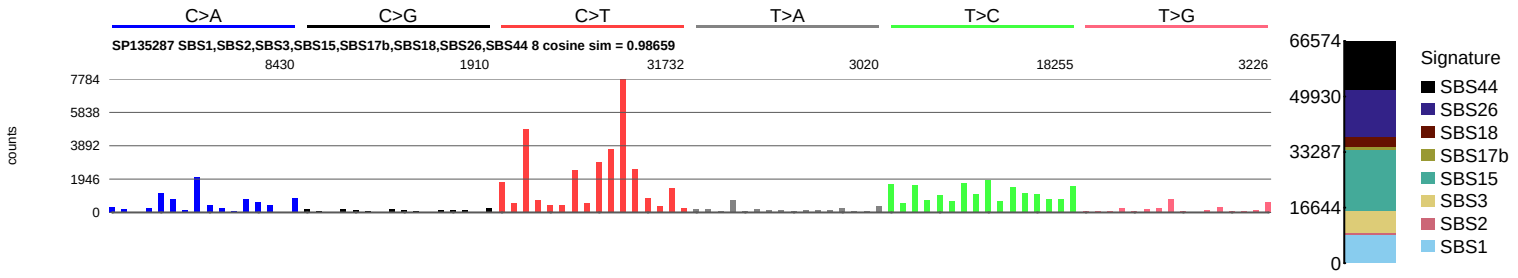
SBS26
SBS21
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS1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SP135287 SBS1,SBS2,SBS3,SBS15,SBS17b,SBS18,SBS26,SBS44 8 cosine sim = 0.98659
8430
1910
31732
3020
18255
3226
counts
7784
5838
3892
1946
0
66574
49930
33287
16644
0
Signature
SBS44
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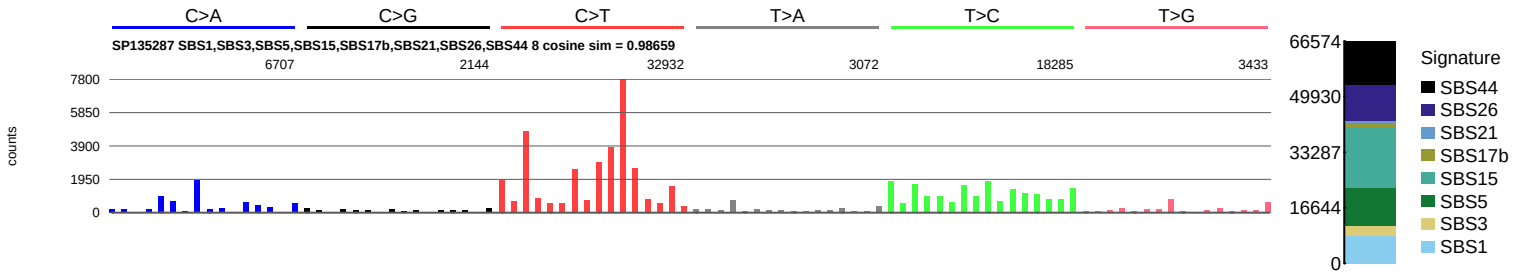
SBS18
SBS17b
SBS15
SBS3
SBS2
SBS1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C>G
C>T
T>A
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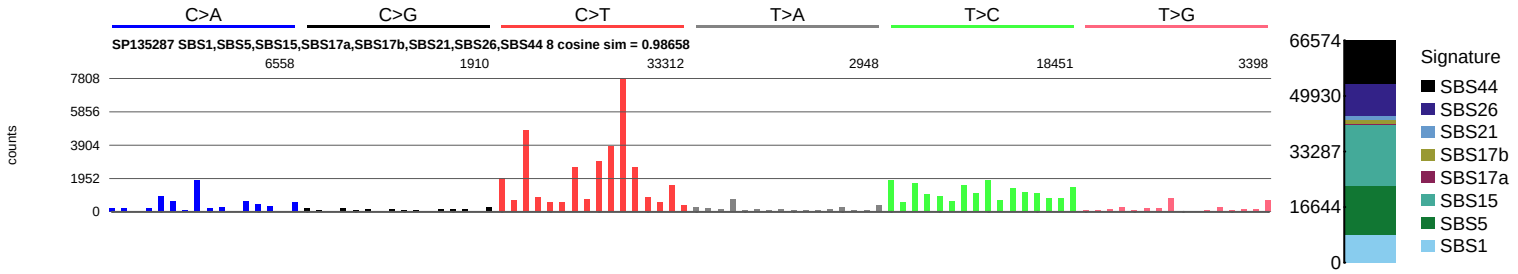
T>G
SP135287 SBS1,SBS3,SBS5,SBS15,SBS17b,SBS21,SBS26,SBS44 8 cosine sim = 0.98659
6707
2144
32932
3072
18285
3433
counts
7800
5850
3900
1950
0
66574
49930
33287
16644
0
Signature
SBS44
SBS26
SBS21
SBS17b
SBS15
SBS5
SBS3
SBS1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C>G
C>T
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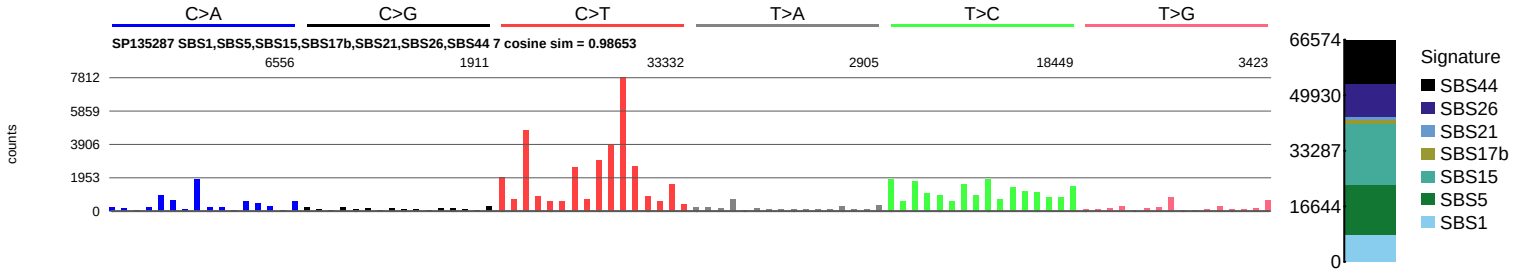
T>G
SP135287 SBS1,SBS5,SBS15,SBS17a,SBS17b,SBS21,SBS26,SBS44 8 cosine sim = 0.98658
6558
1910
33312
2948
18451
3398
counts
7808
5856
3904
1952
0
66574
49930
33287
16644
0
Signature
SBS44
SBS26
SBS21
SBS17b
SBS17a
SBS15
SBS5
SBS1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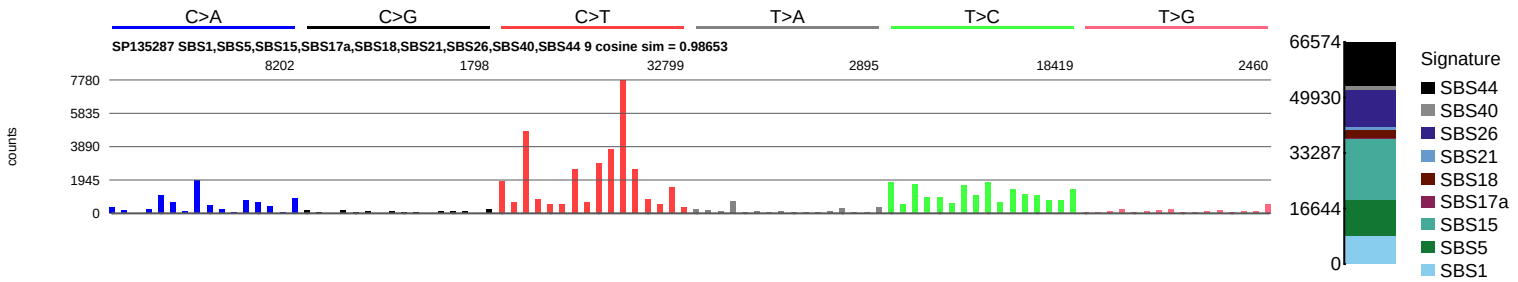
T>G
SP135287 SBS1,SBS5,SBS15,SBS17b,SBS21,SBS26,SBS44 7 cosine sim = 0.98653
6556
1911
33332
2905
18449
3423
counts
7812
5859
3906
1953
0
66574
49930
33287
16644
0
Signature
SBS44
SBS26
SBS21
SBS17b
SBS15
SBS5
SBS1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C>G
C>T
T>A
T>C
T>G
SP135287 SBS1,SBS5,SBS15,SBS17a,SBS18,SBS21,SBS26,SBS40,SBS44 9 cosine sim = 0.98653
8202
1798
32799
2895
18419
2460
counts
7780
5835
3890
1945
0
66574
49930
33287
16644
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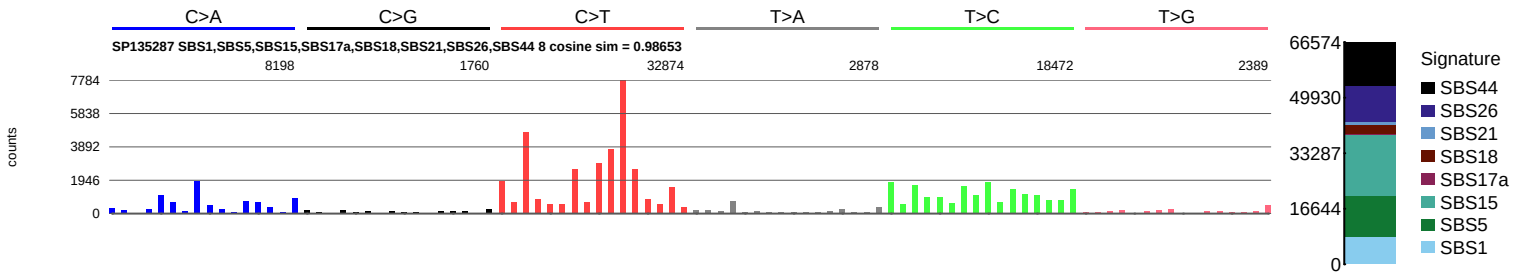
SBS44
SBS40
SBS26
SBS21
SBS18
SBS17a
SBS15
SBS5
SBS1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C>G
C>T
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T>C
T>G
SP135287 SBS1,SBS5,SBS15,SBS17a,SBS18,SBS21,SBS26,SBS44 8 cosine sim = 0.98653
8198
1760
32874
2878
18472
2389
counts
7784
5838
3892
1946
0
66574
49930
33287
16644
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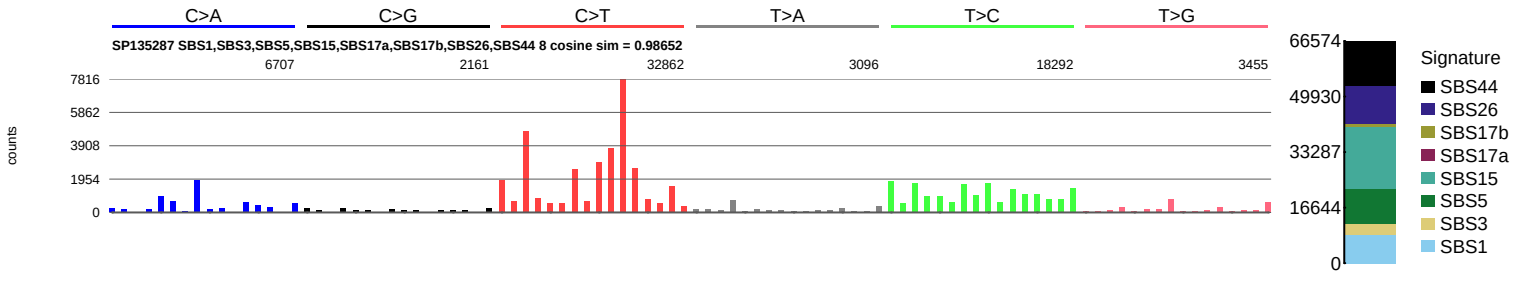
Signature
SBS44
SBS26
SBS21
SBS18
SBS17a
SBS15
SBS5
SBS1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C>G
C>T
T>A
T>C
T>G
SP135287 SBS1,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS26,SBS44 8 cosine sim = 0.98652
6707
2161
32862
3096
18292
3455
counts
7816
5862
3908
1954
0
66574
49930
33287
16644
0
Signature
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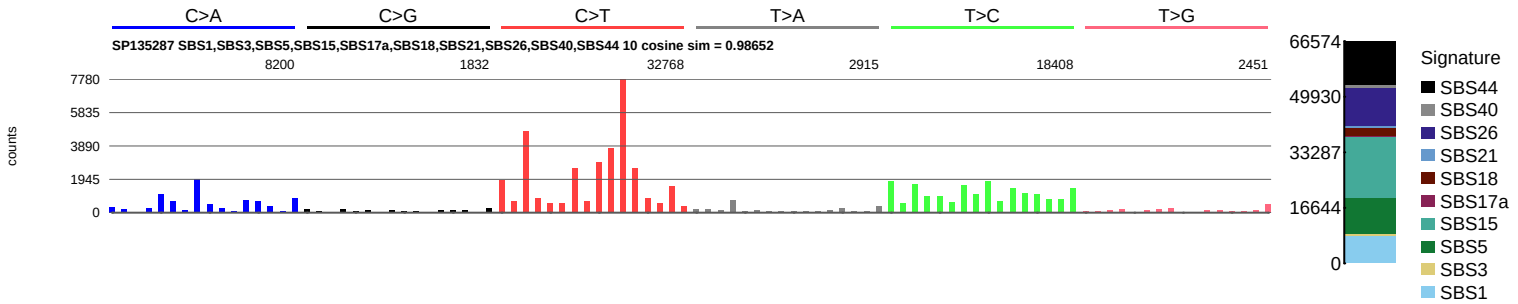
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS1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T>C
T>G
SP135287 SBS1,SBS3,SBS5,SBS15,SBS17a,SBS18,SBS21,SBS26,SBS40,SBS44 10 cosine sim = 0.98652
8200
1832
32768
2915
18408
2451
counts
7780
5835
3890
1945
0
66574
49930
33287
16644
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Signature
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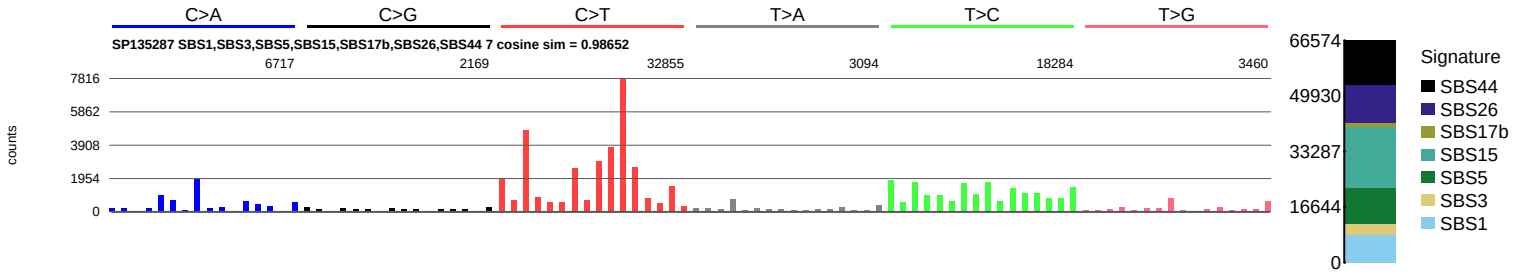
SBS26
SBS21
SBS18
SBS17a
SBS15
SBS5
SBS3
SBS1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C>G
C>T
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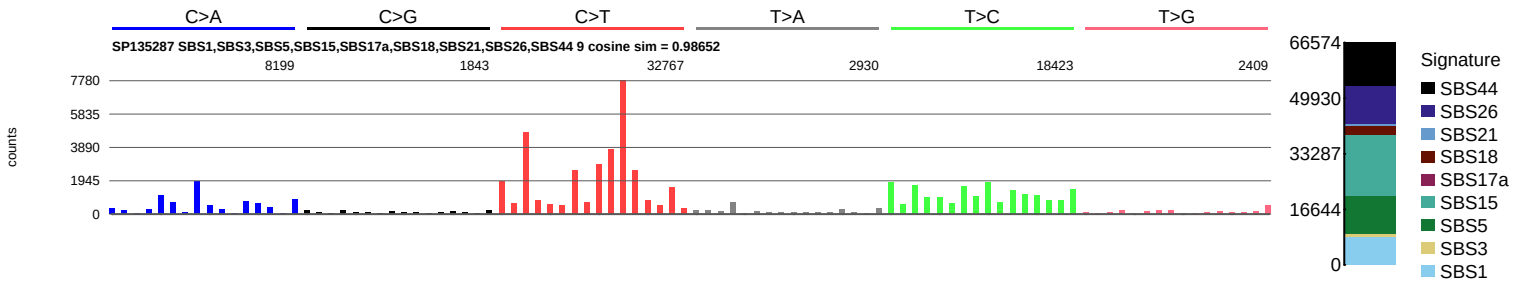
T>C
T>G
SP135287 SBS1,SBS3,SBS5,SBS15,SBS17b,SBS26,SBS44 7 cosine sim = 0.98652
6717
2169
32855
3094
18284
3460
counts
7816
5862
3908
1954
0
66574
49930
33287
16644
0
Signature
SBS44
SBS26
SBS17b
SBS15
SBS5
SBS3
SBS1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C>G
C>T
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T>C
T>G
SP135287 SBS1,SBS3,SBS5,SBS15,SBS17a,SBS18,SBS21,SBS26,SBS44 9 cosine sim = 0.98652
8199
1843
32767
2930
18423
2409
counts
7780
5835
3890
1945
0
66574
49930
33287
16644
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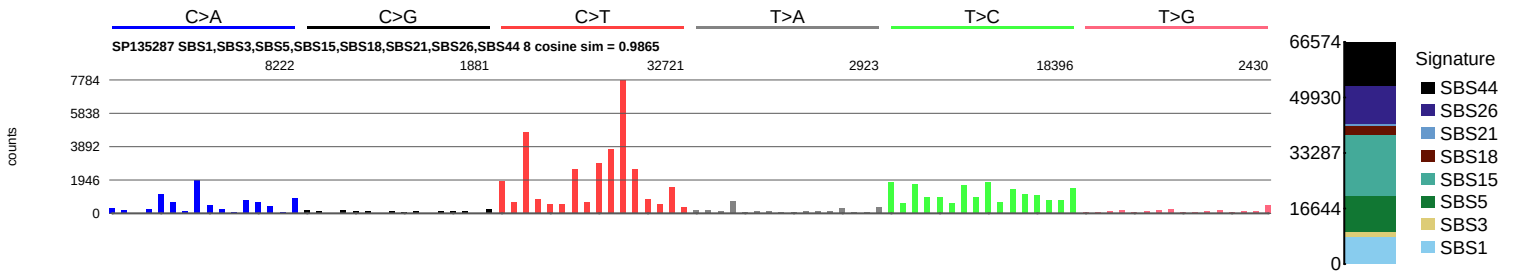
SBS21
SBS18
SBS17a
SBS15
SBS5
SBS3
SBS1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T>C
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SP135287 SBS1,SBS3,SBS5,SBS15,SBS18,SBS21,SBS26,SBS44 8 cosine sim = 0.9865
8222
1881
32721
2923
18396
2430
counts
7784
5838
3892
1946
0
66574
49930
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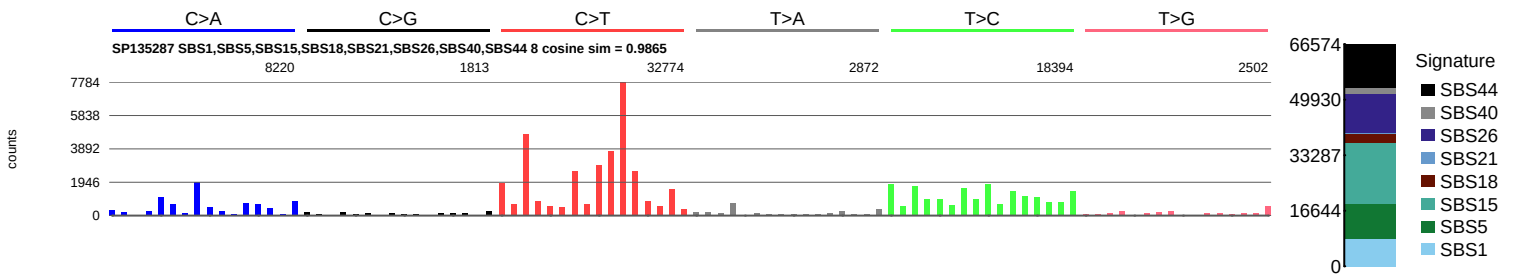
SBS5
SBS3
SBS1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C>G
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T>C
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SP135287 SBS1,SBS5,SBS15,SBS18,SBS21,SBS26,SBS40,SBS44 8 cosine sim = 0.9865
8220
1813
32774
2872
18394
2502
counts
7784
5838
3892
1946
0
66574
49930
33287
16644
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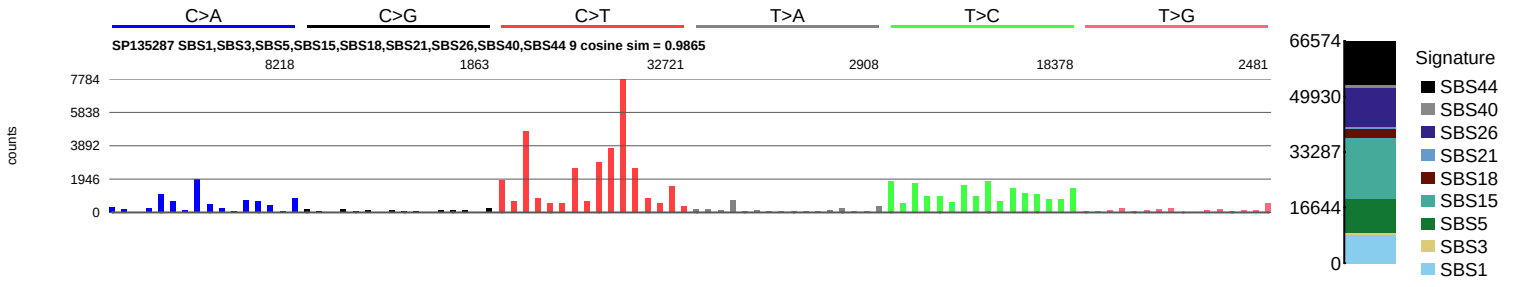
SBS21
SBS18
SBS15
SBS5
SBS1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T>C
T>G
SP135287 SBS1,SBS3,SBS5,SBS15,SBS18,SBS21,SBS26,SBS40,SBS44 9 cosine sim = 0.9865
8218
1863
32721
2908
18378
2481
counts
7784
5838
3892
1946
0
66574
49930
33287
16644
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SBS21
SBS18
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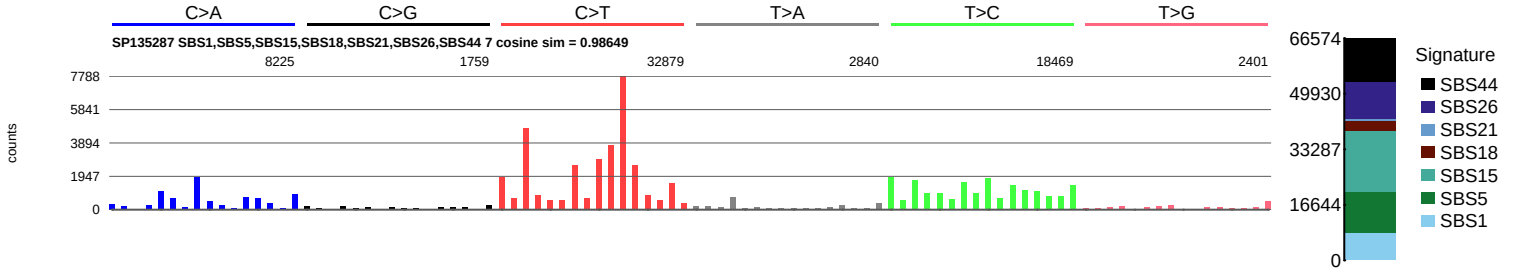
SBS5
SBS3
SBS1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C>G
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T>C
T>G
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8225
1759
32879
2840
18469
2401
counts
7788
5841
3894
1947
0
66574
49930
33287
16644
0
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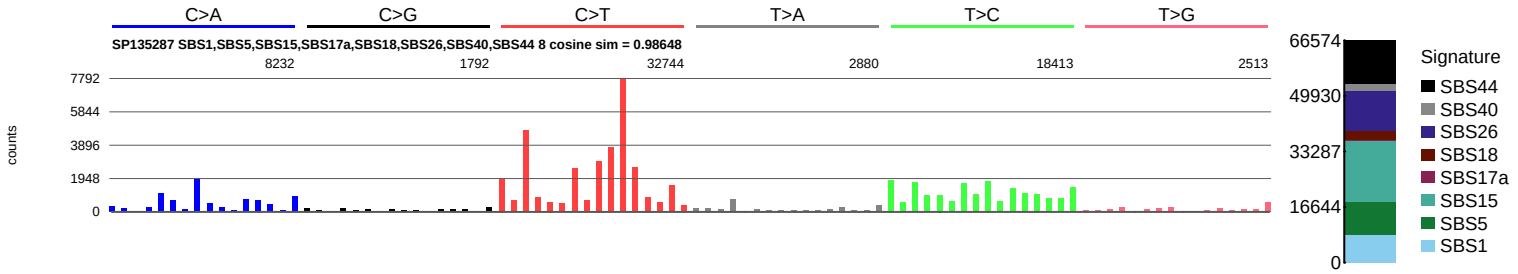
SBS44
SBS26
SBS21
SBS18
SBS15
SBS5
SBS1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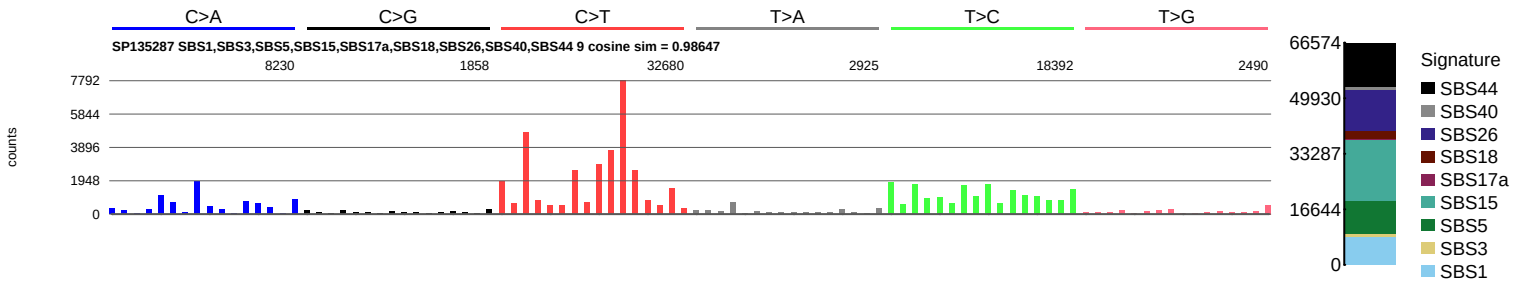
C>G
C>T
T>A
T>C
T>G
SP135287 SBS1,SBS5,SBS15,SBS17a,SBS18,SBS26,SBS40,SBS44 8 cosine sim = 0.98648
8232
1792
32744
2880
18413
2513
counts
7792
5844
3896
1948
0
66574
49930
33287
16644
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS17a
SBS15
SBS5
SBS1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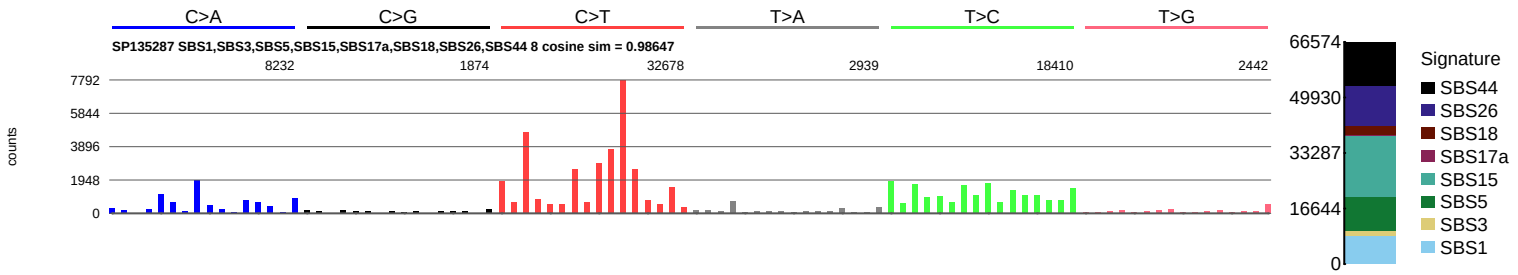
C>G
C>T
T>A
T>C
T>G
SP135287 SBS1,SBS3,SBS5,SBS15,SBS17a,SBS18,SBS26,SBS40,SBS44 9 cosine sim = 0.98647
8230
1858
32680
2925
18392
2490
counts
7792
5844
3896
1948
0
66574
49930
33287
16644
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS17a
SBS15
SBS5
SBS3
SBS1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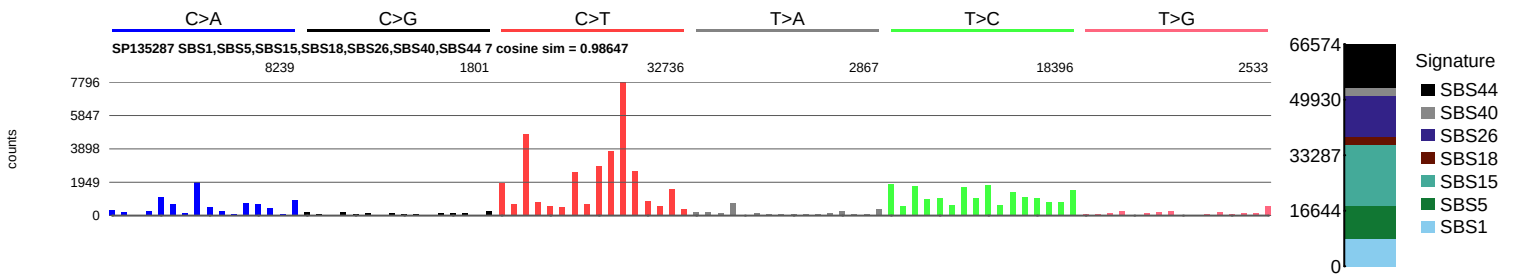
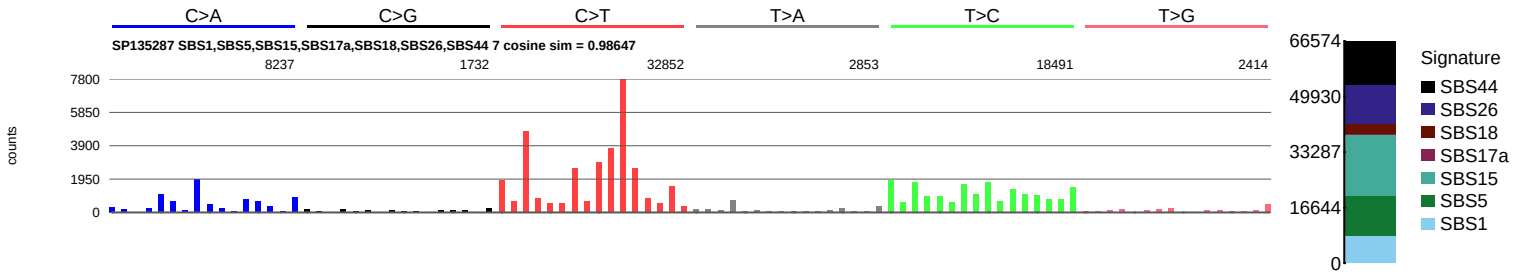
C>G
C>T
T>A
T>C
T>G
SP135287 SBS1,SBS3,SBS5,SBS15,SBS17a,SBS18,SBS26,SBS44 8 cosine sim = 0.98647
8232
1874
32678
2939
18410
2442
counts
7792
5844
3896
1948
0
66574
49930
33287
16644
0
Signature
SBS44
SBS26
SBS18
SBS17a
SBS15
SBS5
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135287 SBS1,SBS5,SBS15,SBS18,SBS26,SBS40,SBS44 7 cosine sim = 0.98647
8239
1801
32736
2867
18396
2533
counts
7796
5847
3898
1949
0
66574
49930
33287
16644
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS15
SBS5
SBS1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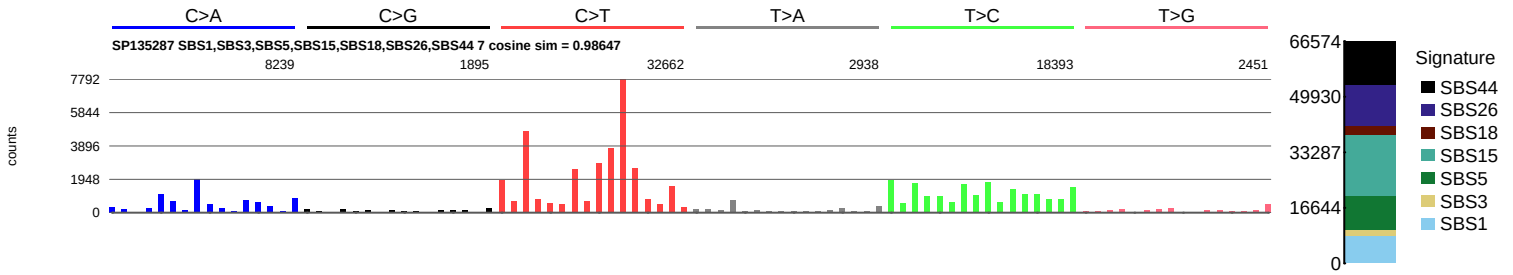
C>G
C>T
T>A
T>C
T>G
SP135287 SBS1,SBS5,SBS15,SBS17a,SBS18,SBS26,SBS44 7 cosine sim = 0.98647
8237
1732
32852
2853
18491
2414
counts
7800
5850
3900
1950
0
66574
49930
33287
16644
0
Signature
SBS44
SBS26
SBS18
SBS17a
SBS15
SBS5
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135287 SBS1,SBS3,SBS5,SBS15,SBS18,SBS26,SBS44 7 cosine sim = 0.98647
8239
1895
32662
2938
18393
2451
counts
7792
5844
3896
1948
0
66574
49930
33287
16644
0
Signature
SBS44
SBS26
SBS18
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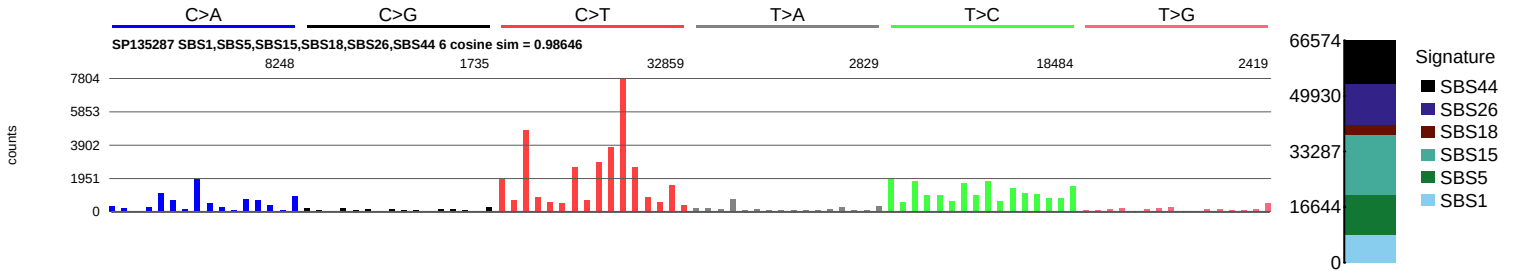
SBS1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SP135287 SBS1,SBS5,SBS15,SBS18,SBS26,SBS44 6 cosine sim = 0.98646
8248
1735
32859
2829
18484
2419
counts
7804
5853
3902
1951
0
66574
49930
33287
16644
0
Signature
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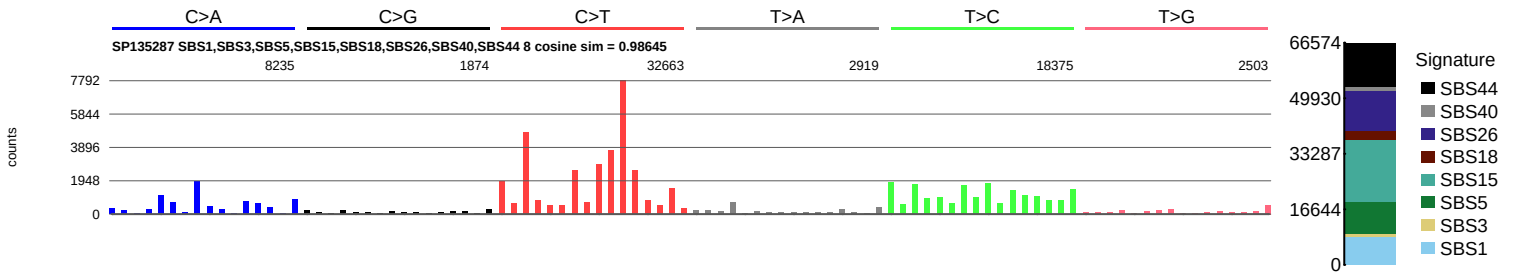
SBS26
SBS18
SBS15
SBS5
SBS1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SP135287 SBS1,SBS3,SBS5,SBS15,SBS18,SBS26,SBS40,SBS44 8 cosine sim = 0.98645
8235
1874
32663
2919
18375
2503
counts
7792
5844
3896
1948
0
66574
49930
33287
16644
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SBS18
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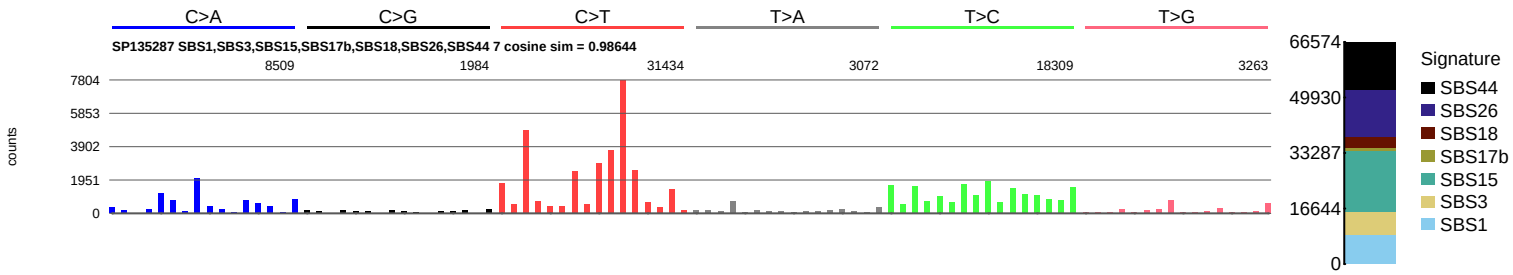
SBS3
SBS1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SP135287 SBS1,SBS3,SBS15,SBS17b,SBS18,SBS26,SBS44 7 cosine sim = 0.98644
8509
1984
31434
3072
18309
3263
counts
7804
5853
3902
1951
0
66574
49930
33287
16644
0
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SBS26
SBS18
SBS17b
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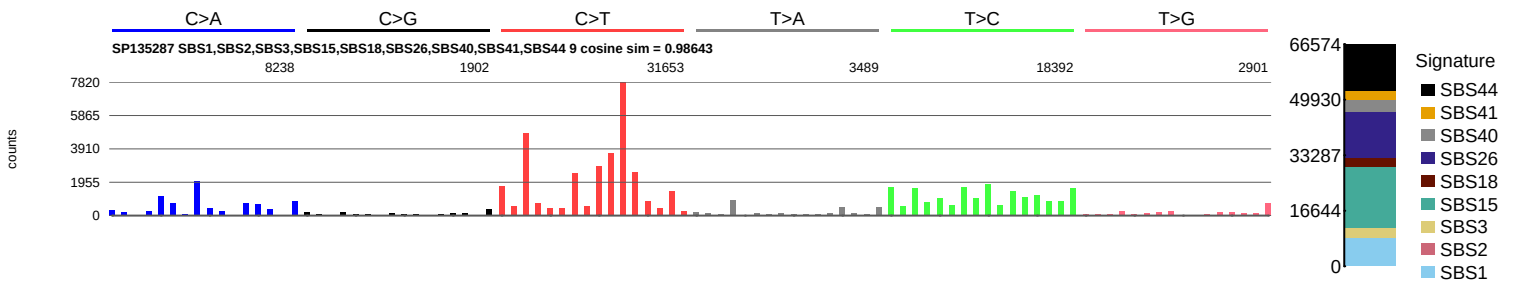
SBS3
SBS1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SP135287 SBS1,SBS2,SBS3,SBS15,SBS18,SBS26,SBS40,SBS41,SBS44 9 cosine sim = 0.98643
8238
1902
31653
3489
18392
2901
counts
7820
5865
3910
1955
0
66574
49930
33287
16644
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Signature
SBS44
SBS41
SBS40
SBS26
SBS18
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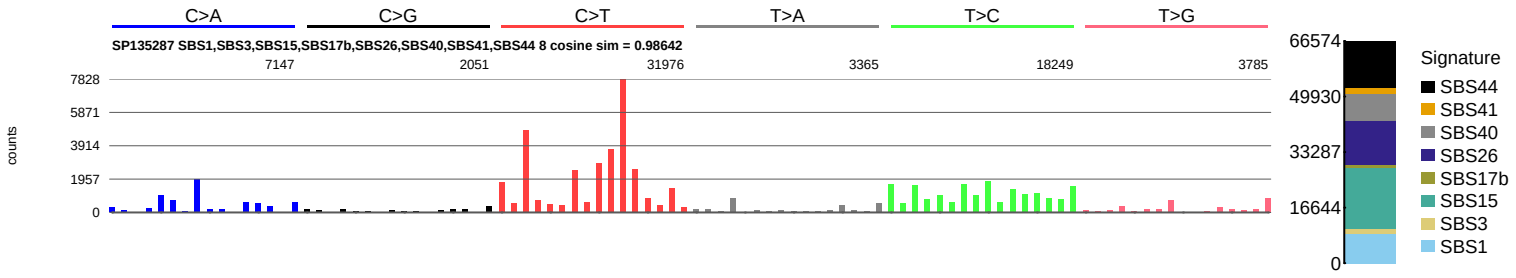
SBS3
SBS2
SBS1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SP135287 SBS1,SBS3,SBS15,SBS17b,SBS26,SBS40,SBS41,SBS44 8 cosine sim = 0.98642
7147
2051
31976
3365
18249
3785
counts
7828
5871
3914
1957
0
66574
49930
33287
16644
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Signature
SBS44
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SBS26
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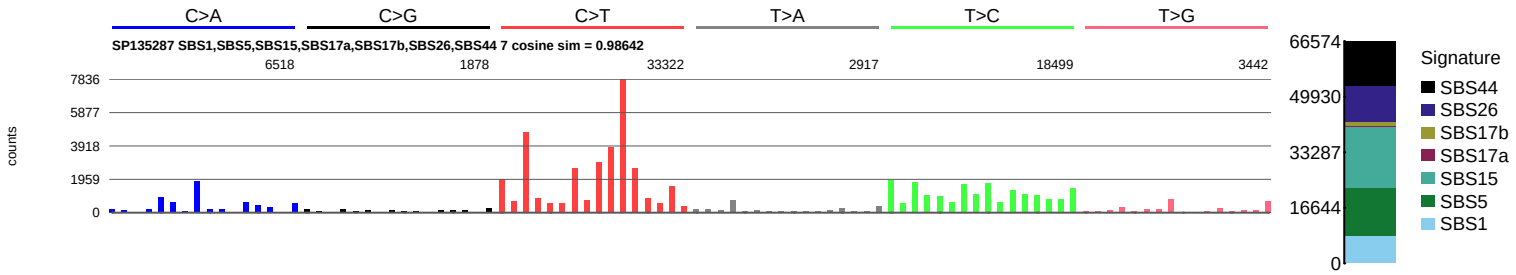
SBS3
SBS1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SP135287 SBS1,SBS5,SBS15,SBS17a,SBS17b,SBS26,SBS44 7 cosine sim = 0.98642
6518
1878
33322
2917
18499
3442
counts
7836
5877
3918
1959
0
66574
49930
33287
16644
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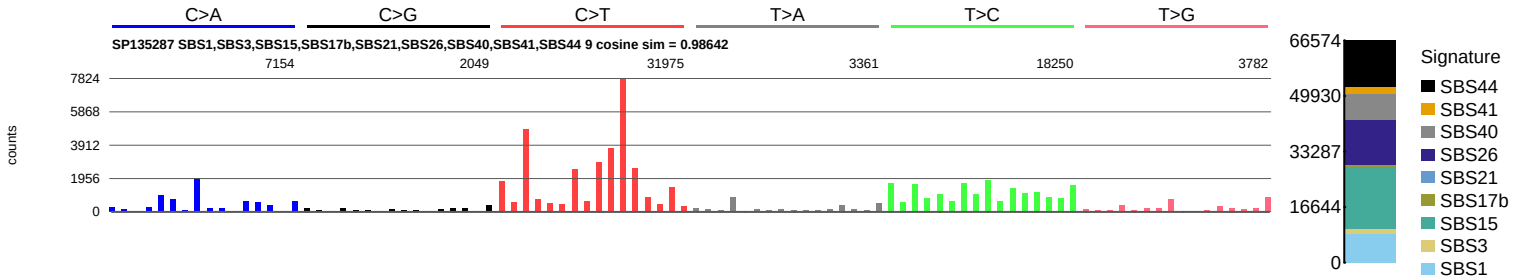
SBS17a
SBS15
SBS5
SBS1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SP135287 SBS1,SBS3,SBS15,SBS17b,SBS21,SBS26,SBS40,SBS41,SBS44 9 cosine sim = 0.98642
7154
2049
31975
3361
18250
3782
counts
7824
5868
3912
1956
0
66574
49930
33287
16644
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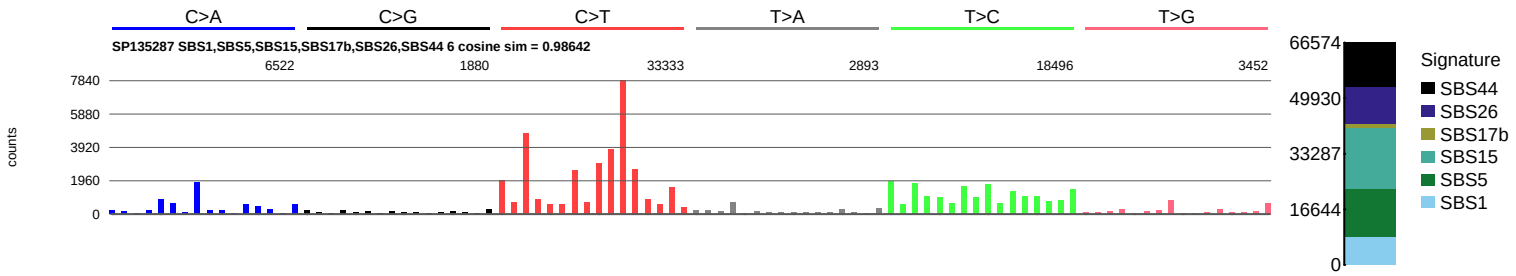
SBS44
SBS41
SBS40
SBS26
SBS21
SBS17b
SBS15
SBS3
SBS1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SP135287 SBS1,SBS5,SBS15,SBS17b,SBS26,SBS44 6 cosine sim = 0.98642
6522
1880
33333
2893
18496
3452
counts
7840
5880
3920
1960
0
66574
49930
33287
16644
0
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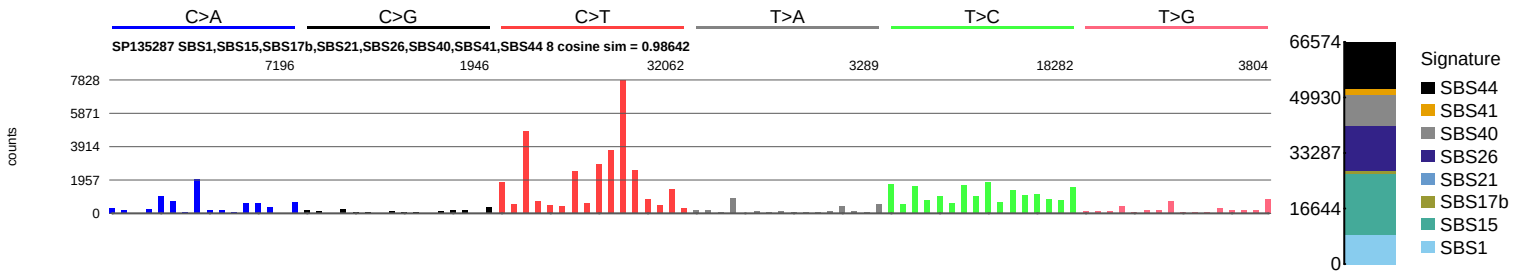
SBS44
SBS26
SBS17b
SBS15
SBS5
SBS1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C>T
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T>C
T>G
SP135287 SBS1,SBS15,SBS17b,SBS21,SBS26,SBS40,SBS41,SBS44 8 cosine sim = 0.98642
7196
1946
32062
3289
18282
3804
counts
7828
5871
3914
1957
0
66574
49930
33287
16644
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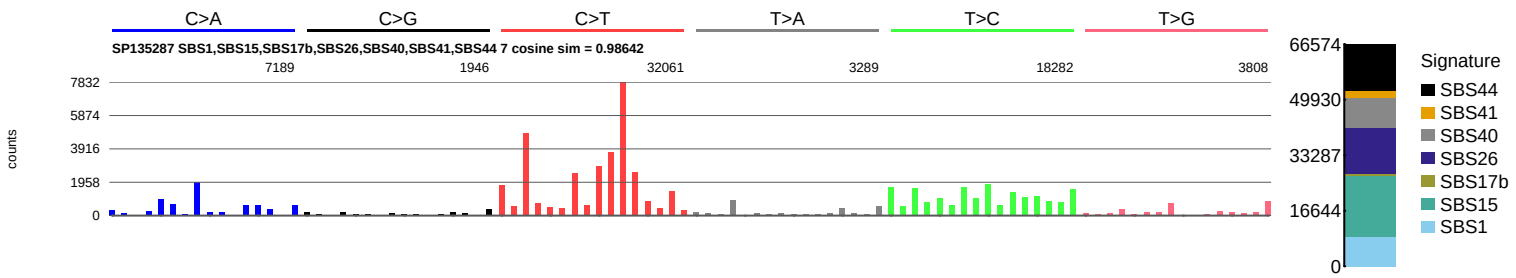
Signature
SBS44
SBS41
SBS40
SBS26
SBS21
SBS17b
SBS15
SBS1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C>G
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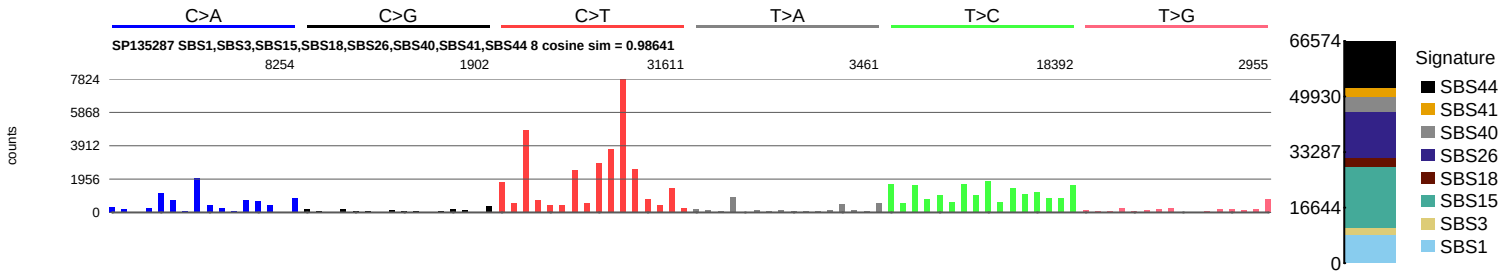
T>A
T>C
T>G
SP135287 SBS1,SBS15,SBS17b,SBS26,SBS40,SBS41,SBS44 7 cosine sim = 0.98642
7189
1946
32061
3289
18282
3808
counts
7832
5874
3916
1958
0
66574
49930
33287
16644
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS17b
SBS15
SBS1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T>C
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SP135287 SBS1,SBS3,SBS15,SBS18,SBS26,SBS40,SBS41,SBS44 8 cosine sim = 0.98641
8254
1902
31611
3461
18392
2955
counts
7824
5868
3912
1956
0
66574
49930
33287
16644
0
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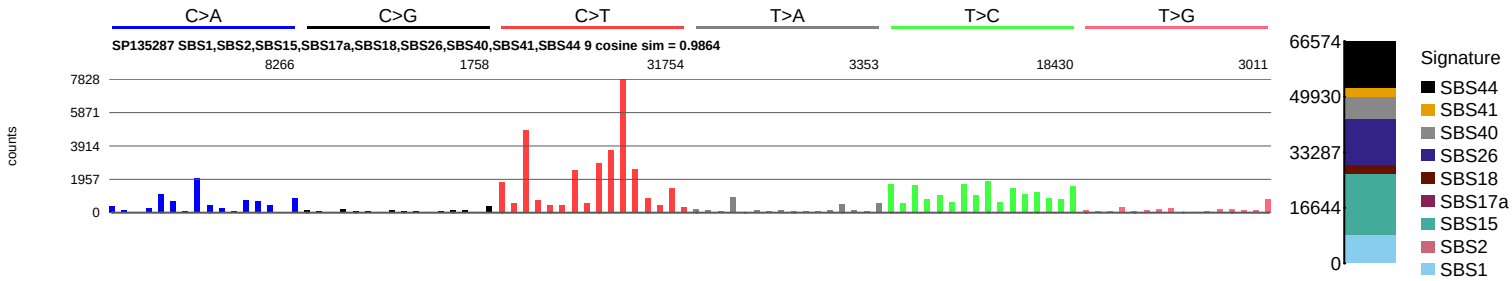
SBS41
SBS40
SBS26
SBS18
SBS15
SBS3
SBS1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SP135287 SBS1,SBS2,SBS15,SBS17a,SBS18,SBS26,SBS40,SBS41,SBS44 9 cosine sim = 0.9864
8266
1758
31754
3353
18430
3011
counts
7828
5871
3914
1957
0
66574
49930
33287
16644
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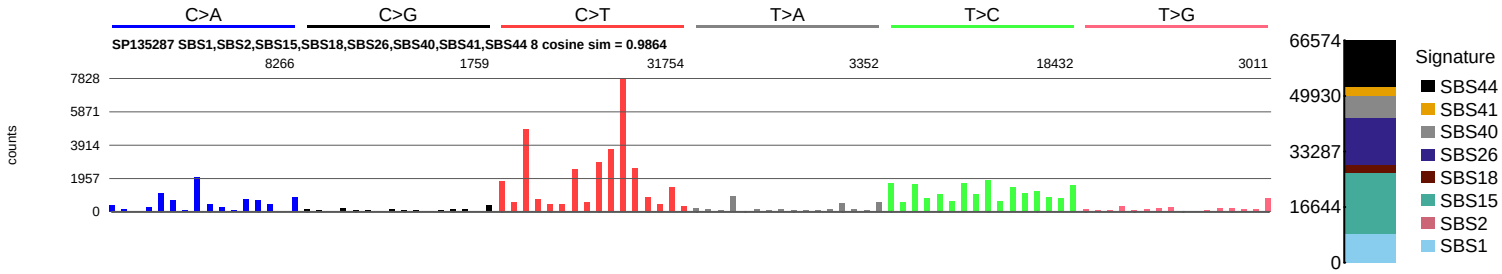
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS17a
SBS15
SBS2
SBS1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C>G
C>T
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T>C
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SP135287 SBS1,SBS2,SBS15,SBS18,SBS26,SBS40,SBS41,SBS44 8 cosine sim = 0.9864
8266
1759
31754
3352
18432
3011
counts
7828
5871
3914
1957
0
66574
49930
33287
16644
0
Signature
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SBS41
SBS40
SBS26
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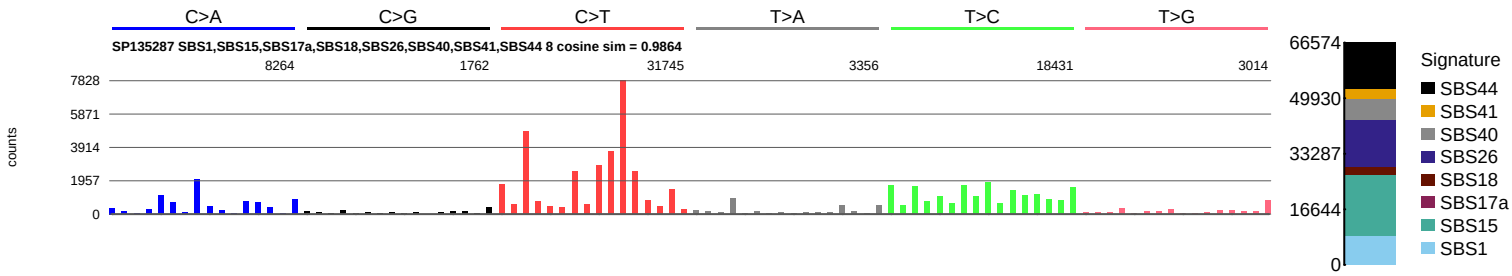
SBS15
SBS2
SBS1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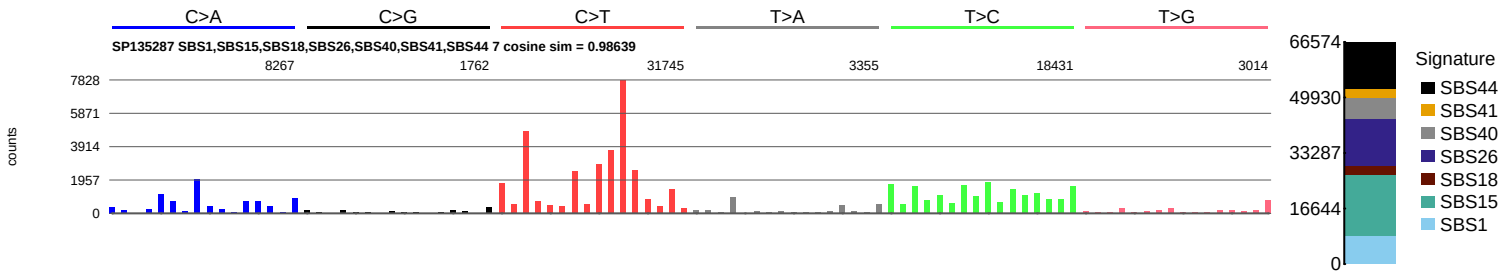
C>G
C>T
T>A
T>C
T>G
SP135287 SBS1,SBS15,SBS17a,SBS18,SBS26,SBS40,SBS41,SBS44 8 cosine sim = 0.9864
8264
1762
31745
3356
18431
3014
counts
7828
5871
3914
1957
0
66574
49930
33287
16644
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS17a
SBS15
SBS1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C>G
C>T
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T>C
T>G
SP135287 SBS1,SBS15,SBS18,SBS26,SBS40,SBS41,SBS44 7 cosine sim = 0.98639
8267
1762
31745
3355
18431
3014
counts
7828
5871
3914
1957
0
66574
49930
33287
16644
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SBS40
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SBS15
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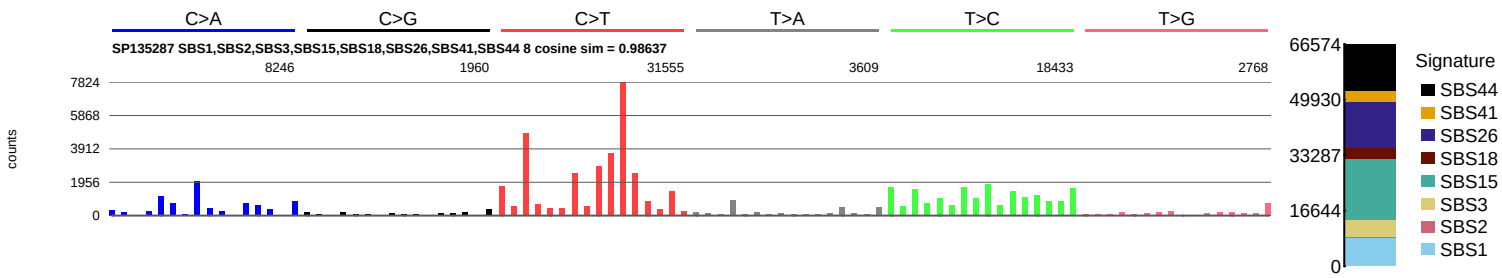
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C>G
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T>C
T>G
SP135287 SBS1,SBS2,SBS3,SBS15,SBS18,SBS26,SBS41,SBS44 8 cosine sim = 0.98637
8246
1960
31555
3609
18433
2768
counts
7824
5868
3912
1956
0
66574
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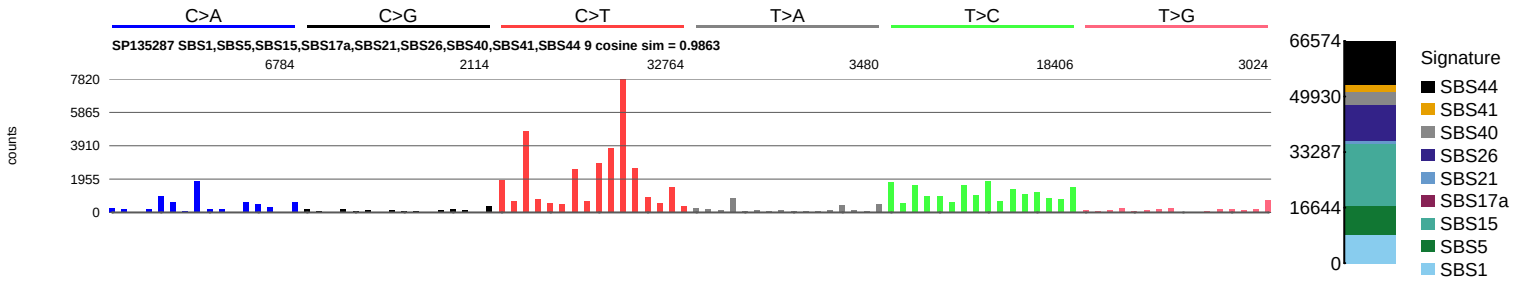
SBS2
SBS1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T>A
T>C
T>G
SP135287 SBS1,SBS5,SBS15,SBS17a,SBS21,SBS26,SBS40,SBS41,SBS44 9 cosine sim = 0.9863
6784
2114
32764
3480
18406
3024
counts
7820
5865
3910
1955
0
66574
49930
33287
16644
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SBS26
SBS21
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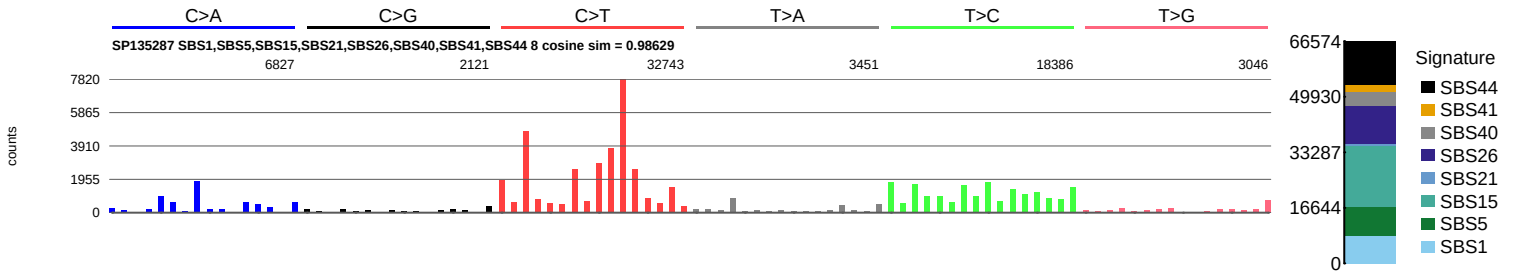
SBS15
SBS5
SBS1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T>A
T>C
T>G
SP135287 SBS1,SBS5,SBS15,SBS21,SBS26,SBS40,SBS41,SBS44 8 cosine sim = 0.98629
6827
2121
32743
3451
18386
3046
counts
7820
5865
3910
1955
0
66574
49930
33287
16644
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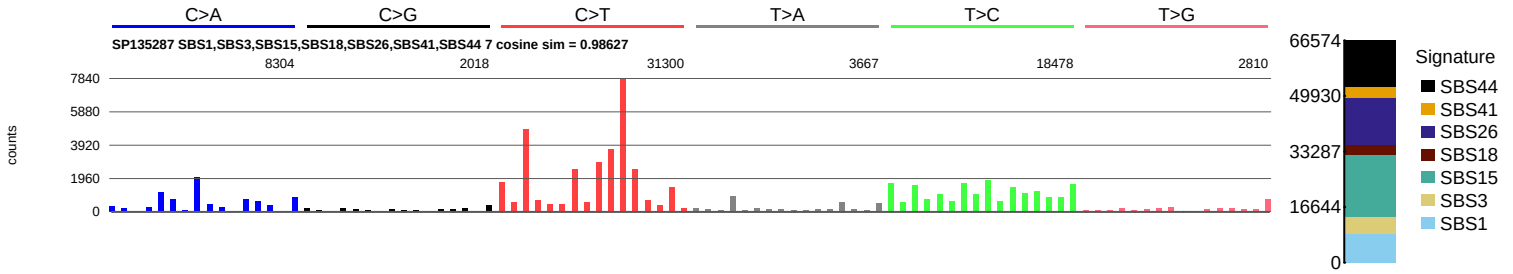
SBS40
SBS26
SBS21
SBS15
SBS5
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135287 SBS1,SBS3,SBS15,SBS18,SBS26,SBS41,SBS44 7 cosine sim = 0.98627
8304
2018
31300
3667
18478
2810
counts
7840
5880
3920
1960
0
66574
49930
33287
16644
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SBS26
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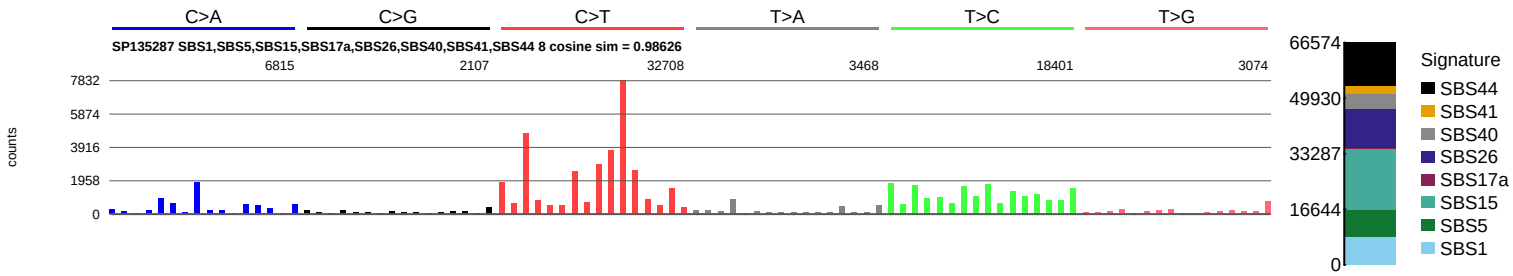
SBS3
SBS1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C>G
C>T
T>A
T>C
T>G
SP135287 SBS1,SBS5,SBS15,SBS17a,SBS26,SBS40,SBS41,SBS44 8 cosine sim = 0.98626
6815
2107
32708
3468
18401
3074
counts
7832
5874
3916
1958
0
66574
49930
33287
16644
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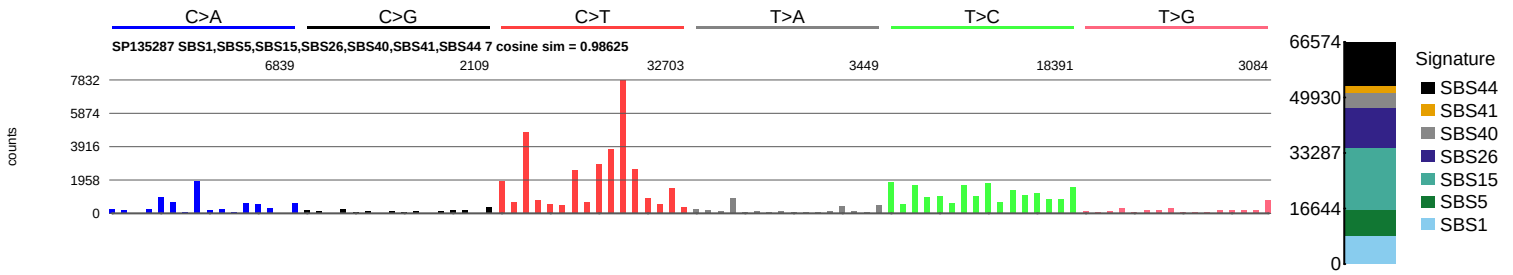
Signature
SBS44
SBS41
SBS40
SBS26
SBS17a
SBS15
SBS5
SBS1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C>G
C>T
T>A
T>C
T>G
SP135287 SBS1,SBS5,SBS15,SBS26,SBS40,SBS41,SBS44 7 cosine sim = 0.98625
6839
2109
32703
3449
18391
3084
counts
7832
5874
3916
1958
0
66574
49930
33287
16644
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SBS40
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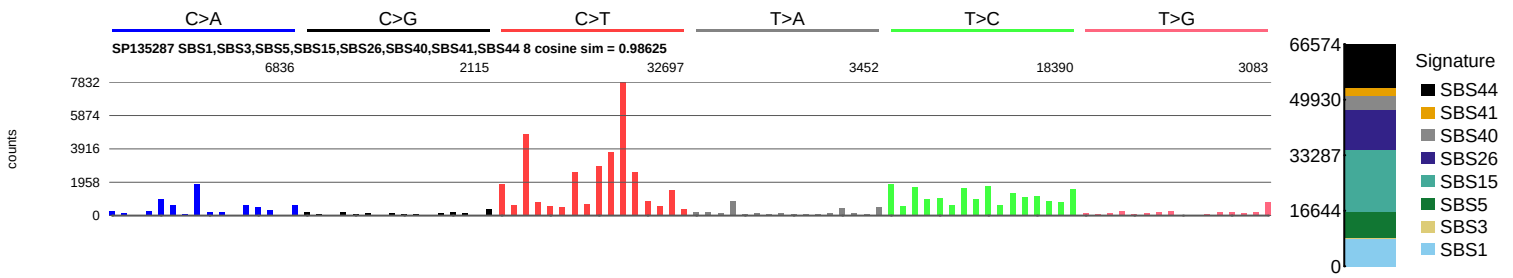
SBS5
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135287 SBS1,SBS3,SBS5,SBS15,SBS26,SBS40,SBS41,SBS44 8 cosine sim = 0.98625
6836
2115
32697
3452
18390
3083
counts
7832
5874
3916
1958
0
66574
49930
33287
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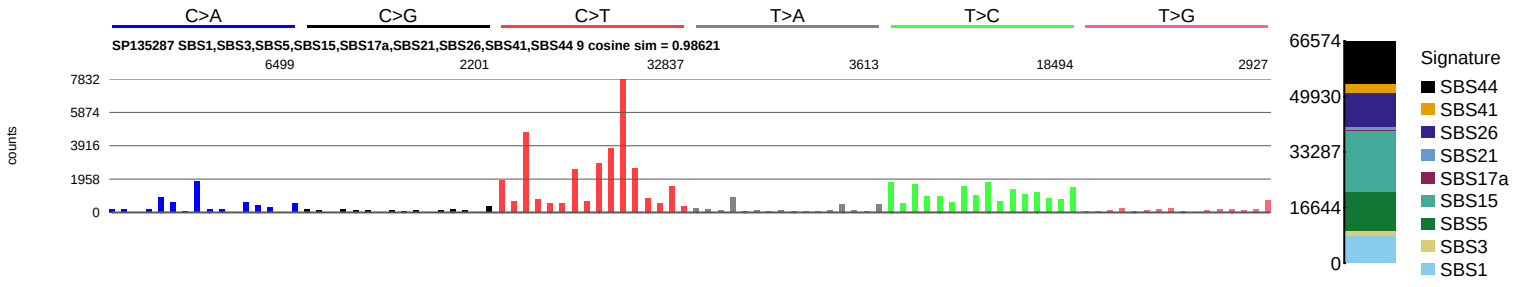
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135287 SBS1,SBS3,SBS5,SBS15,SBS17a,SBS21,SBS26,SBS41,SBS44 9 cosine sim = 0.98621
6499
2201
32837
3613
18494
2927
counts
7832
5874
3916
1958
0
66574
49930
33287
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SBS3
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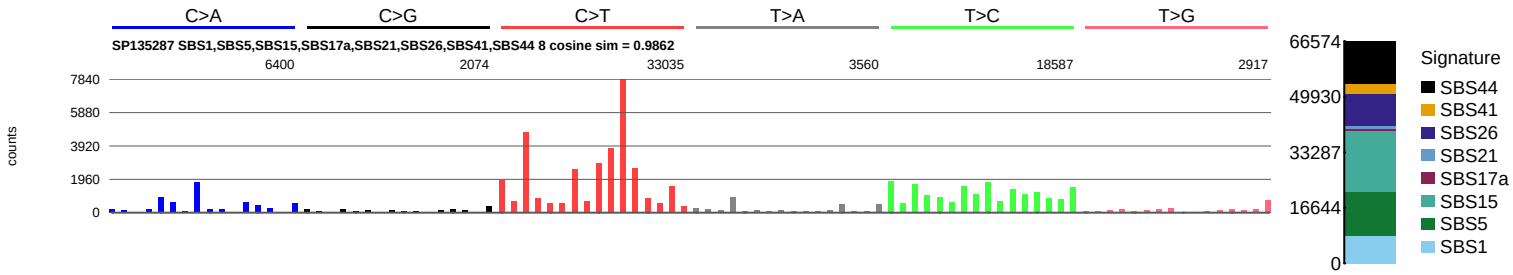
C>A
C>G
C>T
T>A
T>C
T>G
SP135287 SBS1,SBS5,SBS15,SBS17a,SBS21,SBS26,SBS41,SBS44 8 cosine sim = 0.9862
6400
2074
33035
3560
18587
2917
counts
7840
5880
3920
1960
0
66574
49930
33287
16644
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SBS44
SBS41
SBS26
SBS21
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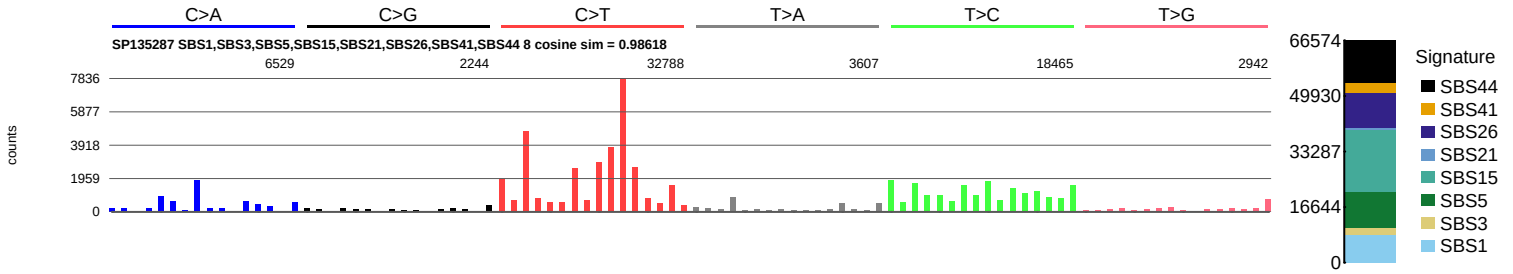
SBS5
SBS1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C>G
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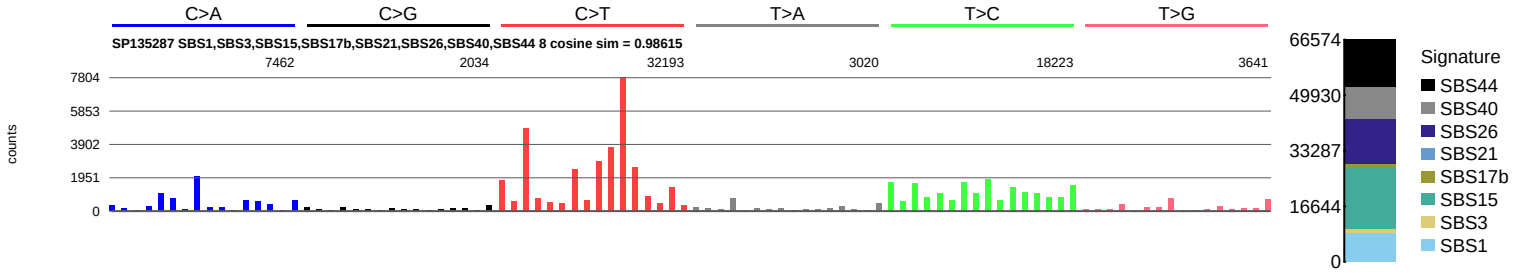
T>A
T>C
T>G
SP135287 SBS1,SBS3,SBS5,SBS15,SBS21,SBS26,SBS41,SBS44 8 cosine sim = 0.98618
6529
2244
32788
3607
18465
2942
counts
7836
5877
3918
1959
0
66574
49930
33287
16644
0
Signature
SBS44
SBS41
SBS26
SBS21
SBS15
SBS5
SBS3
SBS1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SP135287 SBS1,SBS3,SBS15,SBS17b,SBS21,SBS26,SBS40,SBS44 8 cosine sim = 0.98615
7462
2034
32193
3020
18223
3641
counts
7804
5853
3902
1951
0
66574
49930
33287
16644
0
Signature
SBS44
SBS40
SBS26
SBS21
SBS17b
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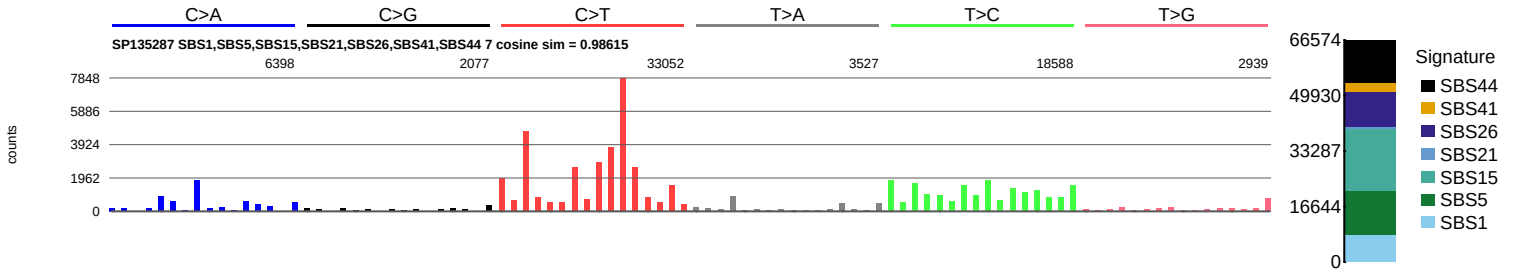
SBS3
SBS1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SP135287 SBS1,SBS5,SBS15,SBS21,SBS26,SBS41,SBS44 7 cosine sim = 0.98615
6398
2077
33052
3527
18588
2939
counts
7848
5886
3924
1962
0
66574
49930
33287
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SBS41
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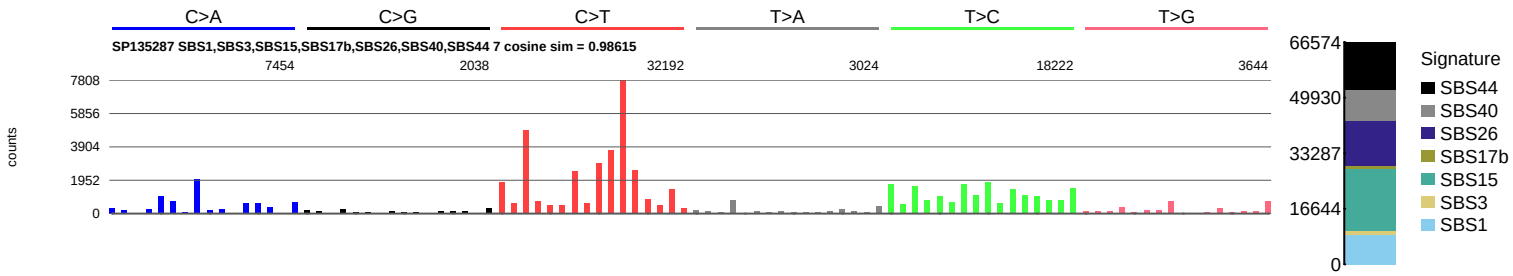
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7454
2038
32192
3024
18222
3644
counts
7808
5856
3904
1952
0
66574
49930
33287
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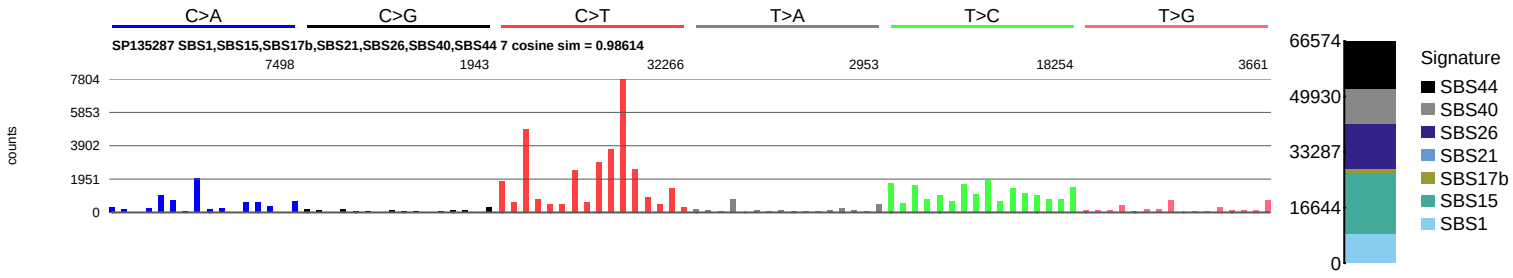
SBS3
SBS1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7498
1943
32266
2953
18254
3661
counts
7804
5853
3902
1951
0
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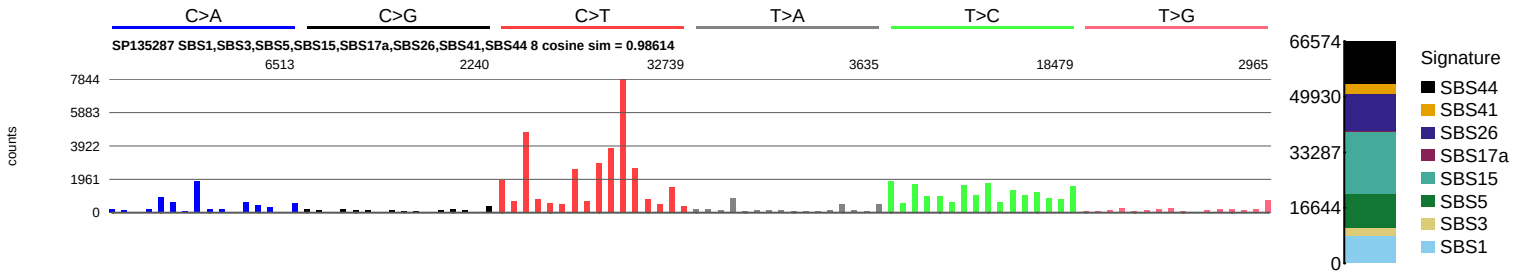
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SP135287 SBS1,SBS3,SBS5,SBS15,SBS17a,SBS26,SBS41,SBS44 8 cosine sim = 0.98614
6513
2240
32739
3635
18479
2965
counts
7844
5883
3922
1961
0
66574
49930
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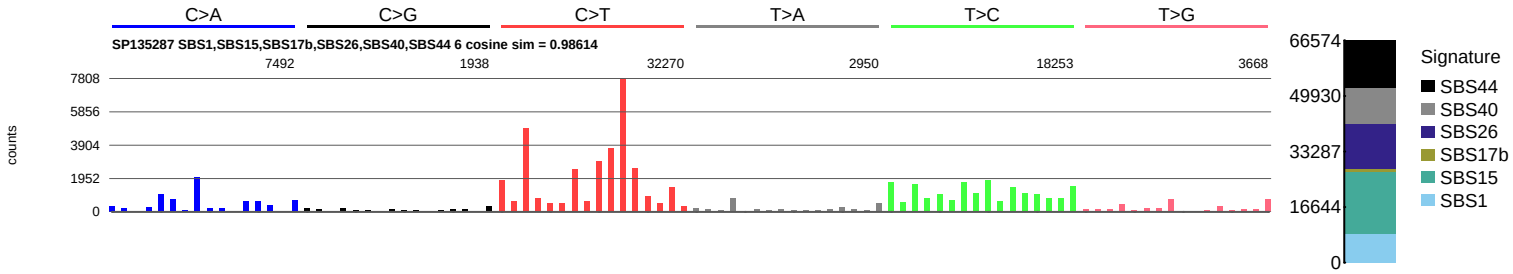
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SP135287 SBS1,SBS15,SBS17b,SBS26,SBS40,SBS44 6 cosine sim = 0.98614
7492
1938
32270
2950
18253
3668
counts
7808
5856
3904
1952
0
66574
49930
33287
16644
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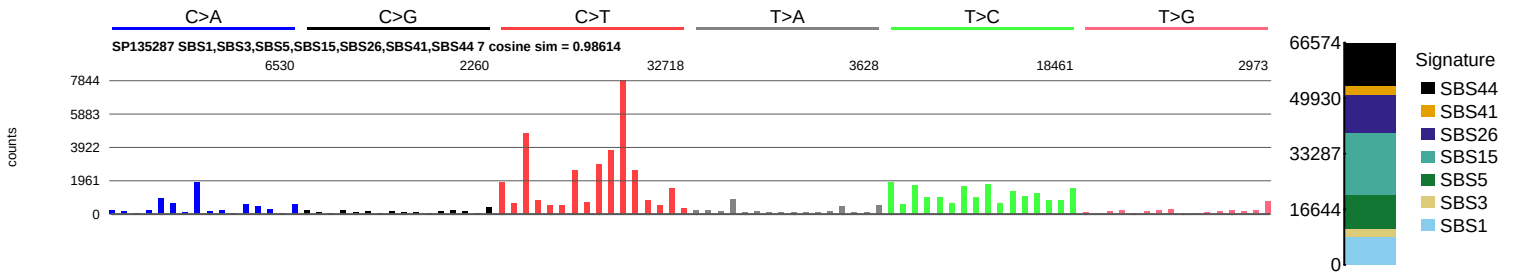
SBS15
SBS1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SP135287 SBS1,SBS3,SBS5,SBS15,SBS26,SBS41,SBS44 7 cosine sim = 0.98614
6530
2260
32718
3628
18461
2973
counts
7844
5883
3922
1961
0
66574
49930
33287
16644
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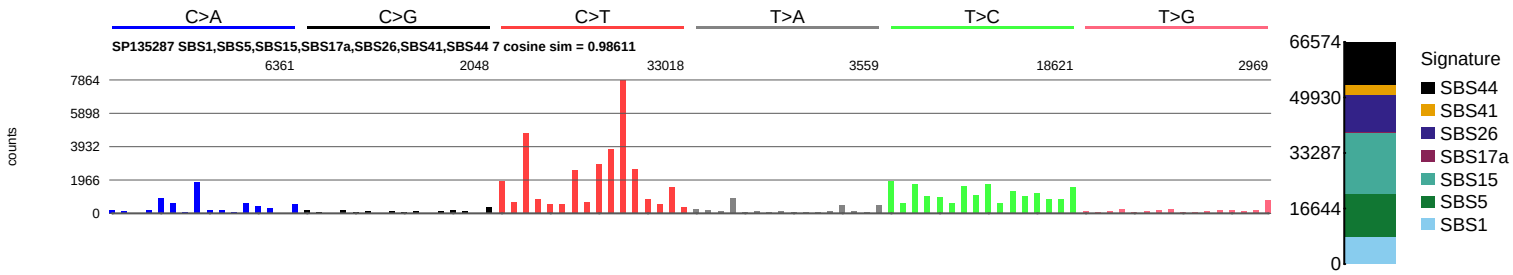
SBS41
SBS26
SBS15
SBS5
SBS3
SBS1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SP135287 SBS1,SBS5,SBS15,SBS17a,SBS26,SBS41,SBS44 7 cosine sim = 0.98611
6361
2048
33018
3559
18621
2969
counts
7864
5898
3932
1966
0
66574
49930
33287
16644
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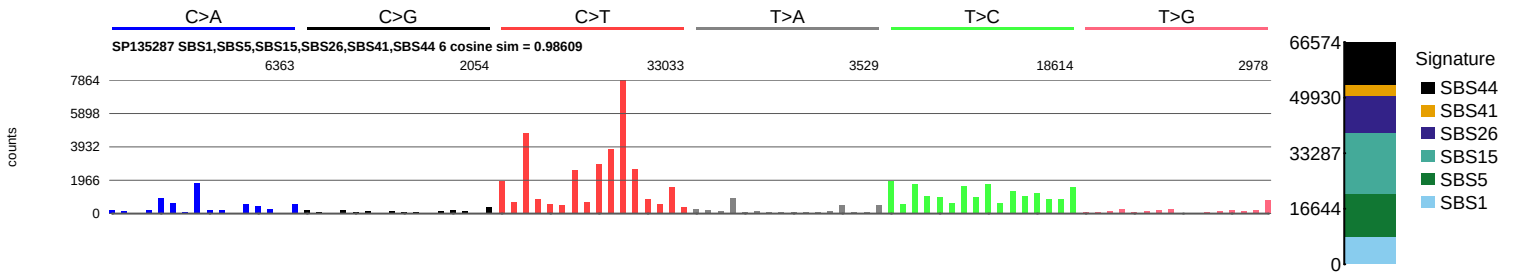
SBS41
SBS26
SBS17a
SBS15
SBS5
SBS1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SP135287 SBS1,SBS5,SBS15,SBS26,SBS41,SBS44 6 cosine sim = 0.98609
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2054
33033
3529
18614
2978
counts
7864
5898
3932
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49930
33287
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SBS41
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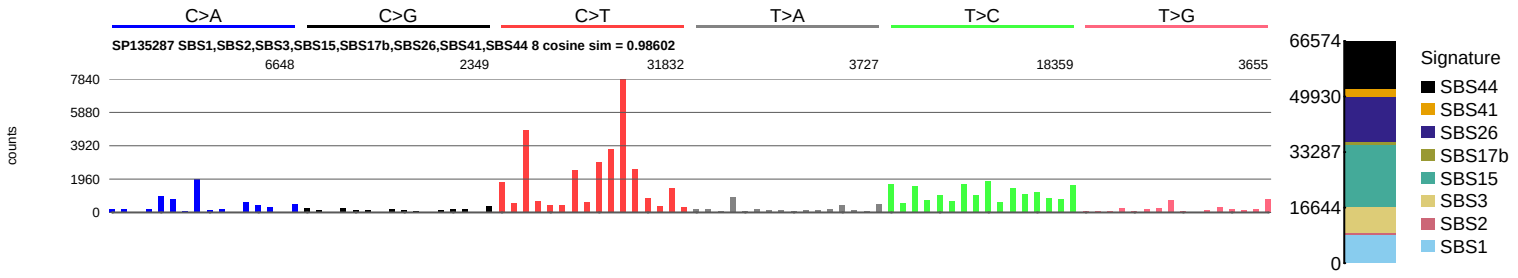
SBS5
SBS1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6648
2349
31832
3727
18359
3655
counts
7840
5880
3920
1960
0
66574
49930
33287
16644
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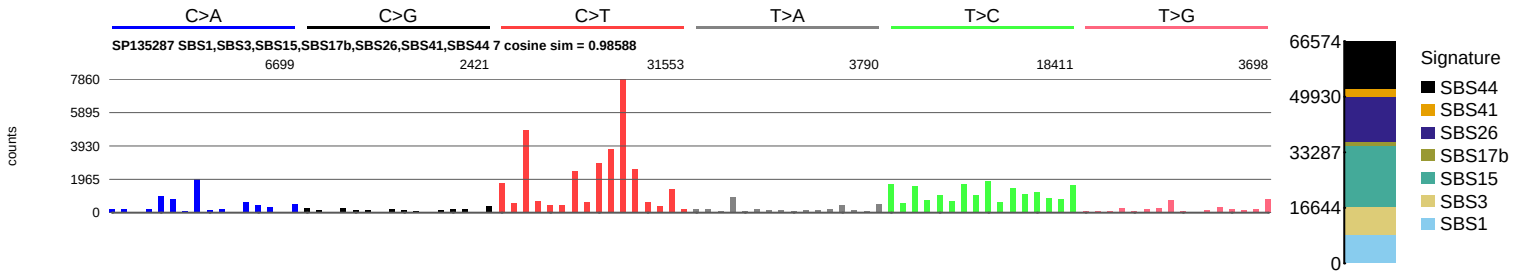
Signature
SBS44
SBS41
SBS26
SBS17b
SBS15
SBS3
SBS2
SBS1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C>G
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SP135287 SBS1,SBS3,SBS15,SBS17b,SBS26,SBS41,SBS44 7 cosine sim = 0.98588
6699
2421
31553
3790
18411
3698
counts
7860
5895
3930
1965
0
66574
49930
33287
16644
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SBS26
SBS17b
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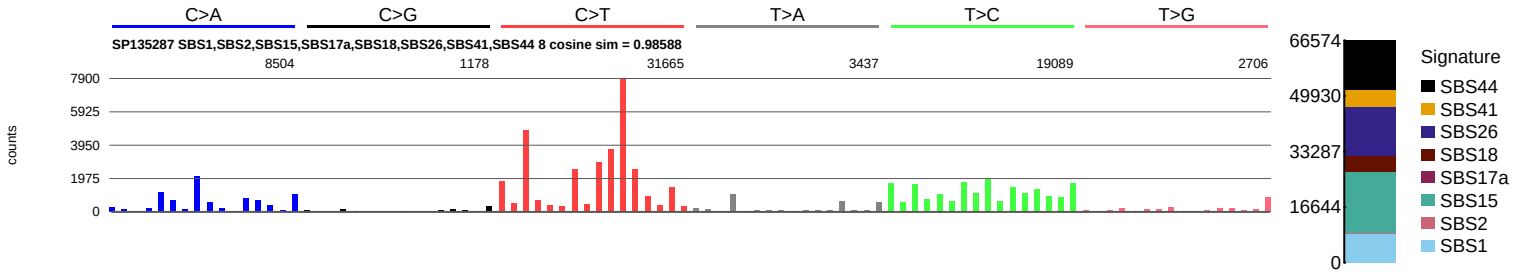
SBS3
SBS1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SP135287 SBS1,SBS2,SBS15,SBS17a,SBS18,SBS26,SBS41,SBS44 8 cosine sim = 0.98588
8504
1178
31665
3437
19089
2706
counts
7900
5925
3950
1975
0
66574
49930
33287
16644
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SBS41
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SBS18
SBS17a
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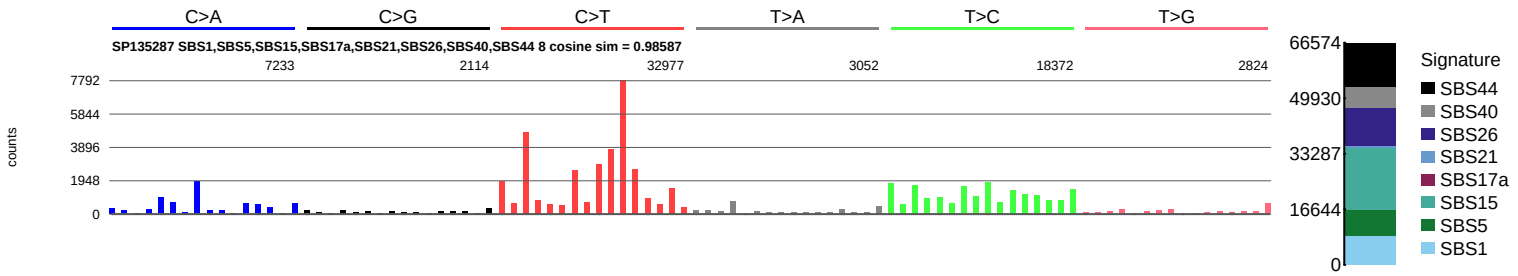
SBS2
SBS1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SP135287 SBS1,SBS5,SBS15,SBS17a,SBS21,SBS26,SBS40,SBS44 8 cosine sim = 0.98587
7233
2114
32977
3052
18372
2824
counts
7792
5844
3896
1948
0
66574
49930
33287
16644
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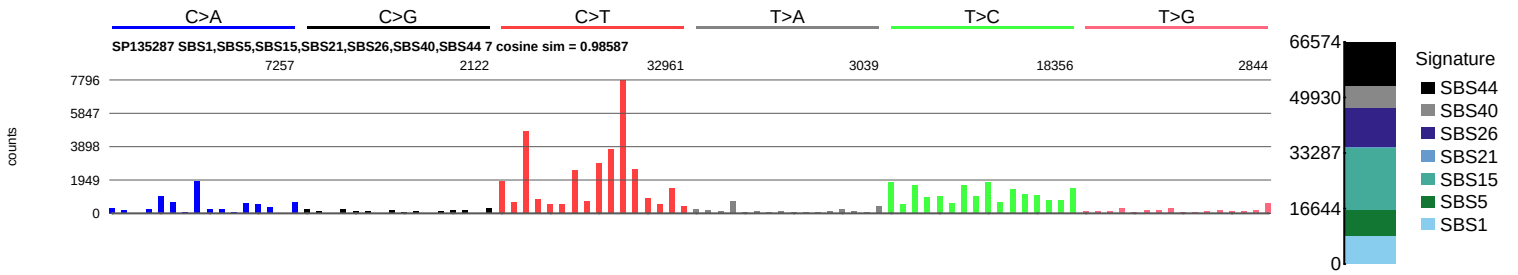
SBS21
SBS17a
SBS15
SBS5
SBS1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SP135287 SBS1,SBS5,SBS15,SBS21,SBS26,SBS40,SBS44 7 cosine sim = 0.98587
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2122
32961
3039
18356
2844
counts
7796
5847
3898
1949
0
66574
49930
33287
16644
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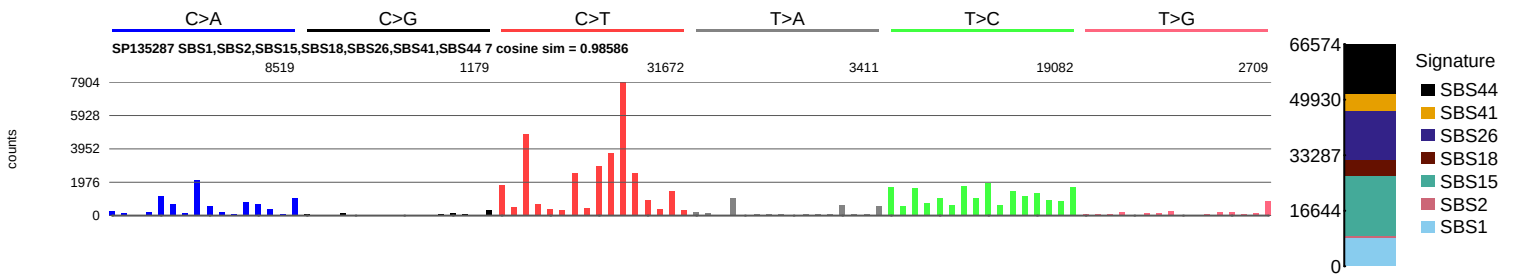
SBS26
SBS21
SBS15
SBS5
SBS1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T>C
T>G
SP135287 SBS1,SBS2,SBS15,SBS18,SBS26,SBS41,SBS44 7 cosine sim = 0.98586
8519
1179
31672
3411
19082
2709
counts
7904
5928
3952
1976
0
66574
49930
33287
16644
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SBS44
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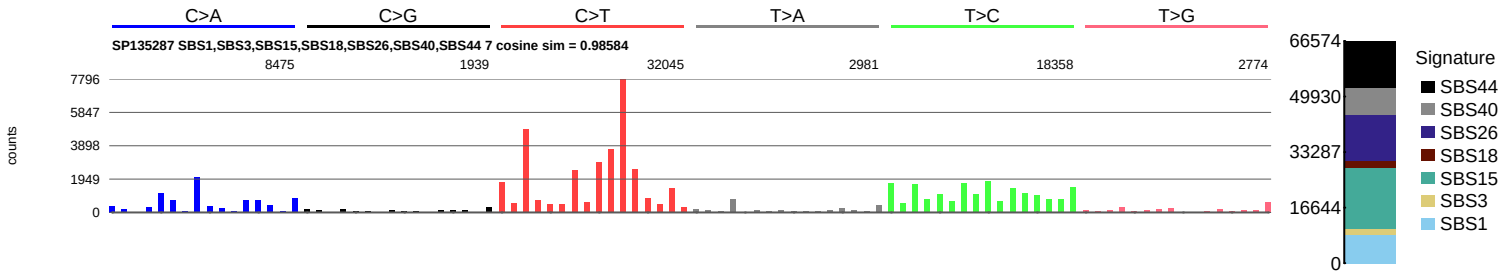
SBS18
SBS15
SBS2
SBS1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SP135287 SBS1,SBS3,SBS15,SBS18,SBS26,SBS40,SBS44 7 cosine sim = 0.98584
8475
1939
32045
2981
18358
2774
counts
7796
5847
3898
1949
0
66574
49930
33287
16644
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SBS18
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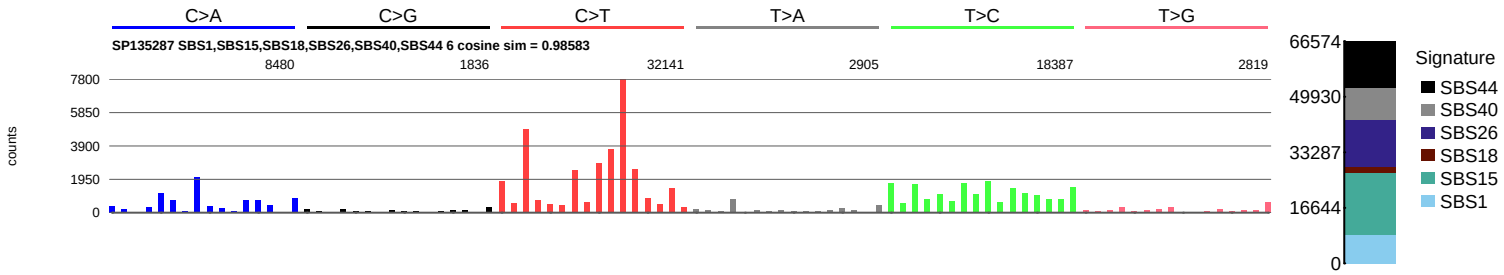
SBS3
SBS1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1836
32141
2905
18387
2819
counts
7800
5850
3900
1950
0
66574
49930
33287
16644
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SBS40
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SBS18
SBS15
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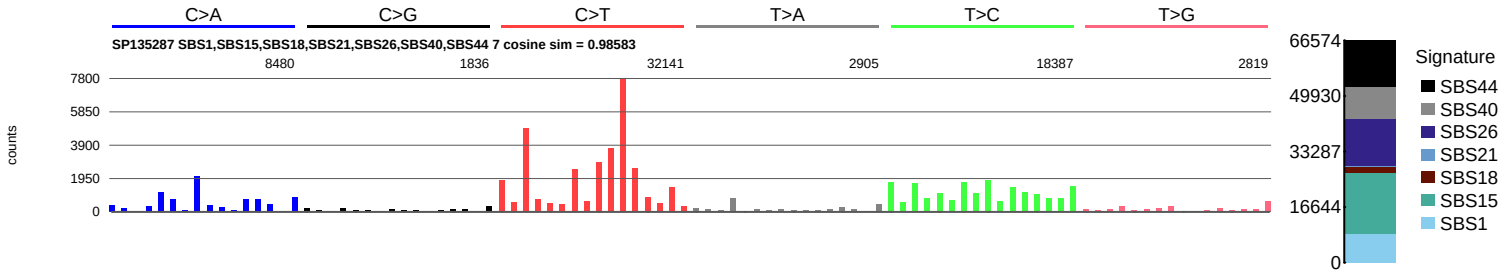
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SP135287 SBS1,SBS15,SBS18,SBS21,SBS26,SBS40,SBS44 7 cosine sim = 0.98583
8480
1836
32141
2905
18387
2819
counts
7800
5850
3900
1950
0
66574
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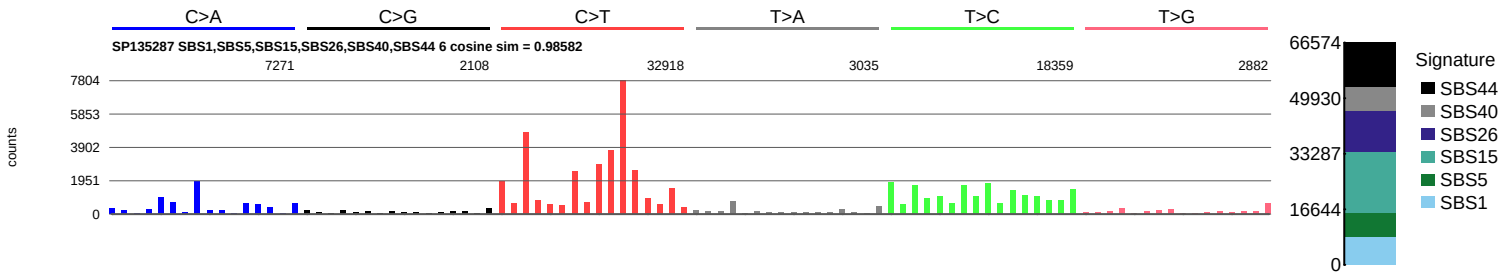
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SP135287 SBS1,SBS5,SBS15,SBS26,SBS40,SBS44 6 cosine sim = 0.98582
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2108
32918
3035
18359
2882
counts
7804
5853
3902
1951
0
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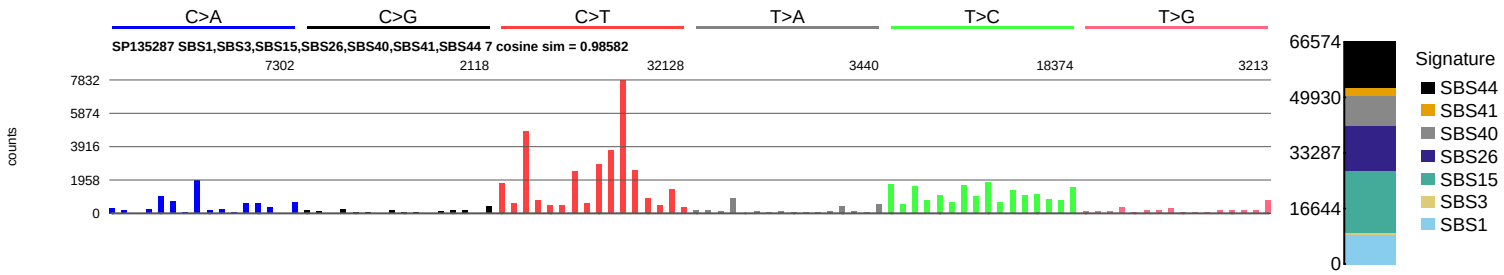
SBS1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T>C
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SP135287 SBS1,SBS3,SBS15,SBS26,SBS40,SBS41,SBS44 7 cosine sim = 0.98582
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2118
32128
3440
18374
3213
counts
7832
5874
3916
1958
0
66574
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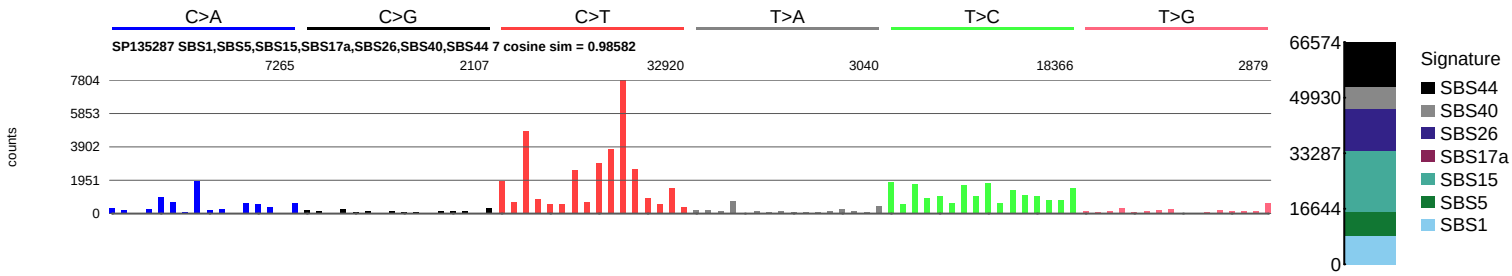
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T>C
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SP135287 SBS1,SBS5,SBS15,SBS17a,SBS26,SBS40,SBS44 7 cosine sim = 0.98582
7265
2107
32920
3040
18366
2879
counts
7804
5853
3902
1951
0
66574
49930
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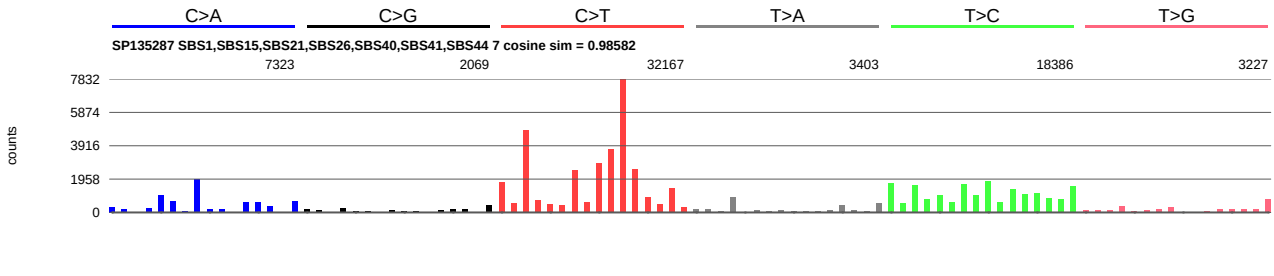
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T>C
T>G
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2069
32167
3403
18386
3227
counts
7832
5874
3916
1958
0
66574
49930
33287
16644
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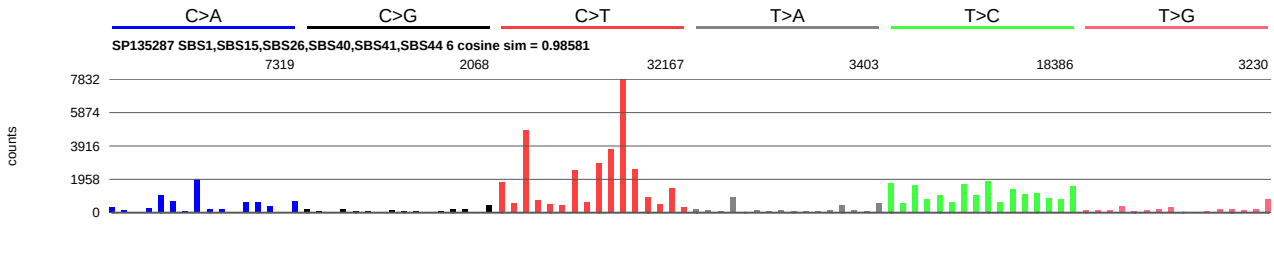
SBS41
SBS40
SBS26
SBS21
SBS15
SBS1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T>A
T>C
T>G
SP135287 SBS1,SBS15,SBS26,SBS40,SBS41,SBS44 6 cosine sim = 0.98581
7319
2068
32167
3403
18386
3230
counts
7832
5874
3916
1958
0
66574
49930
33287
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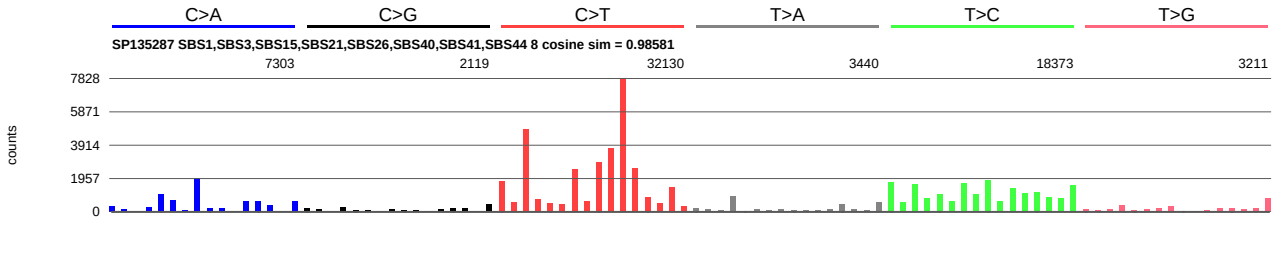
SBS1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T>C
T>G
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7303
2119
32130
3440
18373
3211
counts
7828
5871
3914
1957
0
66574
49930
33287
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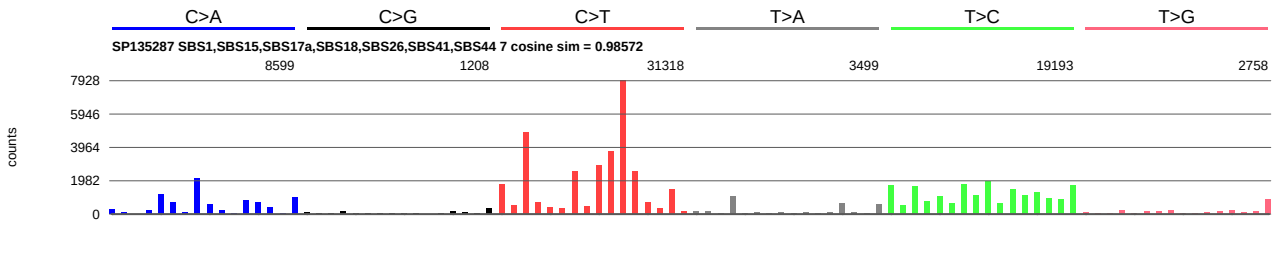
SBS3
SBS1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C>G
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T>C
T>G
SP135287 SBS1,SBS15,SBS17a,SBS18,SBS26,SBS41,SBS44 7 cosine sim = 0.98572
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1208
31318
3499
19193
2758
counts
7928
5946
3964
1982
0
66574
49930
33287
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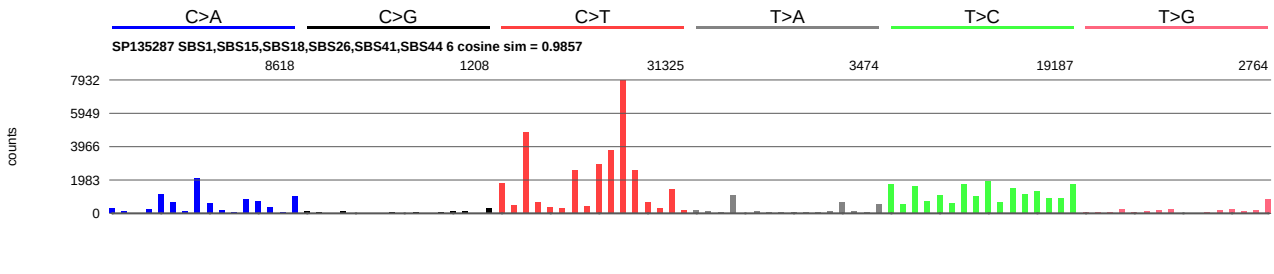
Signature
SBS44
SBS41
SBS26
SBS18
SBS17a
SBS15
SBS1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C>G
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SP135287 SBS1,SBS15,SBS18,SBS26,SBS41,SBS44 6 cosine sim = 0.9857
8618
1208
31325
3474
19187
2764
counts
7932
5949
3966
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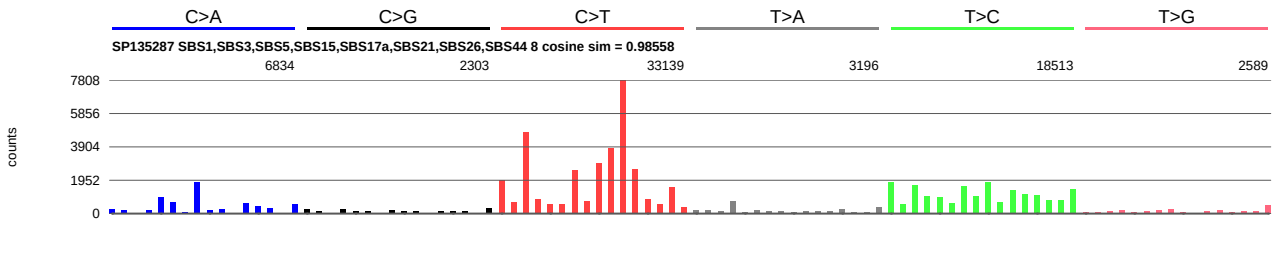
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SP135287 SBS1,SBS3,SBS5,SBS15,SBS17a,SBS21,SBS26,SBS44 8 cosine sim = 0.98558
6834
2303
33139
3196
18513
2589
counts
7808
5856
3904
1952
0
66574
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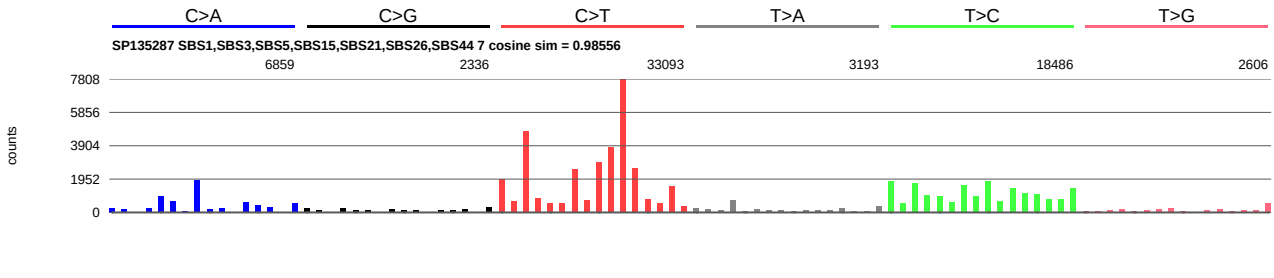
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SP135287 SBS1,SBS3,SBS5,SBS15,SBS21,SBS26,SBS44 7 cosine sim = 0.98556
6859
2336
33093
3193
18486
2606
counts
7808
5856
3904
1952
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33287
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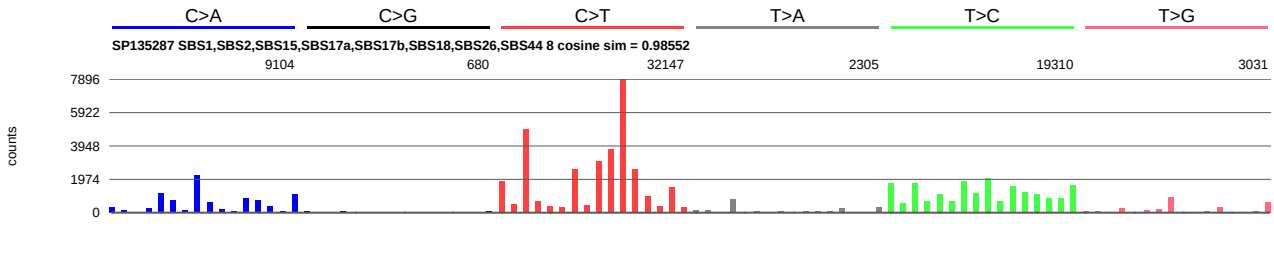
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SP135287 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS44 8 cosine sim = 0.98552
9104
680
32147
2305
19310
3031
counts
7896
5922
3948
1974
0
66574
49930
33287
16644
0
Signature
SBS44
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SBS18
SBS17b
SBS17a
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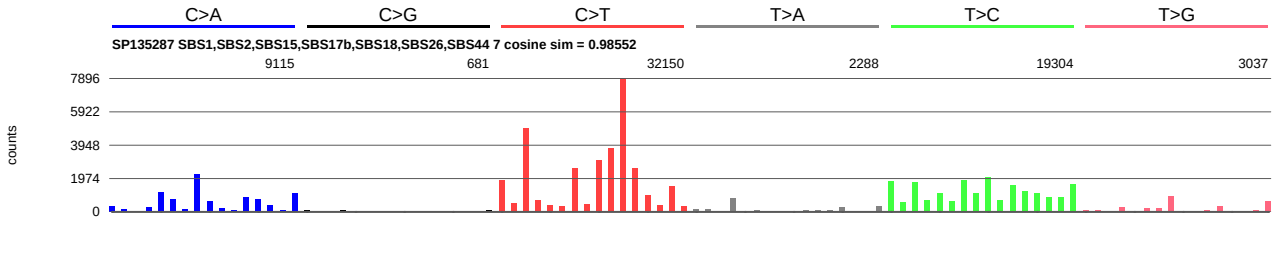
SBS2
SBS1


C>A
C>G
C>T
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T>C
T>G
SP135287 SBS1,SBS2,SBS15,SBS17b,SBS18,SBS26,SBS44 7 cosine sim = 0.98552
9115
681
32150
2288
19304
3037
counts
7896
5922
3948
1974
0
66574
49930
33287
16644
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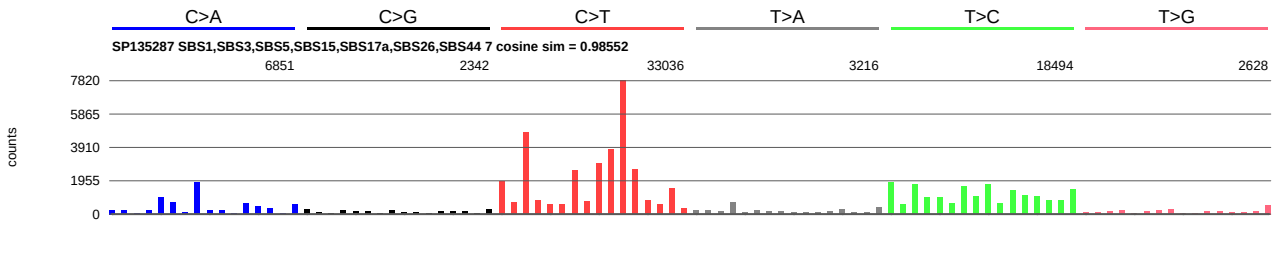
C>A
C>G
C>T
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T>C
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SP135287 SBS1,SBS3,SBS5,SBS15,SBS17a,SBS26,SBS44 7 cosine sim = 0.98552
6851
2342
33036
3216
18494
2628
counts
7820
5865
3910
1955
0
66574
49930
33287
16644
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SBS17a
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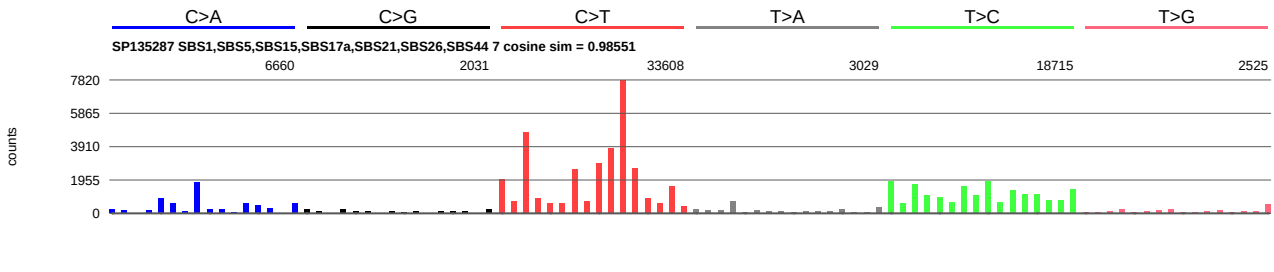
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T>C
T>G
SP135287 SBS1,SBS5,SBS15,SBS17a,SBS21,SBS26,SBS44 7 cosine sim = 0.98551
6660
2031
33608
3029
18715
2525
counts
7820
5865
3910
1955
0
66574
49930
33287
16644
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SBS21
SBS17a
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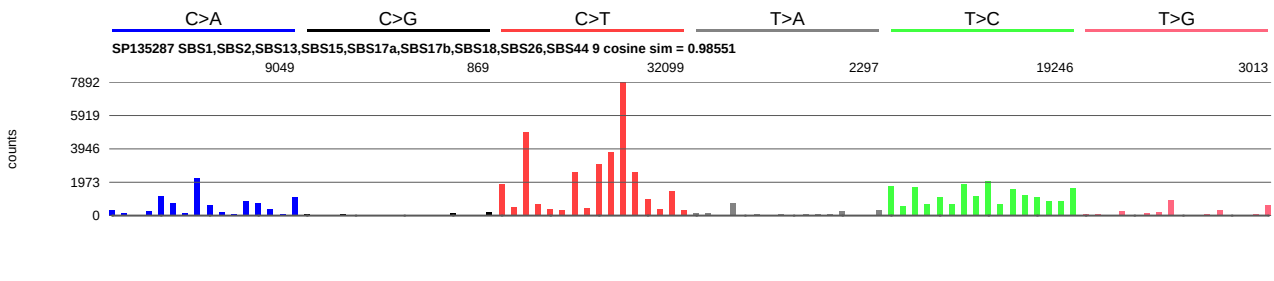
SBS5
SBS1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T>C
T>G
SP135287 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS44 9 cosine sim = 0.98551
9049
869
32099
2297
19246
3013
counts
7892
5919
3946
1973
0
66574
49930
33287
16644
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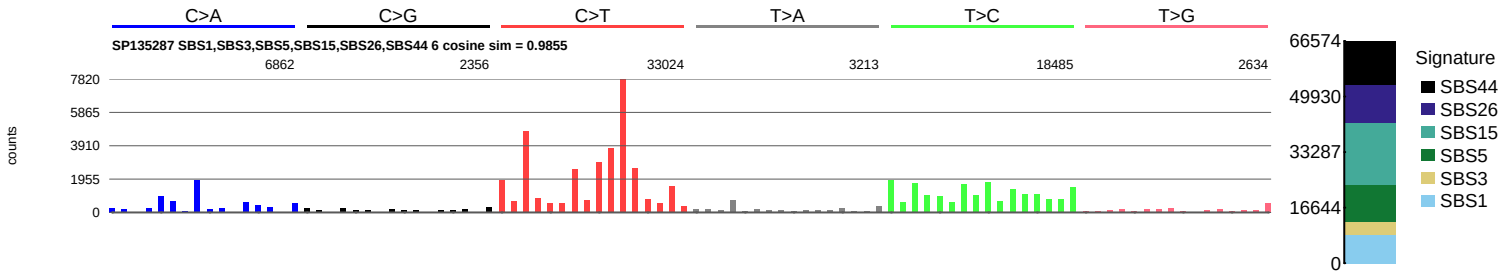
SBS44
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS2
SBS1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T>C
T>G
SP135287 SBS1,SBS3,SBS5,SBS15,SBS26,SBS44 6 cosine sim = 0.9855
6862
2356
33024
3213
18485
2634
counts
7820
5865
3910
1955
0
66574
49930
33287
16644
0
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SBS26
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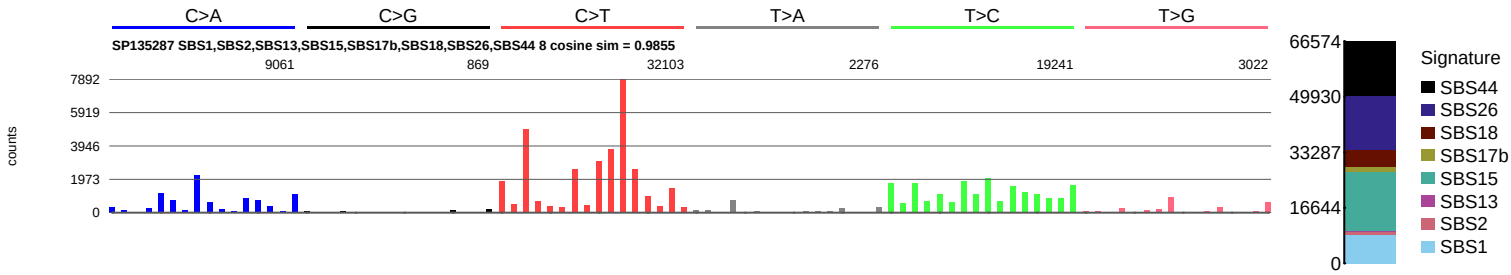
SBS5
SBS3
SBS1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C>T
T>A
T>C
T>G
SP135287 SBS1,SBS2,SBS13,SBS15,SBS17b,SBS18,SBS26,SBS44 8 cosine sim = 0.9855
9061
869
32103
2276
19241
3022
counts
7892
5919
3946
1973
0
66574
49930
33287
16644
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Signature
SBS44
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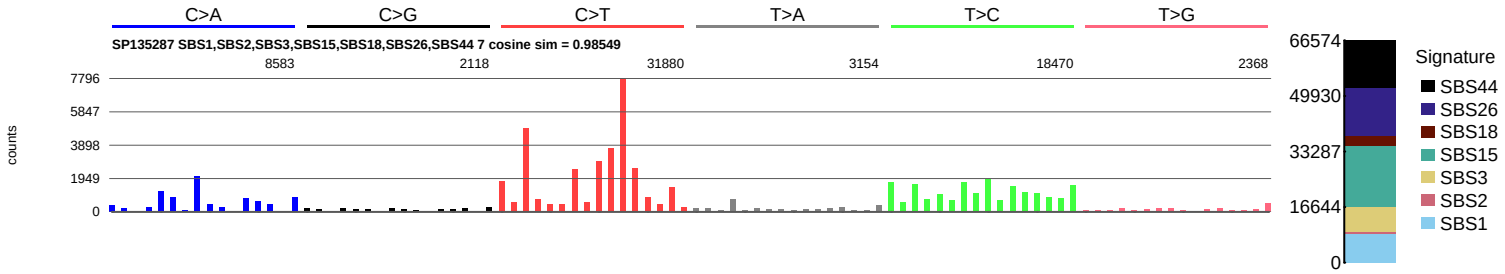
SBS18
SBS17b
SBS15
SBS13
SBS2
SBS1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C>G
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T>A
T>C
T>G
SP135287 SBS1,SBS2,SBS3,SBS15,SBS18,SBS26,SBS44 7 cosine sim = 0.98549
8583
2118
31880
3154
18470
2368
counts
7796
5847
3898
1949
0
66574
49930
33287
16644
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SBS44
SBS26
SBS18
SBS15
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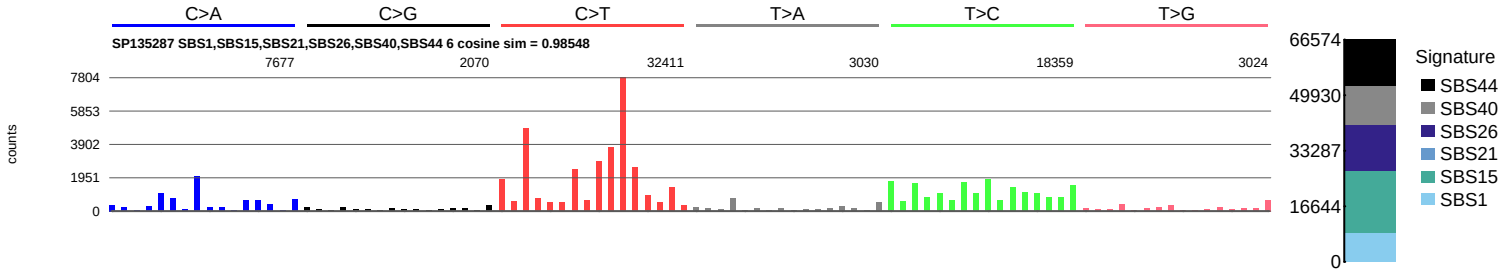
SBS1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T>A
T>C
T>G
SP135287 SBS1,SBS15,SBS21,SBS26,SBS40,SBS44 6 cosine sim = 0.98548
7677
2070
32411
3030
18359
3024
counts
7804
5853
3902
1951
0
66574
49930
33287
16644
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SBS40
SBS26
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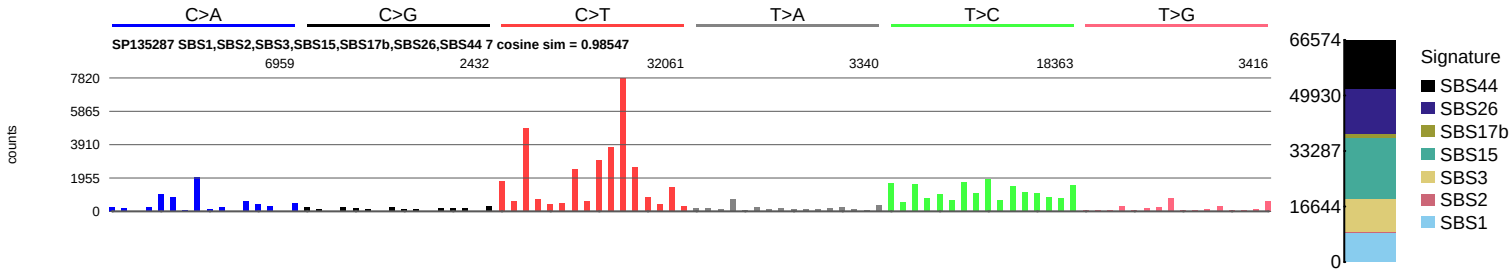
SBS15
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135287 SBS1,SBS2,SBS3,SBS15,SBS17b,SBS26,SBS44 7 cosine sim = 0.98547
6959
2432
32061
3340
18363
3416
counts
7820
5865
3910
1955
0
66574
49930
33287
16644
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SBS26
SBS17b
SBS15
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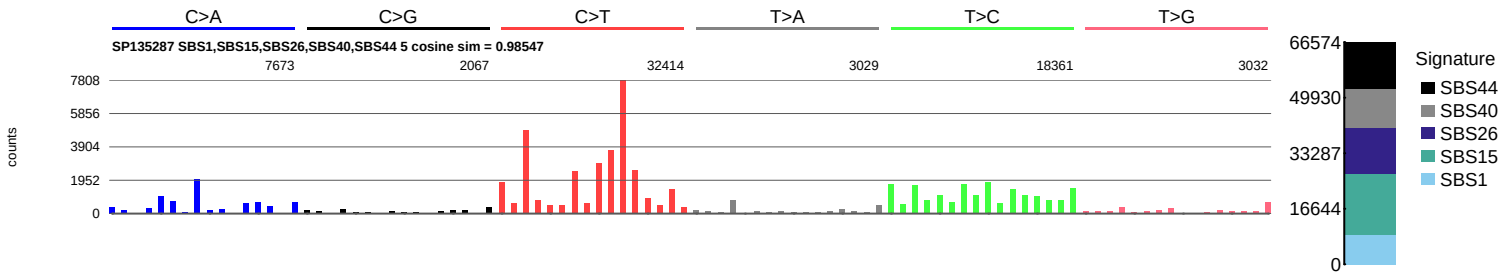
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135287 SBS1,SBS15,SBS26,SBS40,SBS44 5 cosine sim = 0.98547
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2067
32414
3029
18361
3032
counts
7808
5856
3904
1952
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66574
49930
33287
16644
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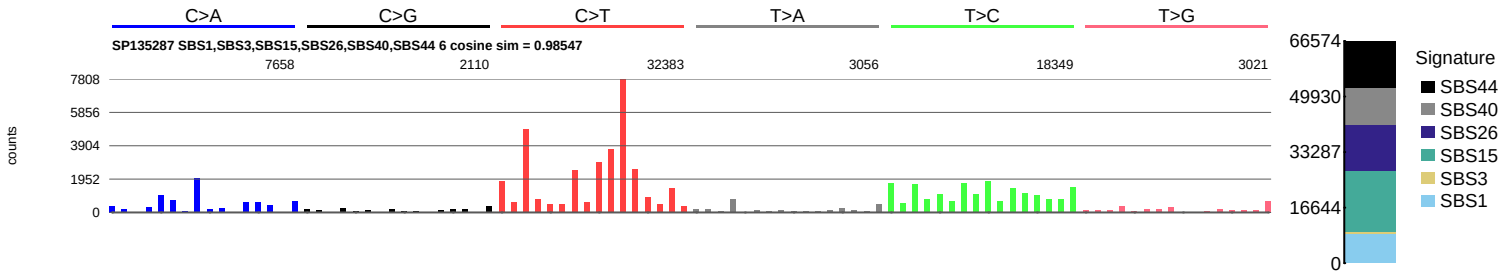
SBS15
SBS1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C>G
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T>A
T>C
T>G
SP135287 SBS1,SBS3,SBS15,SBS26,SBS40,SBS44 6 cosine sim = 0.98547
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2110
32383
3056
18349
3021
counts
7808
5856
3904
1952
0
66574
49930
33287
16644
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SBS3
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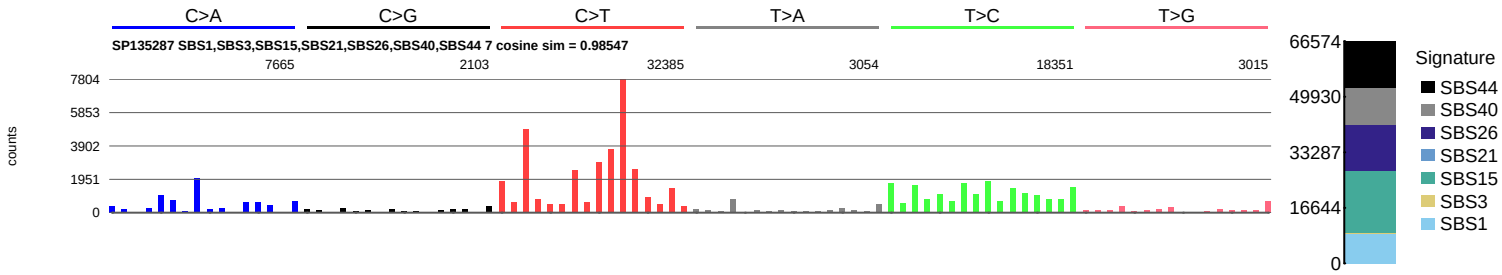
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SP135287 SBS1,SBS3,SBS15,SBS21,SBS26,SBS40,SBS44 7 cosine sim = 0.98547
7665
2103
32385
3054
18351
3015
counts
7804
5853
3902
1951
0
66574
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33287
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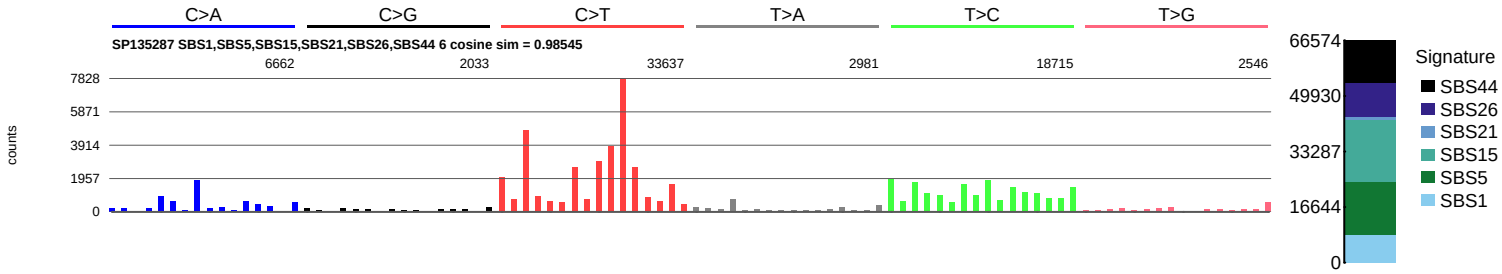
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SP135287 SBS1,SBS5,SBS15,SBS21,SBS26,SBS44 6 cosine sim = 0.98545
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2033
33637
2981
18715
2546
counts
7828
5871
3914
1957
0
66574
49930
33287
16644
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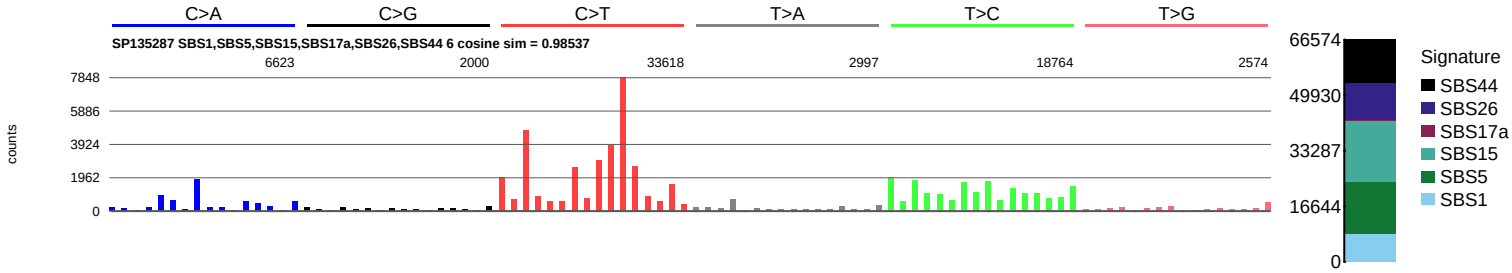
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T>C
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SP135287 SBS1,SBS5,SBS15,SBS17a,SBS26,SBS44 6 cosine sim = 0.98537
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2000
33618
2997
18764
2574
counts
7848
5886
3924
1962
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33287
16644
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SBS17a
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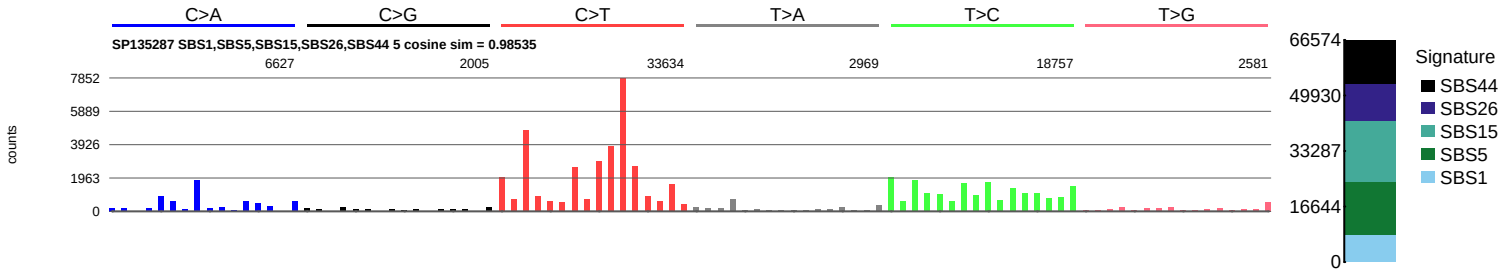
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T>A
T>C
T>G
SP135287 SBS1,SBS5,SBS15,SBS26,SBS44 5 cosine sim = 0.98535
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2005
33634
2969
18757
2581
counts
7852
5889
3926
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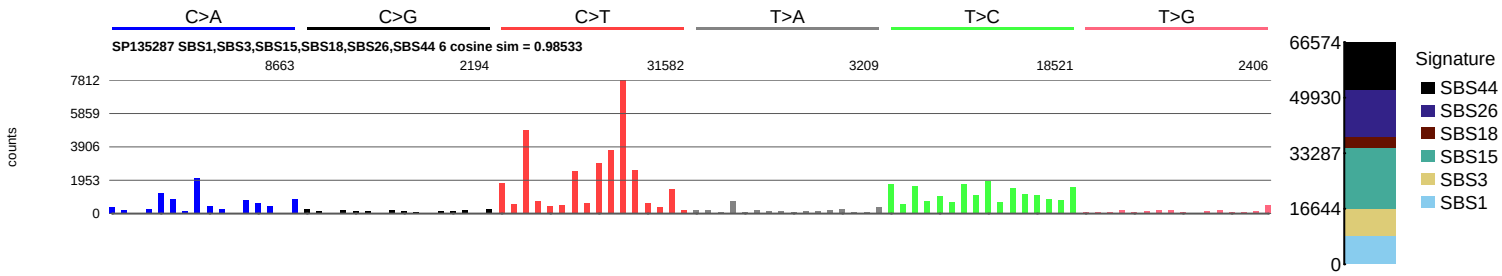
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SP135287 SBS1,SBS3,SBS15,SBS18,SBS26,SBS44 6 cosine sim = 0.98533
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2194
31582
3209
18521
2406
counts
7812
5859
3906
1953
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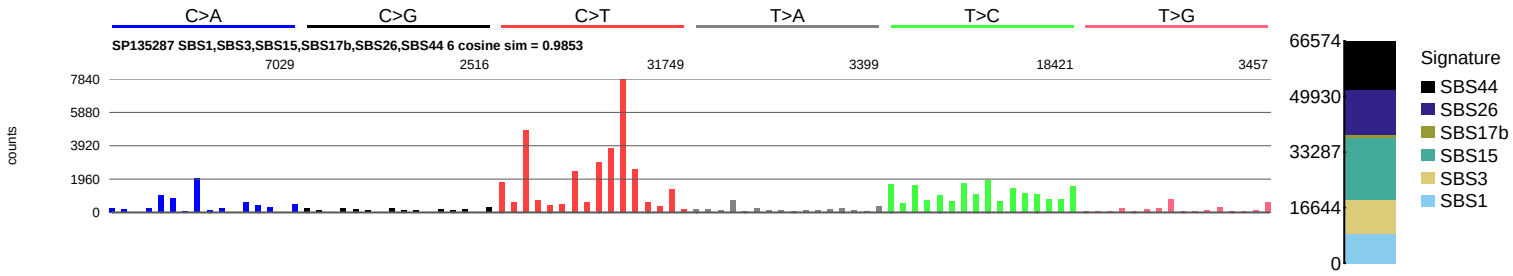
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T>G
SP135287 SBS1,SBS3,SBS15,SBS17b,SBS26,SBS44 6 cosine sim = 0.9853
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2516
31749
3399
18421
3457
counts
7840
5880
3920
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33287
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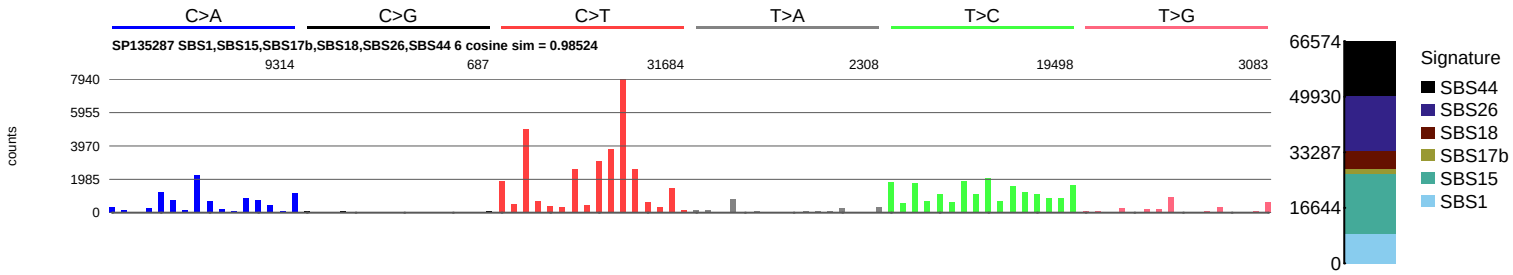
SBS1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T>C
T>G
SP135287 SBS1,SBS15,SBS17b,SBS18,SBS26,SBS44 6 cosine sim = 0.98524
9314
687
31684
2308
19498
3083
counts
7940
5955
3970
1985
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66574
49930
33287
16644
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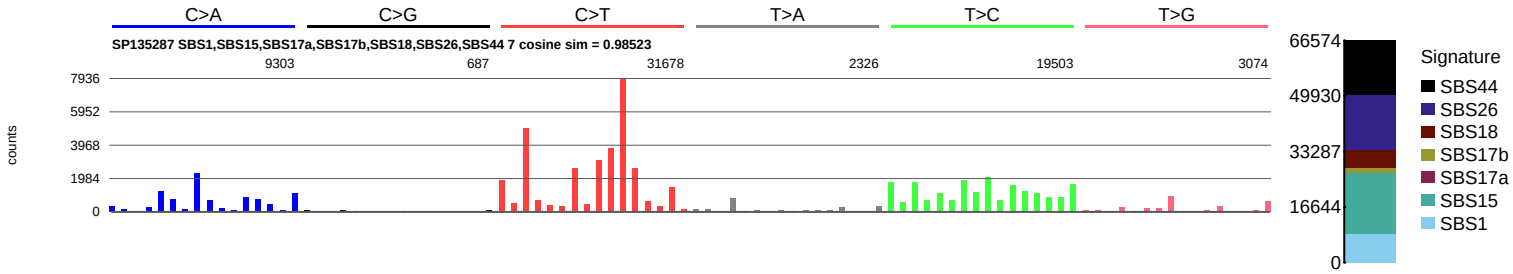
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T>G
SP135287 SBS1,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS44 7 cosine sim = 0.98523
9303
687
31678
2326
19503
3074
counts
7936
5952
3968
1984
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66574
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16644
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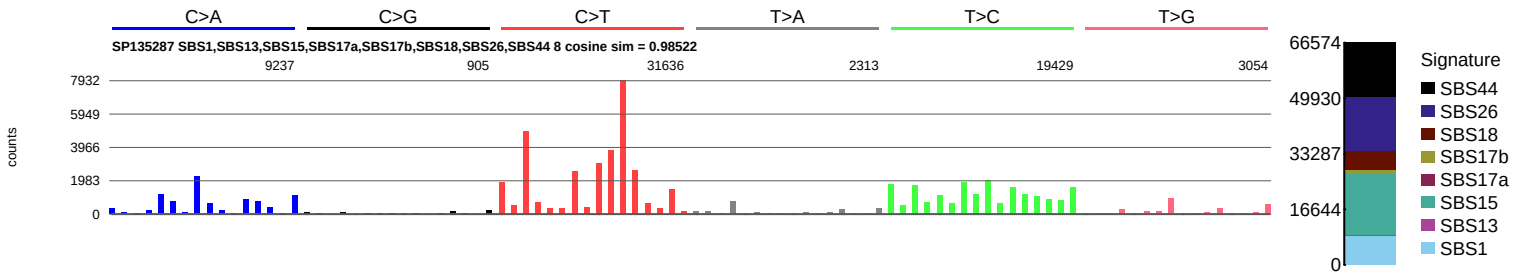
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T>A
T>C
T>G
SP135287 SBS1,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS44 8 cosine sim = 0.98522
9237
905
31636
2313
19429
3054
counts
7932
5949
3966
1983
0
66574
49930
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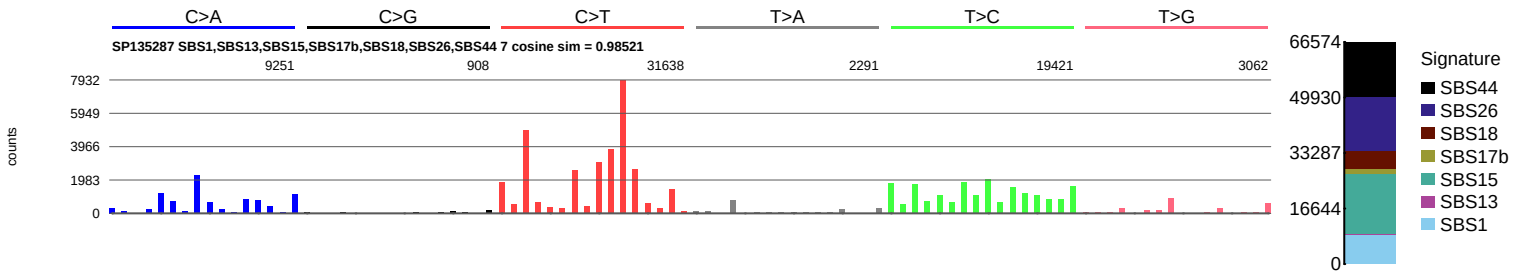
C>A
C>G
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SP135287 SBS1,SBS13,SBS15,SBS17b,SBS18,SBS26,SBS44 7 cosine sim = 0.98521
9251
908
31638
2291
19421
3062
counts
7932
5949
3966
1983
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66574
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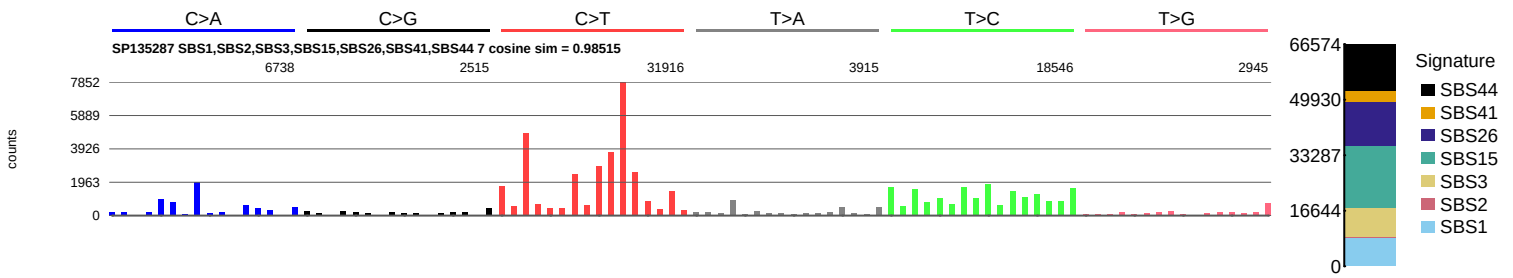
C>A
C>G
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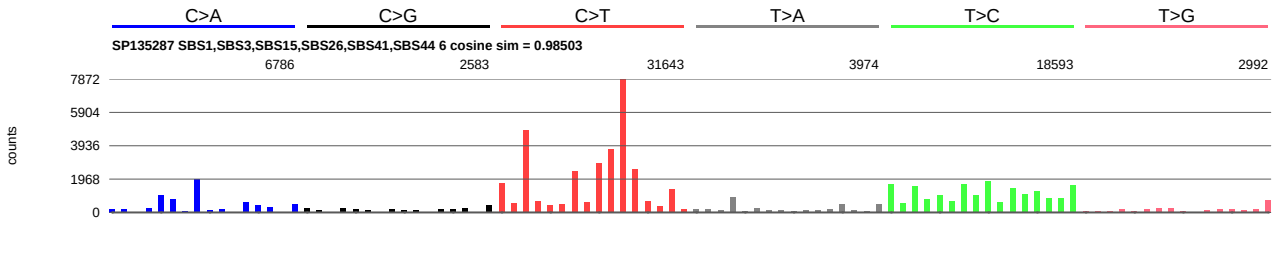
T>A
T>C
T>G
SP135287 SBS1,SBS2,SBS3,SBS15,SBS26,SBS41,SBS44 7 cosine sim = 0.98515
6738
2515
31916
3915
18546
2945
counts
7852
5889
3926
1963
0
66574
49930
33287
16644
0
Signature
SBS44
SBS41
SBS26
SBS15
SBS3
SBS2
SBS1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SP135287 SBS1,SBS3,SBS15,SBS26,SBS41,SBS44 6 cosine sim = 0.98503
6786
2583
31643
3974
18593
2992
counts
7872
5904
3936
1968
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66574
49930
33287
16644
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SBS41
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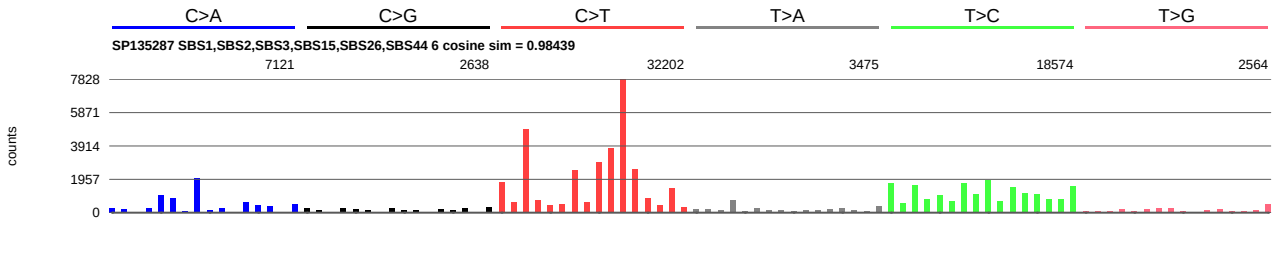
SBS3
SBS1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SP135287 SBS1,SBS2,SBS3,SBS15,SBS26,SBS44 6 cosine sim = 0.98439
7121
2638
32202
3475
18574
2564
counts
7828
5871
3914
1957
0
66574
49930
33287
16644
0
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SBS44
SBS26
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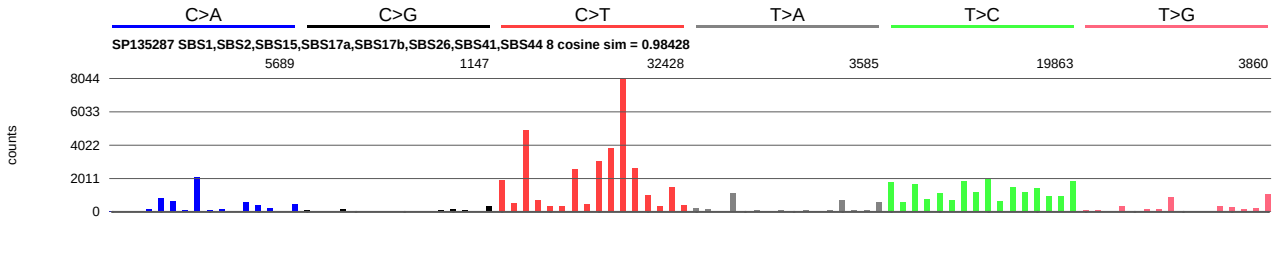
SBS2
SBS1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C>G
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SP135287 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS26,SBS41,SBS44 8 cosine sim = 0.98428
5689
1147
32428
3585
19863
3860
counts
8044
6033
4022
2011
0
66574
49930
33287
16644
0
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SBS44
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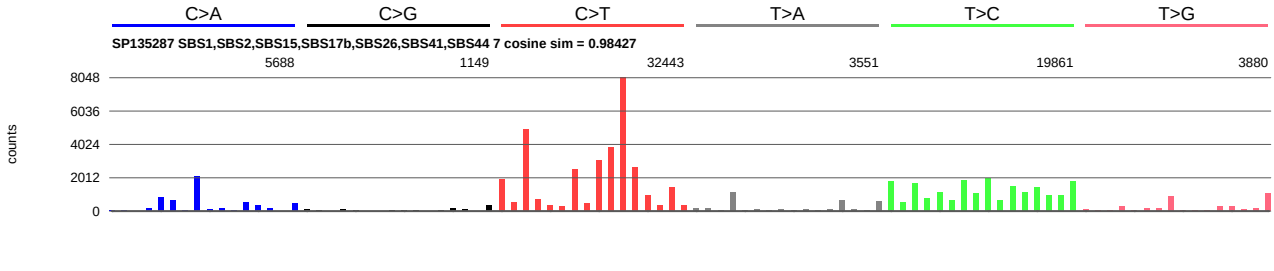
SBS2
SBS1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SP135287 SBS1,SBS2,SBS15,SBS17b,SBS26,SBS41,SBS44 7 cosine sim = 0.98427
5688
1149
32443
3551
19861
3880
counts
8048
6036
4024
2012
0
66574
49930
33287
16644
0
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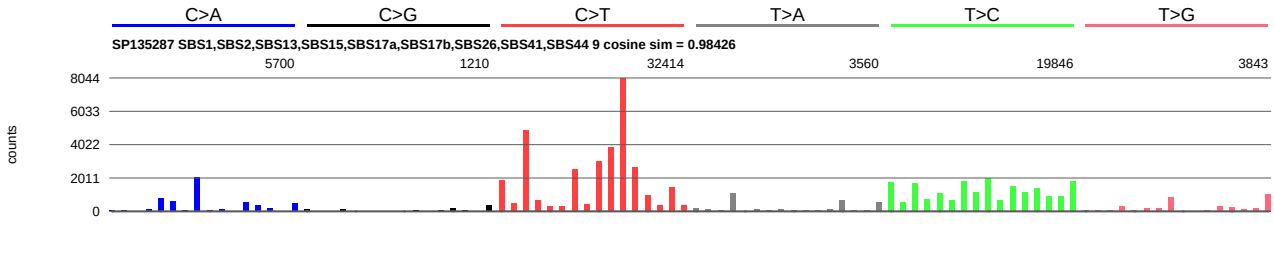
SBS44
SBS41
SBS26
SBS17b
SBS15
SBS2
SBS1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SP135287 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS26,SBS41,SBS44 9 cosine sim = 0.98426
5700
1210
32414
3560
19846
3843
counts
8044
6033
4022
2011
0
66574
49930
33287
16644
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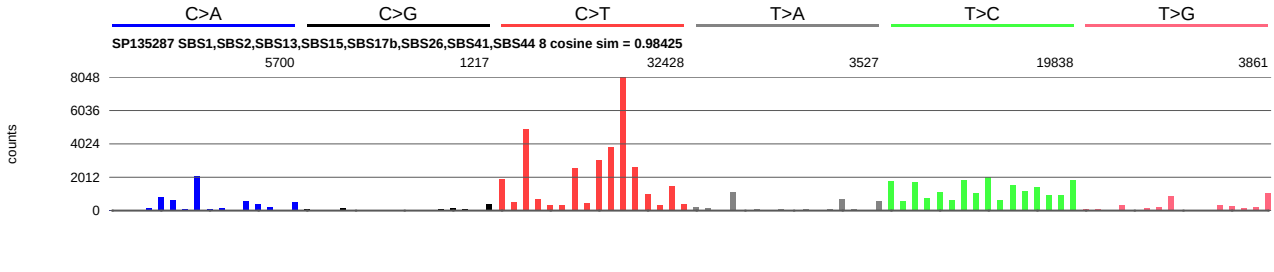
SBS17b
SBS17a
SBS15
SBS13
SBS2
SBS1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SP135287 SBS1,SBS2,SBS13,SBS15,SBS17b,SBS26,SBS41,SBS44 8 cosine sim = 0.98425
5700
1217
32428
3527
19838
3861
counts
8048
6036
4024
2012
0
66574
49930
33287
16644
0
Signature
SBS44
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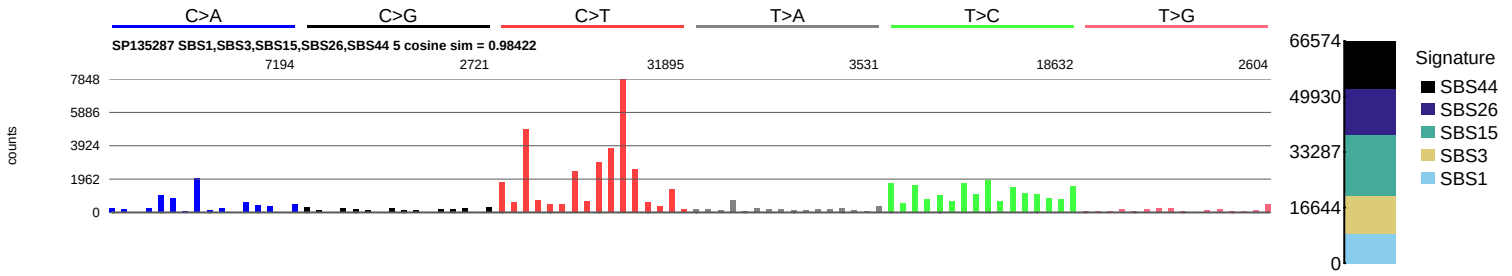
SBS26
SBS17b
SBS15
SBS13
SBS2
SBS1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T>G
SP135287 SBS1,SBS3,SBS15,SBS26,SBS44 5 cosine sim = 0.98422
7194
2721
31895
3531
18632
2604
counts
7848
5886
3924
1962
0
66574
49930
33287
16644
0
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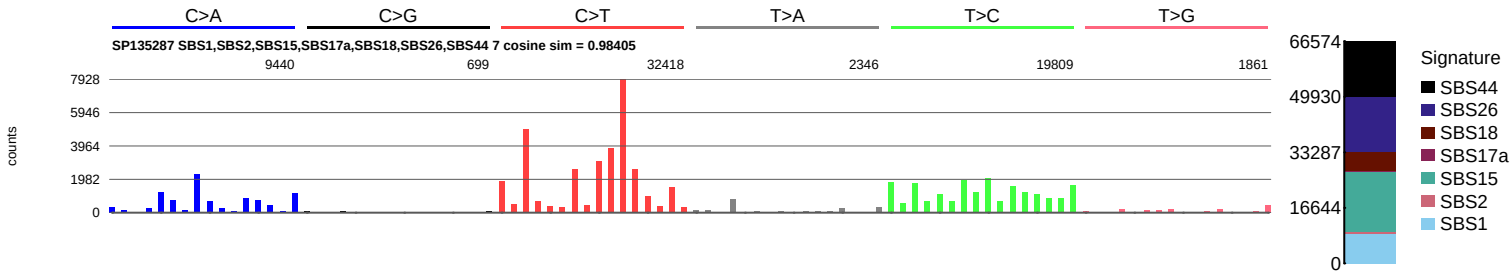
SBS44
SBS26
SBS15
SBS3
SBS1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T>C
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SP135287 SBS1,SBS2,SBS15,SBS17a,SBS18,SBS26,SBS44 7 cosine sim = 0.98405
9440
699
32418
2346
19809
1861
counts
7928
5946
3964
1982
0
66574
49930
33287
16644
0
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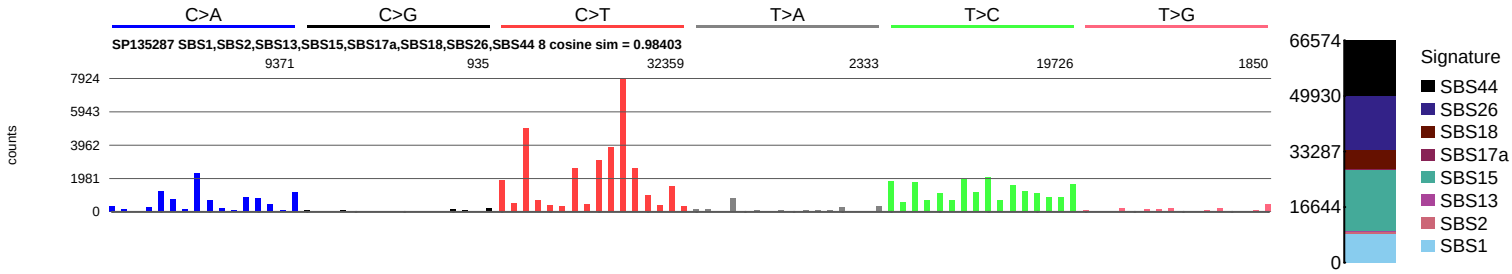
SBS26
SBS18
SBS17a
SBS15
SBS2
SBS1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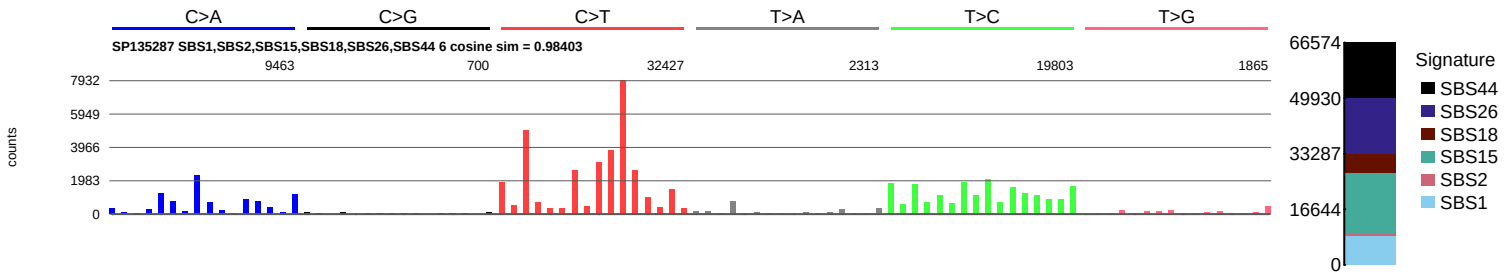
SP135287 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS18,SBS26,SBS44 8 cosine sim = 0.98403
9371
935
32359
2333
19726
1850
counts
7924
5943
3962
1981
0
66574
49930
33287
16644
0
Signature
SBS44
SBS26
SBS18
SBS17a
SBS15
SBS13
SBS2
SBS1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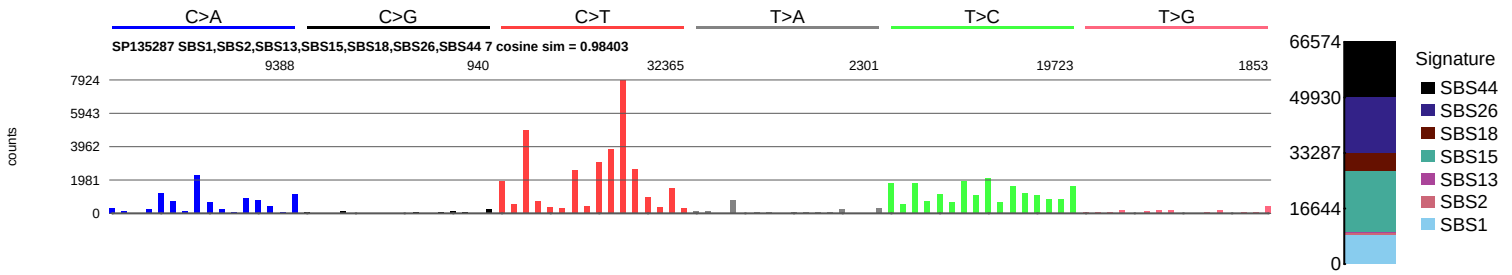
T>A
T>C
T>G
SP135287 SBS1,SBS2,SBS15,SBS18,SBS26,SBS44 6 cosine sim = 0.98403
9463
700
32427
2313
19803
1865
counts
7932
5949
3966
1983
0
66574
49930
33287
16644
0
Signature
SBS44
SBS26
SBS18
SBS15
SBS2
SBS1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C>G
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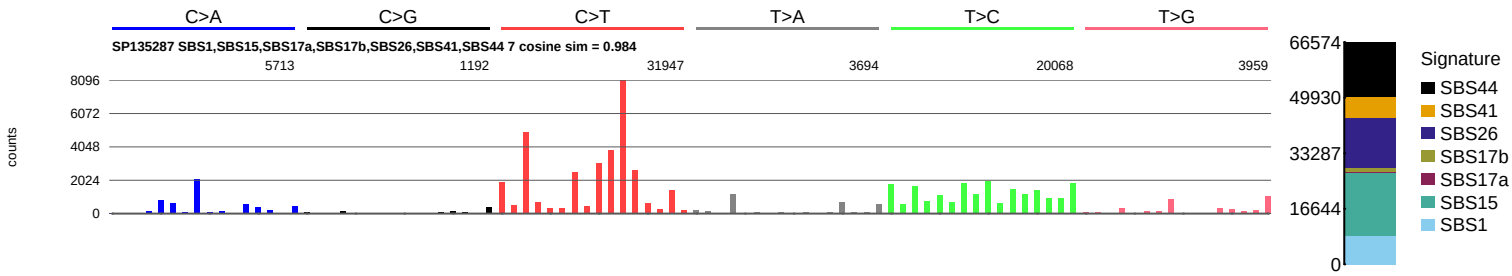
T>A
T>C
T>G
SP135287 SBS1,SBS2,SBS13,SBS15,SBS18,SBS26,SBS44 7 cosine sim = 0.98403
9388
940
32365
2301
19723
1853
counts
7924
5943
3962
1981
0
66574
49930
33287
16644
0
Signature
SBS44
SBS26
SBS18
SBS15
SBS13
SBS2
SBS1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C>G
C>T
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T>C
T>G
SP135287 SBS1,SBS15,SBS17a,SBS17b,SBS26,SBS41,SBS44 7 cosine sim = 0.984
5713
1192
31947
3694
20068
3959
counts
8096
6072
4048
2024
0
66574
49930
33287
16644
0
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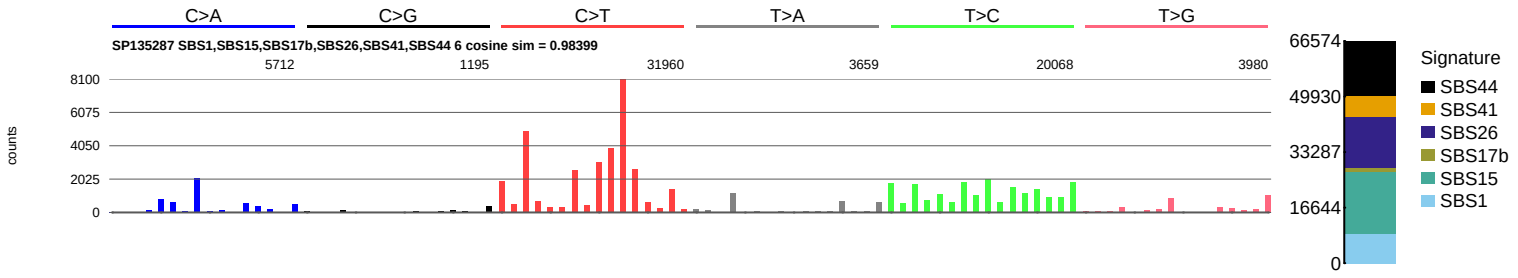
SBS41
SBS26
SBS17b
SBS17a
SBS15
SBS1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C>T
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T>C
T>G
SP135287 SBS1,SBS15,SBS17b,SBS26,SBS41,SBS44 6 cosine sim = 0.98399
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1195
31960
3659
20068
3980
8100
6075
4050
2025
0
counts
66574
49930
33287
16644
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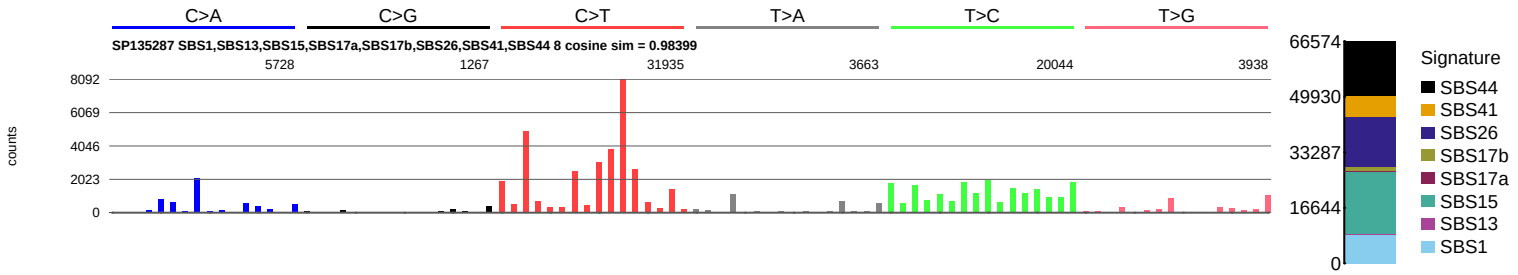
SBS41
SBS26
SBS17b
SBS15
SBS1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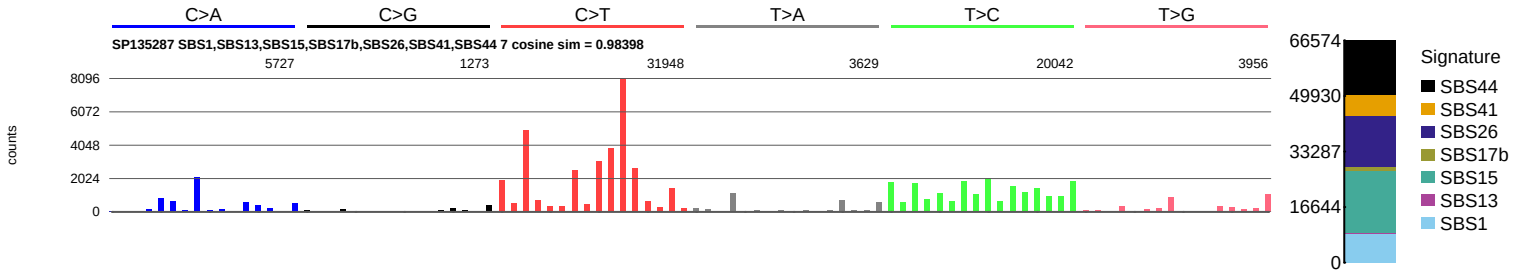
C>G
C>T
T>A
T>C
T>G
SP135287 SBS1,SBS13,SBS15,SBS17a,SBS17b,SBS26,SBS41,SBS44 8 cosine sim = 0.98399
5728
1267
31935
3663
20044
3938
8092
6069
4046
2023
0
counts
66574
49930
33287
16644
0
Signature
SBS44
SBS41
SBS26
SBS17b
SBS17a
SBS15
SBS13
SBS1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C>G
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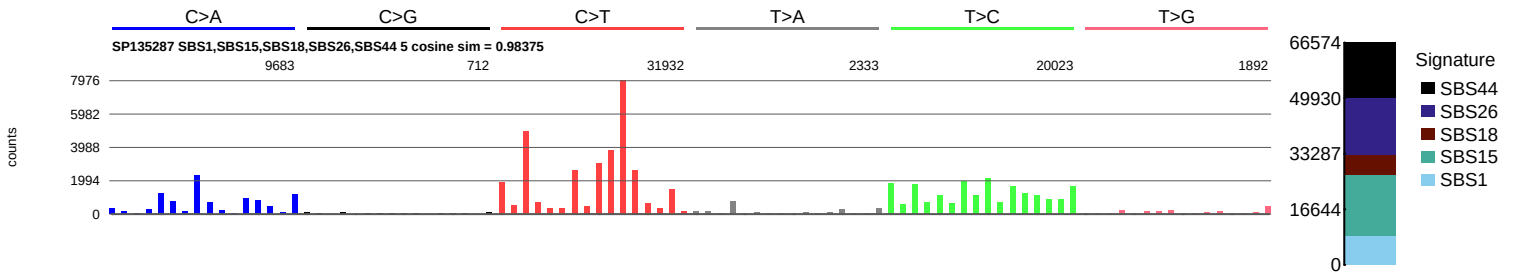
T>A
T>C
T>G
SP135287 SBS1,SBS13,SBS15,SBS17b,SBS26,SBS41,SBS44 7 cosine sim = 0.98398
5727
1273
31948
3629
20042
3956
8096
6072
4048
2024
0
counts
66574
49930
33287
16644
0
Signature
SBS44
SBS41
SBS26
SBS17b
SBS15
SBS13
SBS1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T>A
T>C
T>G
SP135287 SBS1,SBS15,SBS18,SBS26,SBS44 5 cosine sim = 0.98375
9683
712
31932
2333
20023
1892
7976
5982
3988
1994
0
counts
66574
49930
33287
16644
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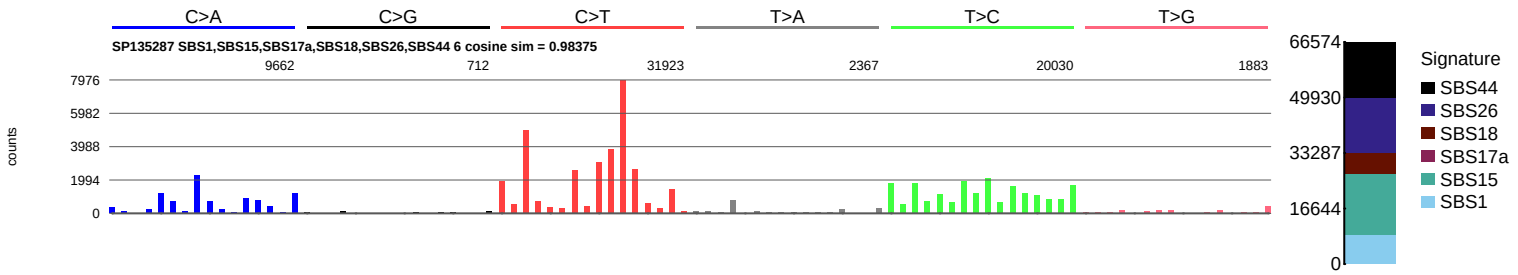
Signature
SBS44
SBS26
SBS18
SBS15
SBS1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C>G
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T>A
T>C
T>G
SP135287 SBS1,SBS15,SBS17a,SBS18,SBS26,SBS44 6 cosine sim = 0.98375
9662
712
31923
2367
20030
1883
7976
5982
3988
1994
0
counts
66574
49930
33287
16644
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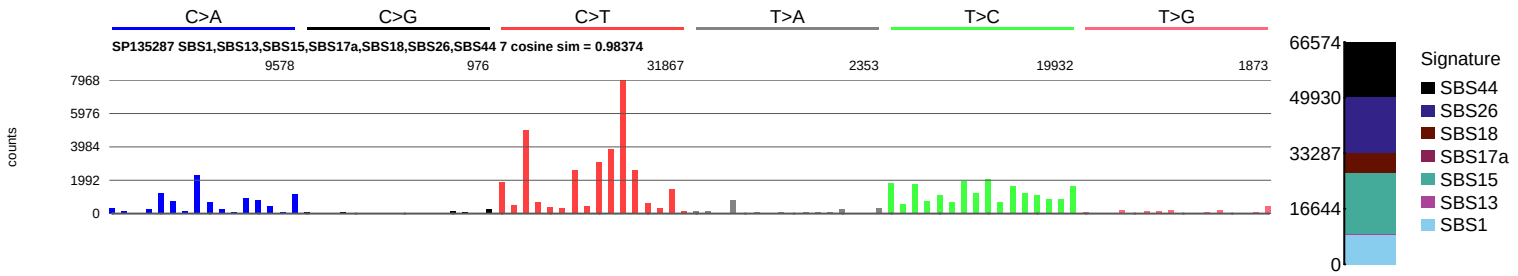
Signature
SBS44
SBS26
SBS18
SBS17a
SBS15
SBS1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C>G
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T>A
T>C
T>G
SP135287 SBS1,SBS13,SBS15,SBS17a,SBS18,SBS26,SBS44 7 cosine sim = 0.98374
9578
976
31867
2353
19932
1873
7968
5976
3984
1992
0
counts
66574
49930
33287
16644
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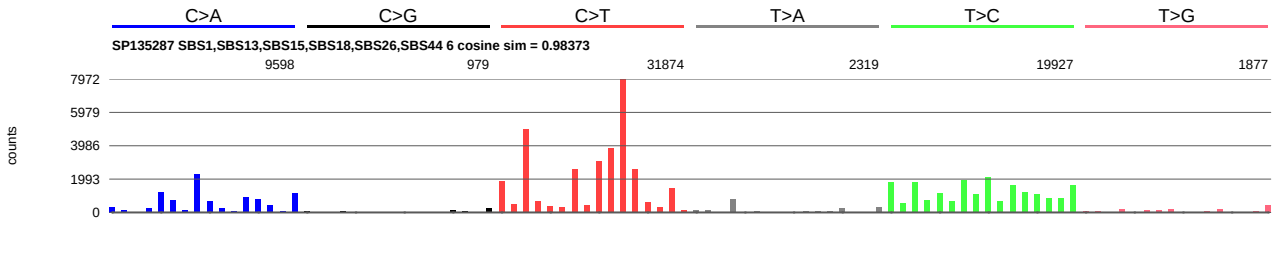
SBS26
SBS18
SBS17a
SBS15
SBS13
SBS1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T>A
T>C
T>G
SP135287 SBS1,SBS13,SBS15,SBS18,SBS26,SBS44 6 cosine sim = 0.98373
9598
979
31874
2319
19927
1877
counts
7972
5979
3986
1993
0
66574
49930
33287
16644
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SBS13
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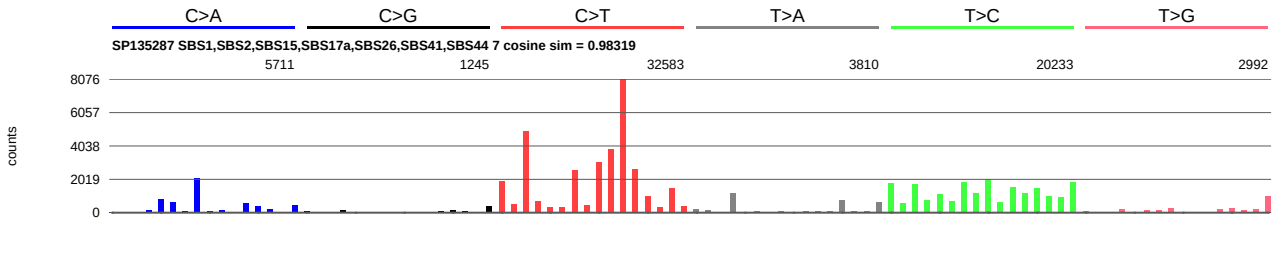
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C>G
C>T
T>A
T>C
T>G
SP135287 SBS1,SBS2,SBS15,SBS17a,SBS26,SBS41,SBS44 7 cosine sim = 0.98319
5711
1245
32583
3810
20233
2992
counts
8076
6057
4038
2019
0
66574
49930
33287
16644
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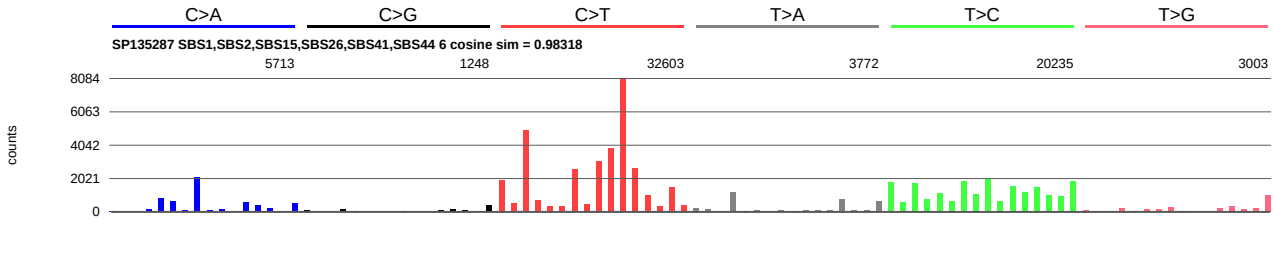
C>A
C>G
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T>A
T>C
T>G
SP135287 SBS1,SBS2,SBS15,SBS26,SBS41,SBS44 6 cosine sim = 0.98318
5713
1248
32603
3772
20235
3003
counts
8084
6063
4042
2021
0
66574
49930
33287
16644
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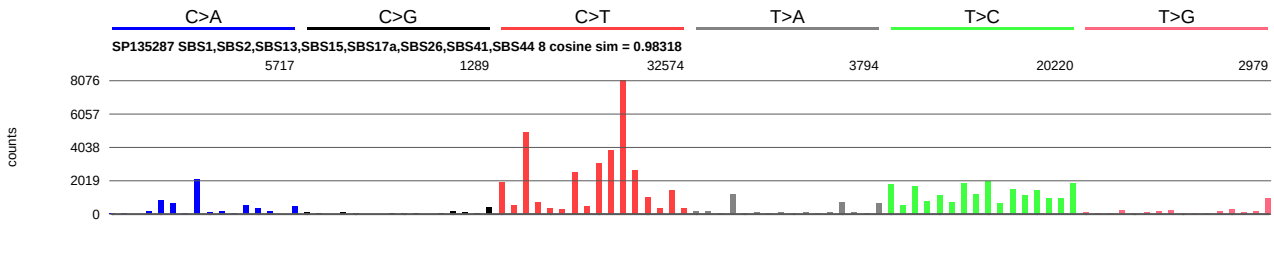
SBS41
SBS26
SBS15
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135287 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS26,SBS41,SBS44 8 cosine sim = 0.98318
5717
1289
32574
3794
20220
2979
counts
8076
6057
4038
2019
0
66574
49930
33287
16644
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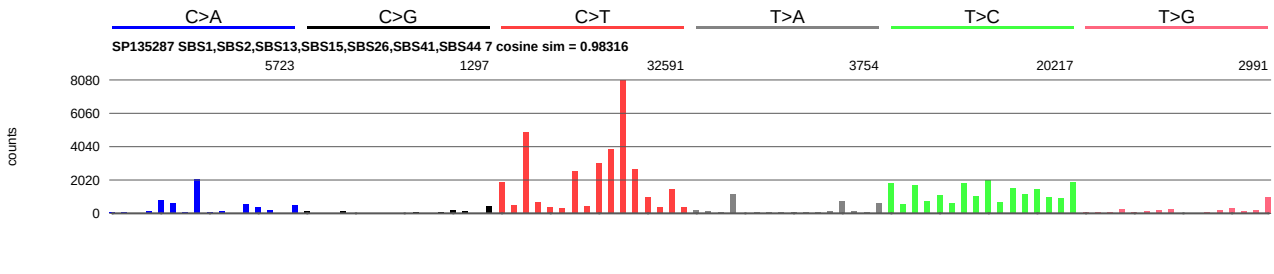
Signature
SBS44
SBS41
SBS26
SBS17a
SBS15
SBS13
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135287 SBS1,SBS2,SBS13,SBS15,SBS26,SBS41,SBS44 7 cosine sim = 0.98316
5723
1297
32591
3754
20217
2991
counts
8080
6060
4040
2020
0
66574
49930
33287
16644
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SBS44
SBS41
SBS26
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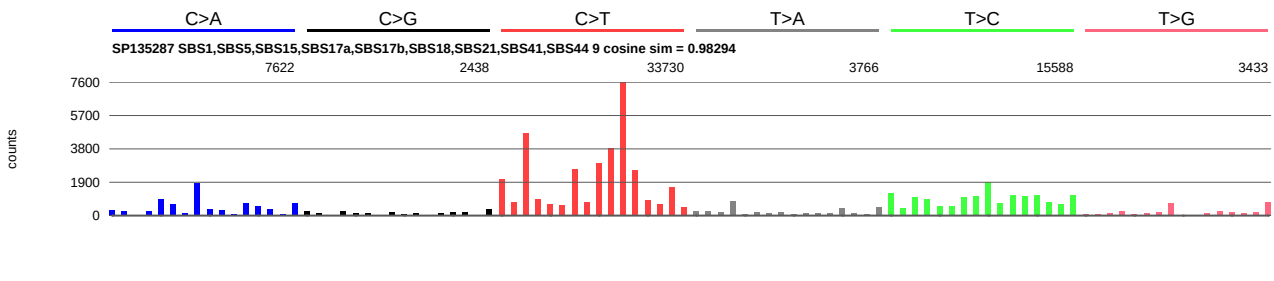
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135287 SBS1,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41,SBS44 9 cosine sim = 0.98294
7622
2438
33730
3766
15588
3433
counts
7600
5700
3800
1900
0
66574
49930
33287
16644
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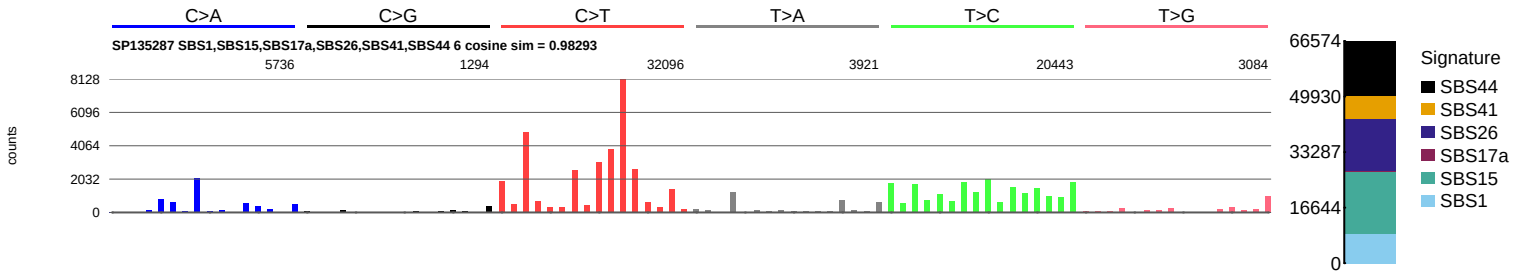
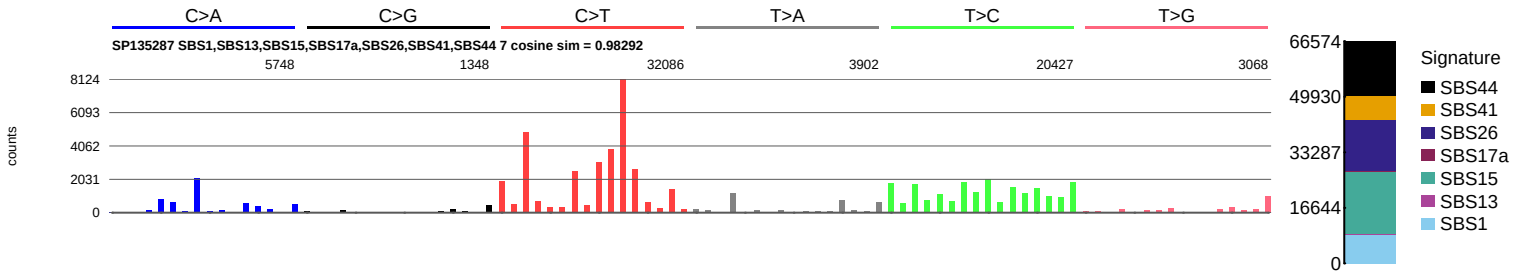
Signature
SBS44
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135287 SBS1,SBS15,SBS17a,SBS26,SBS41,SBS44 6 cosine sim = 0.98293
5736
1294
32096
3921
20443
3084
counts
8128
6096
4064
2032
0
66574
49930
33287
16644
0
Signature
SBS44
SBS41
SBS26
SBS17a
SBS15
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135287 SBS1,SBS13,SBS15,SBS17a,SBS26,SBS41,SBS44 7 cosine sim = 0.98292
5748
1348
32086
3902
20427
3068
counts
8124
6093
4062
2031
0
66574
49930
33287
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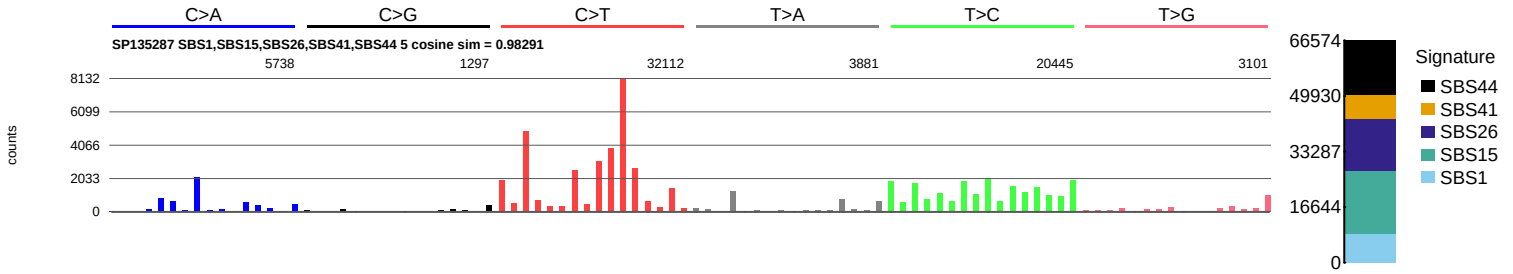
SBS13
SBS1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C>G
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T>G
SP135287 SBS1,SBS15,SBS26,SBS41,SBS44 5 cosine sim = 0.98291
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1297
32112
3881
20445
3101
counts
8132
6099
4066
2033
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66574
49930
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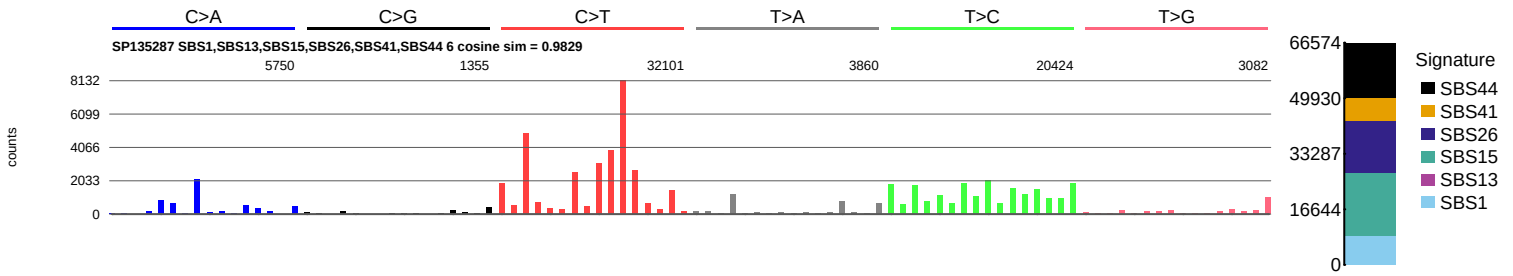
SBS1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SP135287 SBS1,SBS13,SBS15,SBS26,SBS41,SBS44 6 cosine sim = 0.9829
5750
1355
32101
3860
20424
3082
counts
8132
6099
4066
2033
0
66574
49930
33287
16644
0
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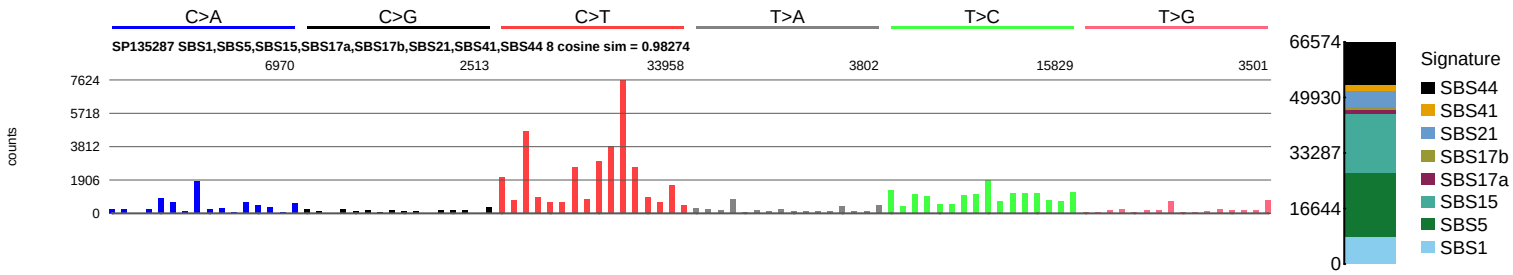
SBS41
SBS26
SBS15
SBS13
SBS1


C>A
C>G
C>T
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T>C
T>G
SP135287 SBS1,SBS5,SBS15,SBS17a,SBS17b,SBS21,SBS41,SBS44 8 cosine sim = 0.98274
6970
2513
33958
3802
15829
3501
counts
7624
5718
3812
1906
0
66574
49930
33287
16644
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Signature
SBS44
SBS41
SBS21
SBS17b
SBS17a
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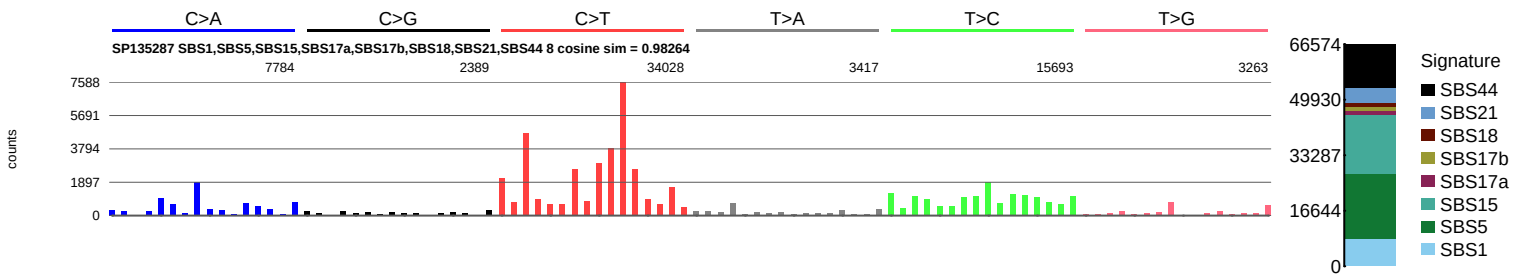
SBS5
SBS1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C>G
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T>G
SP135287 SBS1,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS44 8 cosine sim = 0.98264
7784
2389
34028
3417
15693
3263
counts
7588
5691
3794
1897
0
66574
49930
33287
16644
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Signature
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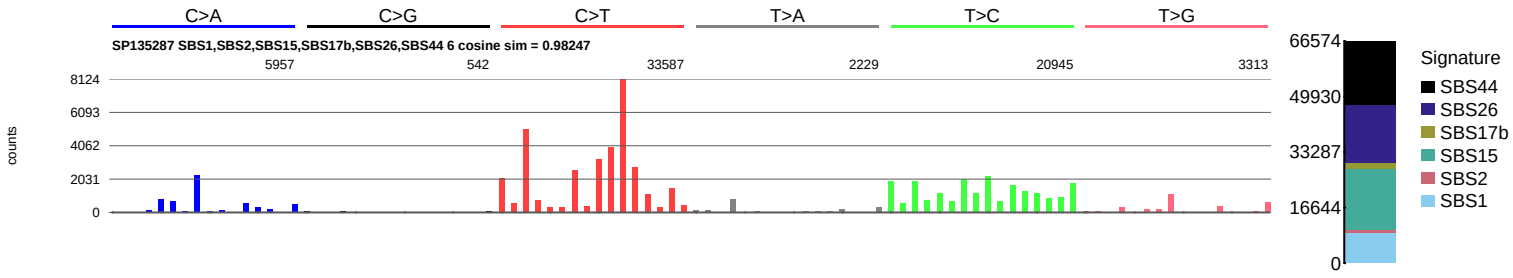
SBS1


C>A
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C>T
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T>C
T>G
SP135287 SBS1,SBS2,SBS15,SBS17b,SBS26,SBS44 6 cosine sim = 0.98247
5957
542
33587
2229
20945
3313
8124
6093
4062
2031
0
counts
66574
49930
33287
16644
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Signature
SBS44
SBS26
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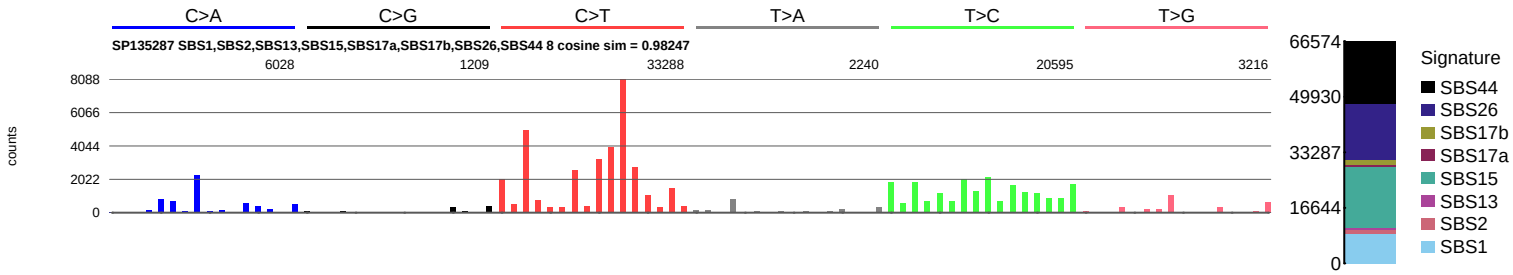
SBS2
SBS1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C>G
C>T
T>A
T>C
T>G
SP135287 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS26,SBS44 8 cosine sim = 0.98247
6028
1209
33288
2240
20595
3216
8088
6066
4044
2022
0
counts
66574
49930
33287
16644
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Signature
SBS44
SBS26
SBS17b
SBS17a
SBS15
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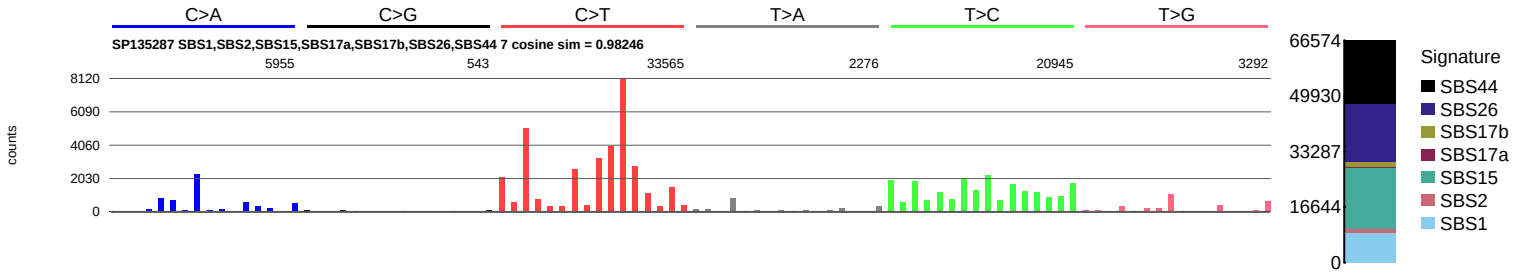
SBS2
SBS1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C>G
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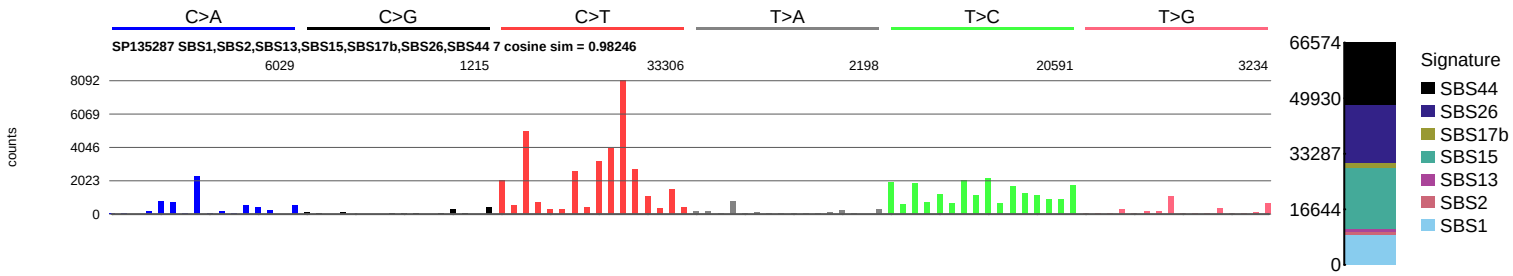
T>C
T>G
SP135287 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS26,SBS44 7 cosine sim = 0.98246
5955
543
33565
2276
20945
3292
8120
6090
4060
2030
0
counts
66574
49930
33287
16644
0
Signature
SBS44
SBS26
SBS17b
SBS17a
SBS15
SBS2
SBS1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C>G
C>T
T>A
T>C
T>G
SP135287 SBS1,SBS2,SBS13,SBS15,SBS17b,SBS26,SBS44 7 cosine sim = 0.98246
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1215
33306
2198
20591
3234
8092
6069
4046
2023
0
counts
66574
49930
33287
16644
0
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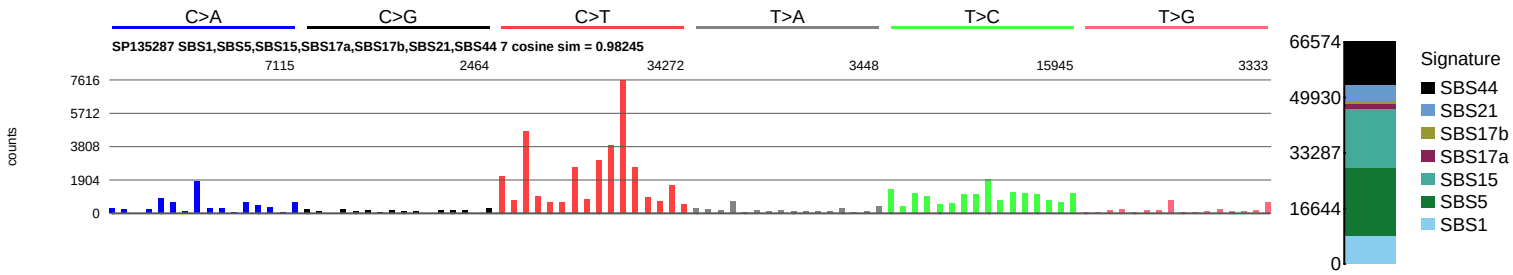
SBS26
SBS17b
SBS15
SBS13
SBS2
SBS1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T>G
SP135287 SBS1,SBS5,SBS15,SBS17a,SBS17b,SBS21,SBS44 7 cosine sim = 0.98245
7115
2464
34272
3448
15945
3333
7616
5712
3808
1904
0
counts
66574
49930
33287
16644
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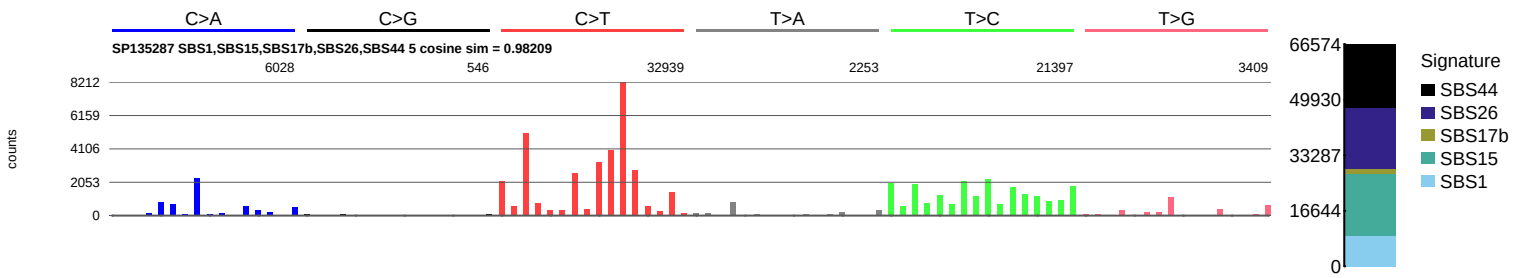
SBS21
SBS17b
SBS17a
SBS15
SBS5
SBS1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C>G
C>T
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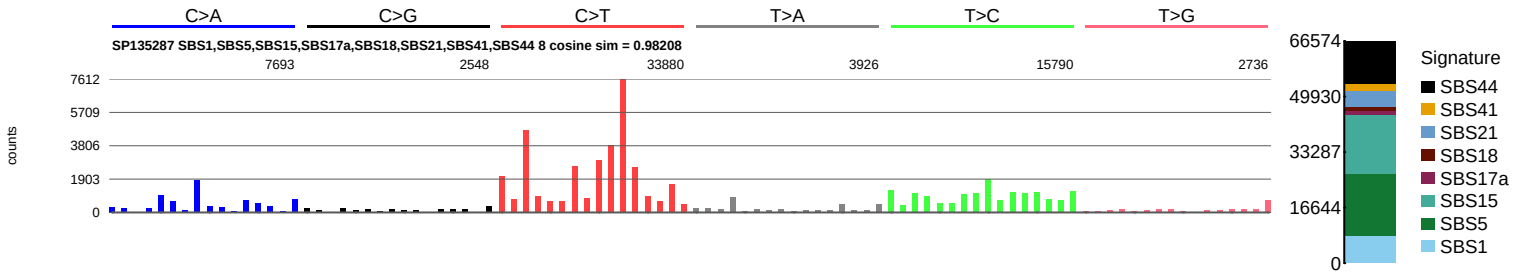
T>C
T>G
SP135287 SBS1,SBS15,SBS17b,SBS26,SBS44 5 cosine sim = 0.98209
6028
546
32939
2253
21397
3409
8212
6159
4106
2053
0
counts
66574
49930
33287
16644
0
Signature
SBS44
SBS26
SBS17b
SBS15
SBS1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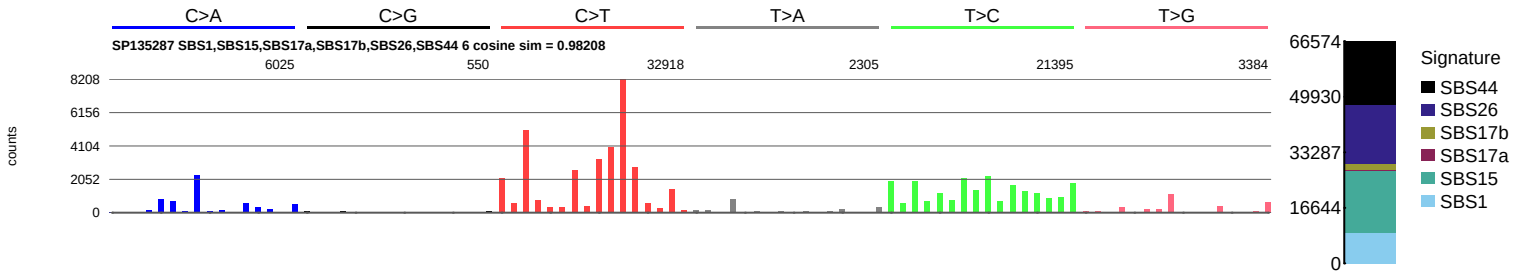
C>G
C>T
T>A
T>C
T>G
SP135287 SBS1,SBS5,SBS15,SBS17a,SBS18,SBS21,SBS41,SBS44 8 cosine sim = 0.98208
7693
2548
33880
3926
15790
2736
counts
7612
5709
3806
1903
0
66574
49930
33287
16644
0
Signature
SBS44
SBS41
SBS21
SBS18
SBS17a
SBS15
SBS5
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135287 SBS1,SBS15,SBS17a,SBS17b,SBS26,SBS44 6 cosine sim = 0.98208
6025
550
32918
2305
21395
3384
counts
8208
6156
4104
2052
0
66574
49930
33287
16644
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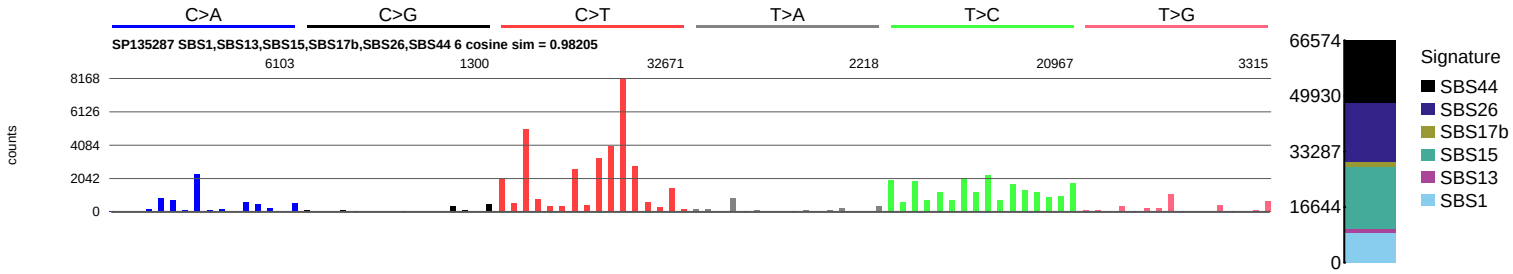
Signature
SBS44
SBS26
SBS17b
SBS17a
SBS15
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135287 SBS1,SBS13,SBS15,SBS17b,SBS26,SBS44 6 cosine sim = 0.98205
6103
1300
32671
2218
20967
3315
counts
8168
6126
4084
2042
0
66574
49930
33287
16644
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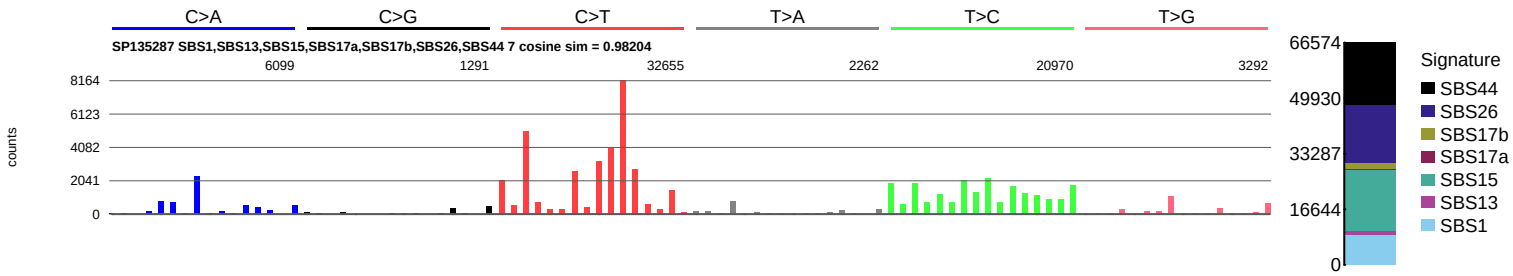
SBS44
SBS26
SBS17b
SBS15
SBS13
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135287 SBS1,SBS13,SBS15,SBS17a,SBS17b,SBS26,SBS44 7 cosine sim = 0.98204
6099
1291
32655
2262
20970
3292
counts
8164
6123
4082
2041
0
66574
49930
33287
16644
0
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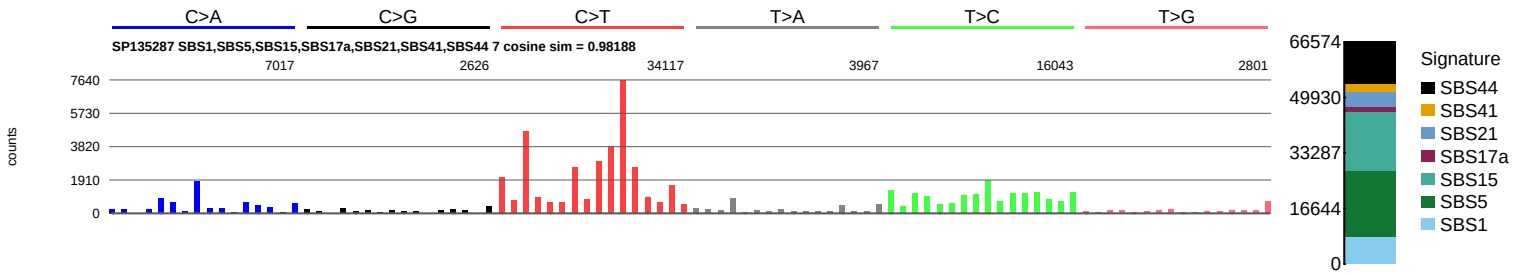
SBS44
SBS26
SBS17b
SBS17a
SBS15
SBS13
SBS1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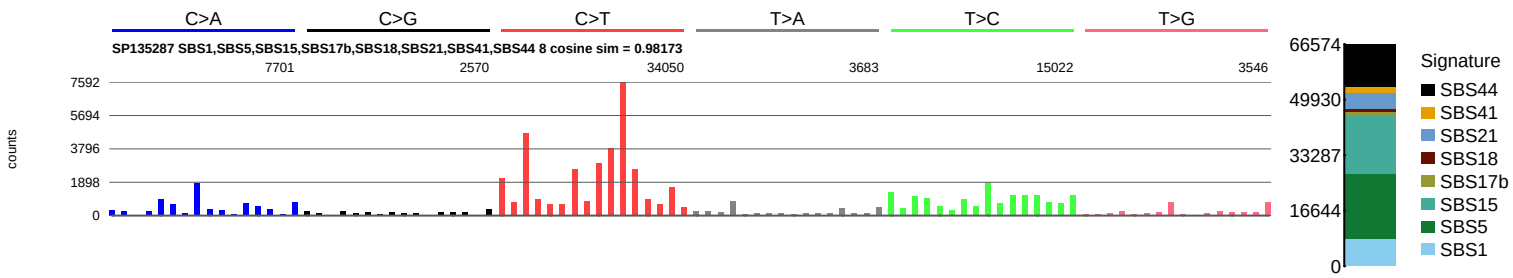
C>G
C>T
T>A
T>C
T>G
SP135287 SBS1,SBS5,SBS15,SBS17a,SBS21,SBS41,SBS44 7 cosine sim = 0.98188
7017
2626
34117
3967
16043
2801
counts
7640
5730
3820
1910
0
66574
49930
33287
16644
0
Signature
SBS44
SBS41
SBS21
SBS17a
SBS15
SBS5
SBS1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C>G
C>T
T>A
T>C
T>G
SP135287 SBS1,SBS5,SBS15,SBS17b,SBS18,SBS21,SBS41,SBS44 8 cosine sim = 0.98173
7701
2570
34050
3683
15022
3546
counts
7592
5694
3796
1898
0
66574
49930
33287
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Signature
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SBS18
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SBS5
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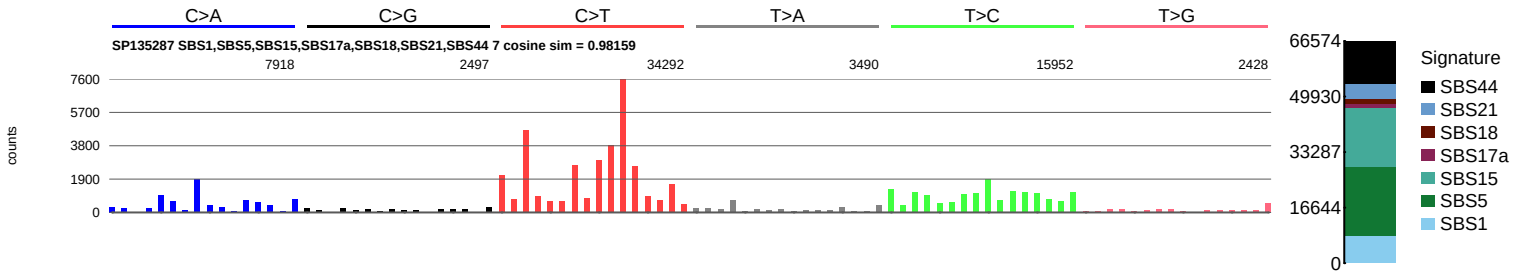
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T>A
T>C
T>G
SP135287 SBS1,SBS5,SBS15,SBS17a,SBS18,SBS21,SBS44 7 cosine sim = 0.98159
7918
2497
34292
3490
15952
2428
counts
7600
5700
3800
1900
0
66574
49930
33287
16644
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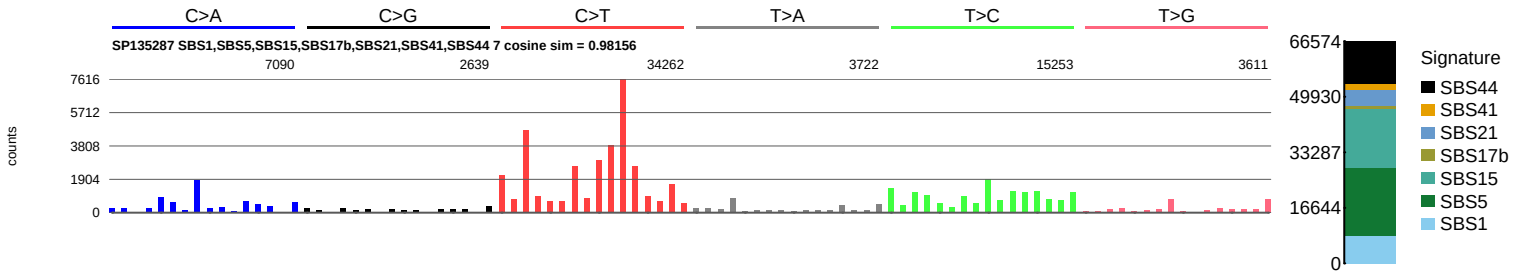
Signature
SBS44
SBS21
SBS18
SBS17a
SBS15
SBS5
SBS1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C>G
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T>A
T>C
T>G
SP135287 SBS1,SBS5,SBS15,SBS17b,SBS21,SBS41,SBS44 7 cosine sim = 0.98156
7090
2639
34262
3722
15253
3611
counts
7616
5712
3808
1904
0
66574
49930
33287
16644
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SBS41
SBS21
SBS17b
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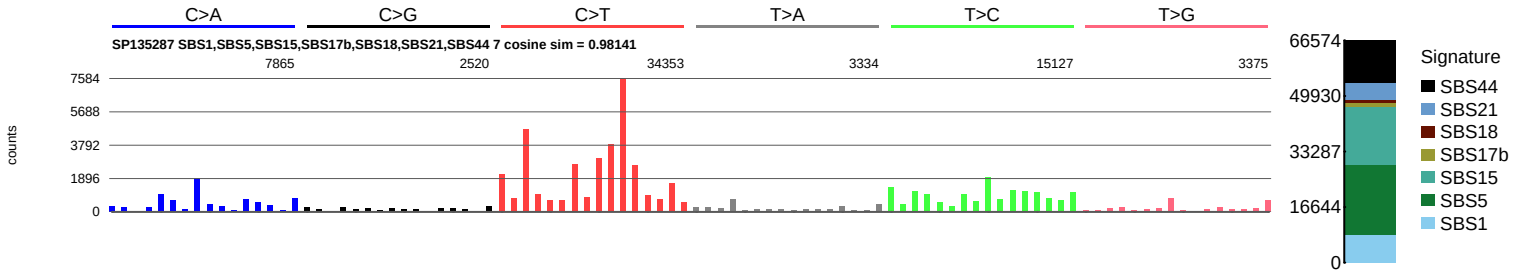
C>A
C>G
C>T
T>A
T>C
T>G
SP135287 SBS1,SBS5,SBS15,SBS17b,SBS18,SBS21,SBS44 7 cosine sim = 0.98141
7865
2520
34353
3334
15127
3375
counts
7584
5688
3792
1896
0
66574
49930
33287
16644
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SBS21
SBS18
SBS17b
SBS15
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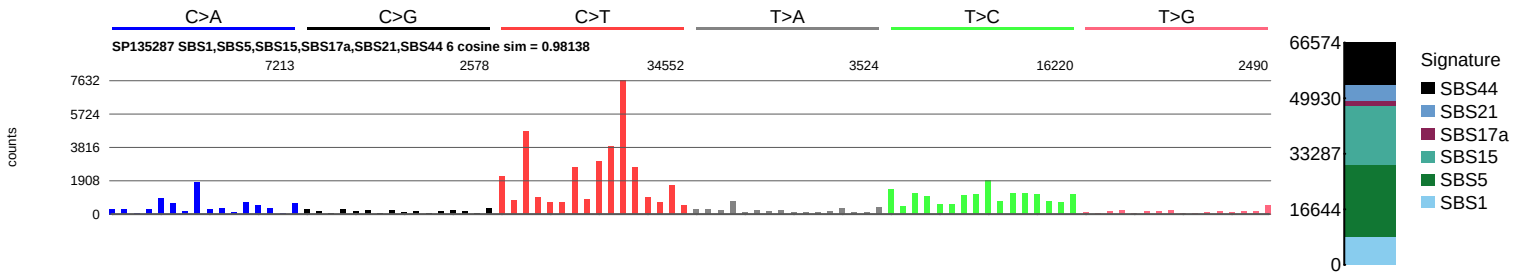
SBS1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C>G
C>T
T>A
T>C
T>G
SP135287 SBS1,SBS5,SBS15,SBS17a,SBS21,SBS44 6 cosine sim = 0.98138
7213
2578
34552
3524
16220
2490
counts
7632
5724
3816
1908
0
66574
49930
33287
16644
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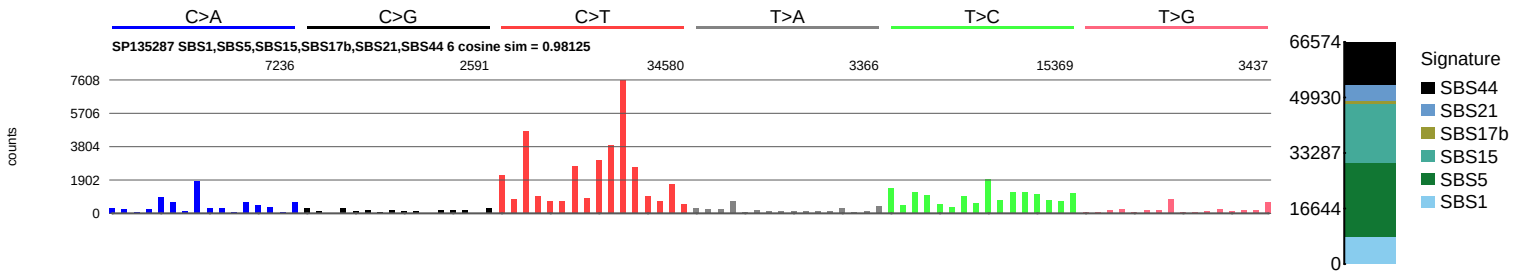
SBS44
SBS21
SBS17a
SBS15
SBS5
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135287 SBS1,SBS5,SBS15,SBS17b,SBS21,SBS44 6 cosine sim = 0.98125
7236
2591
34580
3366
15369
3437
counts
7608
5706
3804
1902
0
66574
49930
33287
16644
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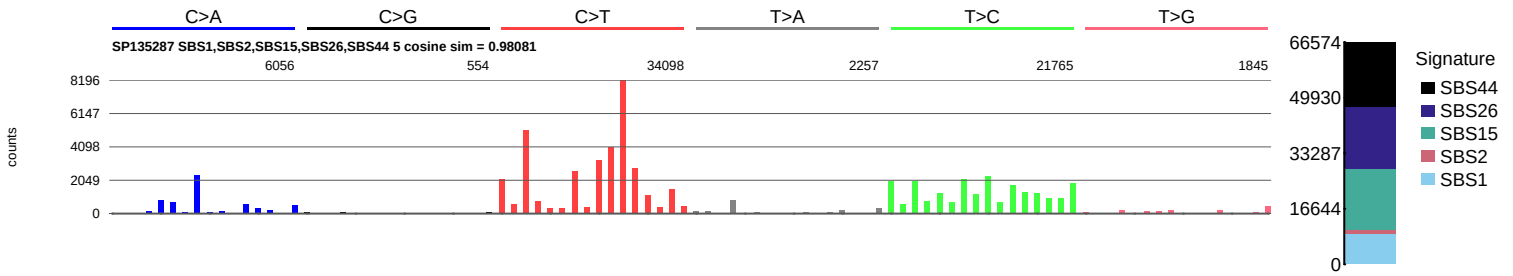
SBS44
SBS21
SBS17b
SBS15
SBS5
SBS1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C>G
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T>C
T>G
SP135287 SBS1,SBS2,SBS15,SBS26,SBS44 5 cosine sim = 0.98081
6056
554
34098
2257
21765
1845
counts
8196
6147
4098
2049
0
66574
49930
33287
16644
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SBS44
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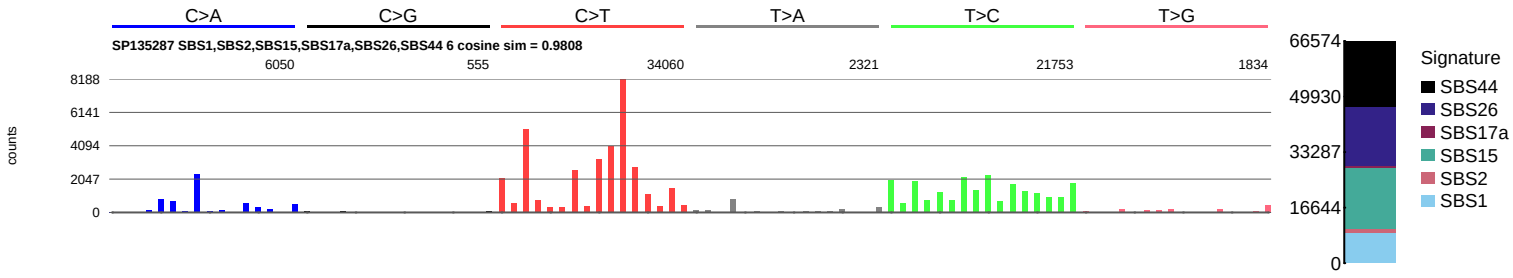
SBS2
SBS1


C>A
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T>A
T>C
T>G
SP135287 SBS1,SBS2,SBS15,SBS17a,SBS26,SBS44 6 cosine sim = 0.9808
6050
555
34060
2321
21753
1834
8188
6141
4094
2047
0
counts
66574
49930
33287
16644
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Signature
SBS44
SBS26
SBS17a
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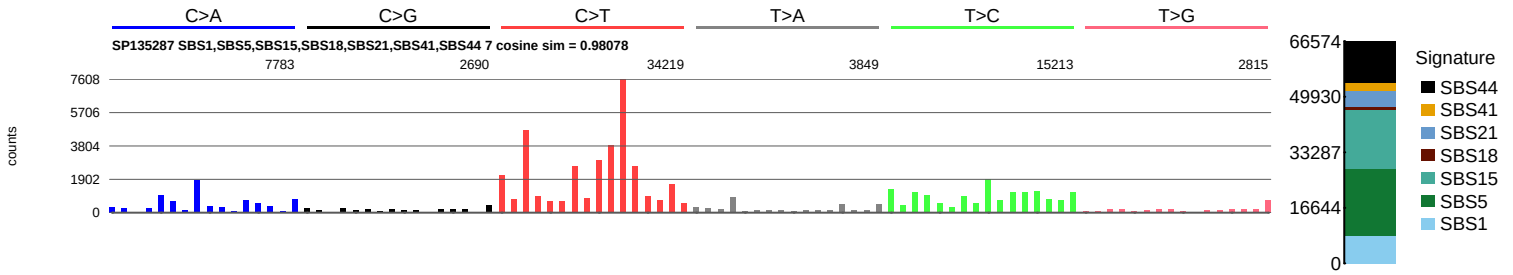
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135287 SBS1,SBS5,SBS15,SBS18,SBS21,SBS41,SBS44 7 cosine sim = 0.98078
7783
2690
34219
3849
15213
2815
7608
5706
3804
1902
0
counts
66574
49930
33287
16644
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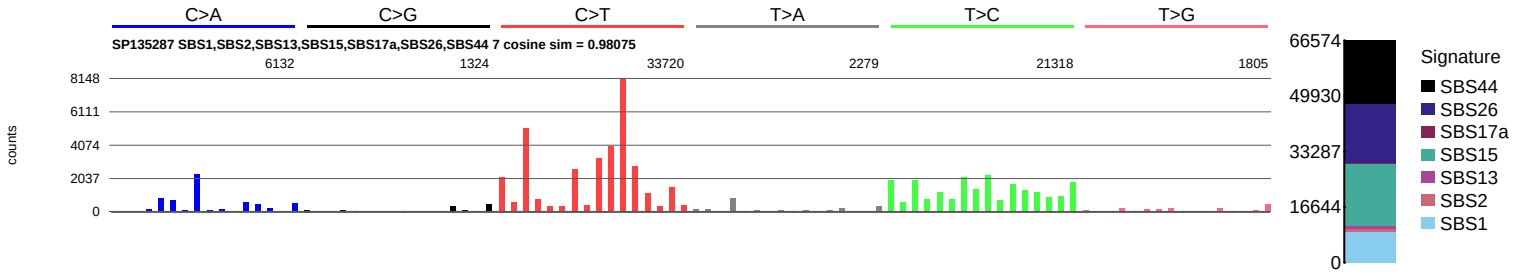
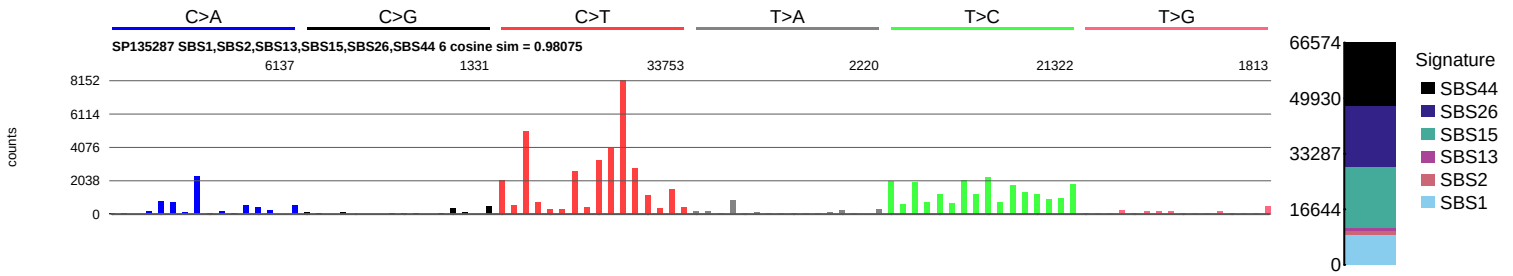
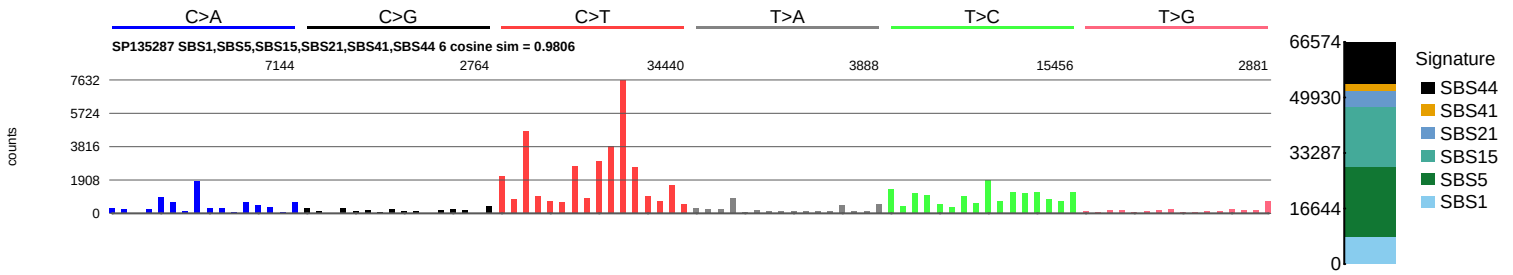
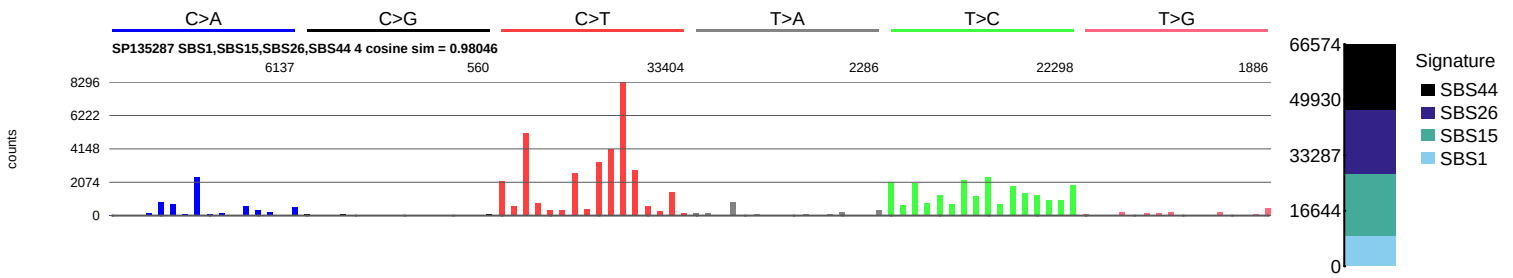
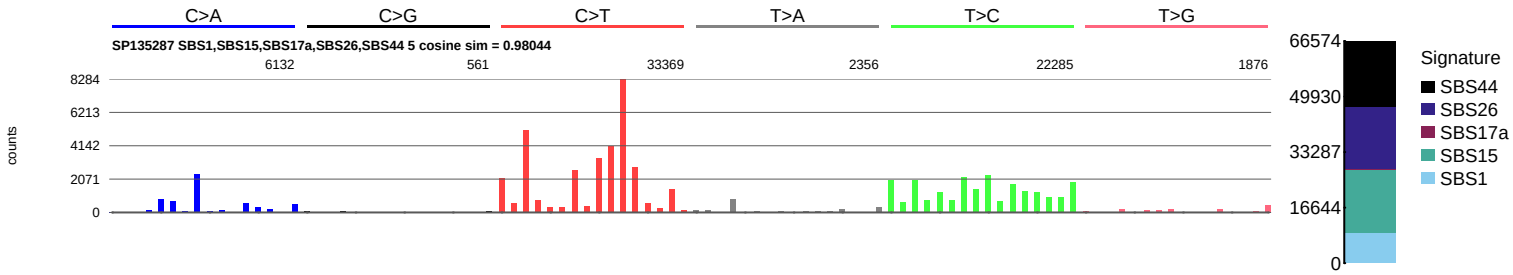
SBS5
SBS1

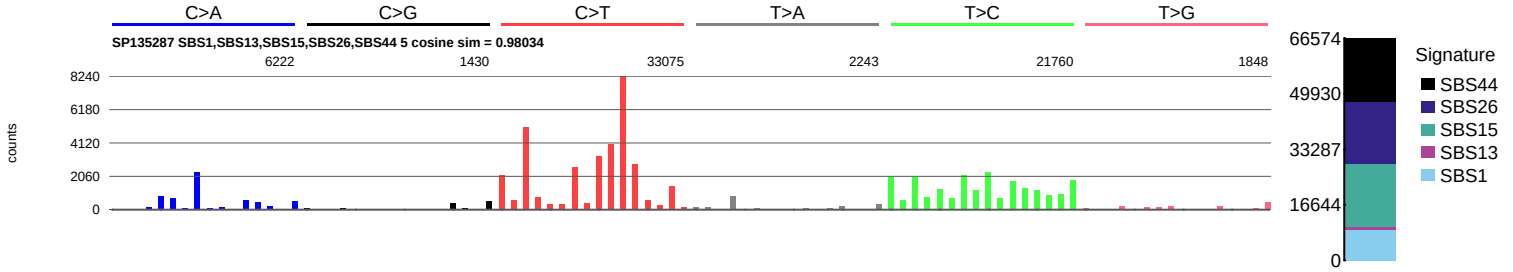

C>A
C>G
C>T
T>A
T>C
T>G
SP135287 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS26,SBS44 7 cosine sim = 0.98075
6132
1324
33720
2279
21318
1805
8148
6111
4074
2037
0
counts
66574
49930
33287
16644
0
Signature
SBS44
SBS26
SBS17a
SBS15
SBS13
SBS2
SBS1

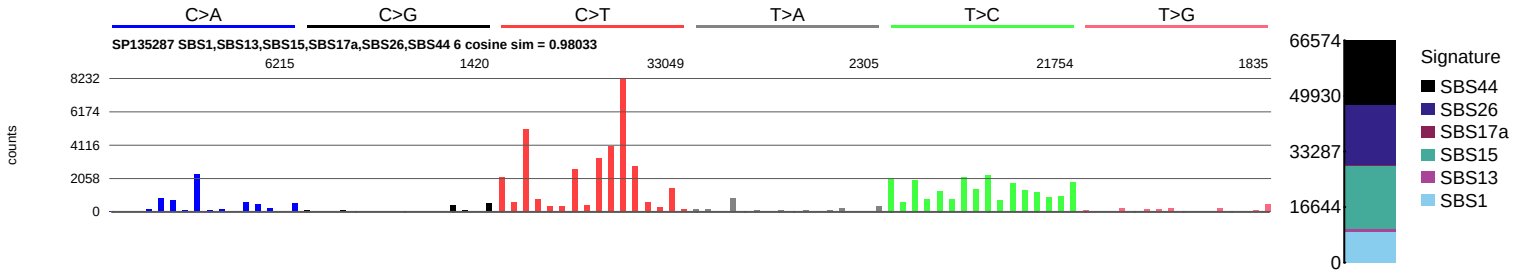

C>A
C>G
C>T
T>A
T>C
T>G
SP135287 SBS1,SBS2,SBS13,SBS15,SBS26,SBS44 6 cosine sim = 0.98075
6137
1331
33753
2220
21322
1813
8152
6114
4076
2038
0
counts
66574
49930
33287
16644
0
Signature
SBS44
SBS26
SBS15
SBS13
SBS2
SBS1

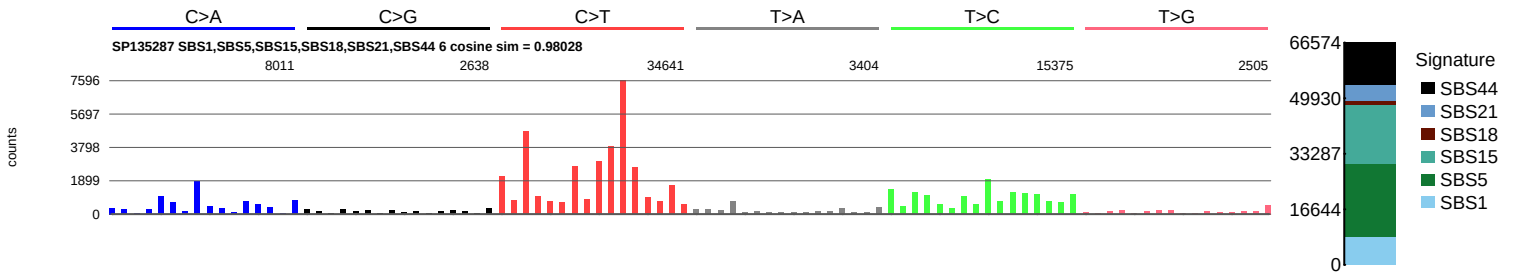

C>A
C>G
C>T
T>A
T>C
T>G
SP135287 SBS1,SBS5,SBS15,SBS21,SBS41,SBS44 6 cosine sim = 0.9806
7144
2764
34440
3888
15456
2881
7632
5724
3816
1908
0
counts
66574
49930
33287
16644
0
Signature
SBS44
SBS41
SBS21
SBS15
SBS5
SBS1

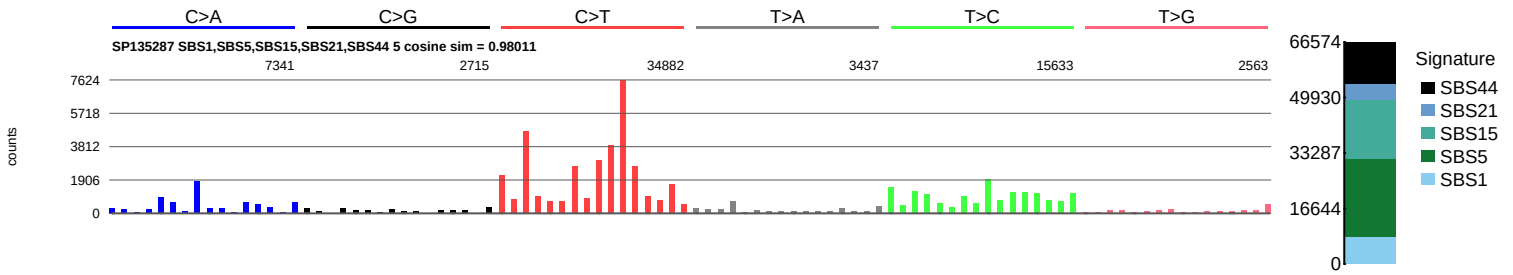

C>A
C>G
C>T
T>A
T>C
T>G
SP135287 SBS1,SBS15,SBS26,SBS44 4 cosine sim = 0.98046
6137
560
33404
2286
22298
1886
8296
6222
4148
2074
0
counts
66574
49930
33287
16644
0
Signature
SBS44
SBS26
SBS15
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135287 SBS1,SBS15,SBS17a,SBS26,SBS44 5 cosine sim = 0.98044
6132
561
33369
2356
22285
1876
counts
8284
6213
4142
2071
0
66574
49930
33287
16644
0
Signature
SBS44
SBS26
SBS17a
SBS15
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135287 SBS1,SBS13,SBS15,SBS26,SBS44 5 cosine sim = 0.98034
6222
1430
33075
2243
21760
1848
counts
8240
6180
4120
2060
0
66574
49930
33287
16644
0
Signature
SBS44
SBS26
SBS15
SBS13
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135287 SBS1,SBS13,SBS15,SBS17a,SBS26,SBS44 6 cosine sim = 0.98033
6215
1420
33049
2305
21754
1835
counts
8232
6174
4116
2058
0
66574
49930
33287
16644
0
Signature
SBS44
SBS26
SBS17a
SBS15
SBS13
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135287 SBS1,SBS5,SBS15,SBS18,SBS21,SBS44 6 cosine sim = 0.98028
8011
2638
34641
3404
15375
2505
counts
7596
5697
3798
1899
0
66574
49930
33287
16644
0
Signature
SBS44
SBS21
SBS18
SBS15
SBS5
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135287 SBS1,SBS5,SBS15,SBS21,SBS44 5 cosine sim = 0.98011
7341
2715
34882
3437
15633
2563
counts
7624
5718
3812
1906
0
66574
49930
33287
16644
0
Signature
SBS44
SBS21
SBS15
SBS5
SBS1